Electoral Studies 104 (2026) 103172

Contents lists available at ScienceDirect

Electoral Studies

journal homepage: www.elsevier.com/locate/electstud

Accuracy goals and wishful thinking in electoral expectations

Hannah Rajski

Mannheim Centre for European Social Research (MZES), University of Mannheim, A5 6, 68159 Mannheim, Germany

ARTICLE INFO

Keywords:
Electoral expectations
Vote expectations
Monetary reward
Wishful thinking
Accuracy goal
German election

ABSTRACT

Citizens' expectations about election outcomes shape strategic voting, election forecasting, and citizens' reactions to electoral outcomes. Yet, research shows that expectations are often distorted by wishful thinking, i.e., they are biased by voters' partisan preferences. While this phenomenon is well established, less is known about whether its magnitude reflects only citizens' partisan predispositions or also depends on the conditions under which electoral expectations are elicited. This paper addresses that question using monetary incentives as an experimental tool to induce accuracy goals. Building on dual-process theory and motivated reasoning, I argue that financial incentives increase cognitive effort, promote accuracy goals over directional goals, and thereby improve forecasting accuracy while attenuating wishful thinking. The analysis draws on two survey experiments conducted ahead of the 2025 German federal election and the subnational 2024 Saxon state election, involving more than 28,000 and 5000 eligible voters, respectively. Respondents who are offered monetary incentives spend more time answering the expectation question, provide more accurate vote share forecasts, and exhibit weaker wishful thinking. These findings suggest that observed wishful thinking reflects not only citizens' partisan predispositions but also respondents' engagement with the forecasting task, thereby qualifying the interpretation that it solely reflects stable partisan bias.

1. Introduction

Citizens' expectations about election outcomes play a central role in electoral politics. They shape strategic voting, campaign participation, and citizens' reactions to election outcomes. Moreover, aggregating electoral expectations into citizen forecasts has become an important source of election forecasting and has often been shown to rival or even outperform alternative forecasting approaches. Understanding how citizens form electoral expectations, and how accurate these expectations are, is therefore important for both theories of voting behavior and election forecasting. Yet research consistently shows that electoral expectations are systematically distorted by wishful thinking: citizens tend to overestimate the electoral prospects of their preferred party or candidate (see, e.g., Babad, 1995; Dolan and Holbrook, 2001; Ganser and Riordan, 2015; Granberg and Brent, 1983; Lewis-Beck and Skalaban, 1989; Lewis-Beck and Tien, 1999; Meffert et al., 2011; Miller et al., 2012; Mongrain, 2021a, 2023b; Morisi and Leeper, 2024; Sjöberg, 2009; Stiers and Dassonneville, 2018). Such distortions may lead citizens to vote strategically based on inaccurate beliefs and may also contribute to dissatisfaction with democratic processes when election outcomes differ from expectations (Mongrain, 2023a).

While the existence of wishful thinking is well established, much less is known about the conditions under which it arises. Existing research has primarily documented that supporters systematically overestimate their preferred party's electoral prospects. This paper instead asks whether the magnitude of observed wishful thinking depends not only on citizens' partisan predispositions but also on the conditions under which electoral expectations are elicited. If respondents devote little effort to a cognitively demanding forecasting task or prioritize directional over accuracy goals, the magnitude of observed wishful thinking may depend not only on what citizens believe, but also on how they engage with the task of forming and reporting those beliefs. Understanding whether motivational factors shape reported electoral expectations matters because it helps distinguish between partisan distortions that are largely stable and those that can be attenuated when respondents are motivated to answer accurately. More generally, it speaks to how researchers should interpret electoral expectations as measures of citizens' beliefs about future election outcomes.

This paper addresses this question by experimentally inducing accuracy goals through monetary incentives, designed to increase respondents' motivation to provide accurate forecasts while leaving their

Email address: hannah.rajski@uni-mannheim.de.

https://doi.org/10.1016/j.electstud.2026.103172
Received 6 February 2026; Received in revised form 6 October 2026; Accepted 9 October 2026
Available online 31 October 2026
0261-3794/© 2026 The Author. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).
Open access funding enabled and organized by Projekt DEAL.

underlying political knowledge unchanged. Importantly, monetary incentives are not proposed as a practical intervention for election surveys. Rather, they serve as an experimental tool to examine whether greater motivation to answer accurately changes respondents' engagement with the forecasting task. Building on dual-process theory and motivated reasoning, I argue that incentives increase cognitive effort, promote accuracy goals over directional goals, and thereby improve forecasting accuracy while attenuating wishful thinking. This design allows me to test not only whether citizens can provide more accurate forecasts, but also whether the wishful thinking observed in electoral surveys depends, at least in part, on the conditions under which electoral expectations are elicited.

To date, only two studies, both from Israel, have examined whether monetary incentives reduce wishful thinking in electoral expectations. Babad (1997) treated the monetary reward as a motivational remedy, but based on a between-subjects design found that it did not meaningfully reduce wishful thinking, measured by the predicted margin between two candidates compared across groups of differing partisan strength. More recently, Tikochinski and Babad (2023) showed that offering financial incentives shifted respondents' initial expectations closer to contemporaneous polls, thereby reducing wishful thinking, and interpreted this as evidence that voters are capable of more rational reasoning when sufficiently motivated.

Besides showing that the magnitude of wishful thinking depends not only on citizens' partisan predispositions but also on the conditions under which electoral expectations are elicited, this paper contributes to the literature in three further ways. First, this paper benchmarks electoral expectations against the actually realized future election outcomes about which the expectations were formed. Whereas Babad (1997) disregards the actual election outcome entirely and instead compares the predicted margin between two candidates across partisan preference groups, and Tikochinski and Babad (2023) assess wishful thinking relative to contemporaneous polls, this study is interested in the actual precision of citizens' predictive ability, i.e., how closely individuals' expectations match the eventual election result. Second, Tikochinski and Babad (2023) assume, but do not directly test, that incentivized respondents invest more effort into the task. This study addresses that gap by incorporating response times, an established behavioral measure of effort (Mayerl, 2013), thereby allowing a direct test of the assumed incentive–effort–accuracy chain. Third, this paper draws on two different electoral contexts: a highly salient first-order federal election and a lower-salience second-order state election that differ substantially in political importance, polling availability, and information environments. In doing so, the paper shows that the relationship between motivation, forecasting accuracy, and wishful thinking is not confined to a single type of election, and the consistency of the findings across these contrasting settings increases confidence in the robustness of the proposed relationship.

In more detail, I analyze two original survey experiments conducted before the 2025 German federal election and the 2024 Saxon state election. More than 28,000 and 5000 eligible voters, respectively, were recruited through an online access panel and social media advertisements. Respondents were randomly assigned to a control condition or a monetary incentive treatment before reporting expected vote shares for all major parties. The analyses support the expectations: Respondents who were offered a monetary reward spend more time answering the expectation question, provide more accurate vote share forecasts, and exhibit substantially weaker wishful thinking toward their preferred party, even though such distortions remain evident overall. Importantly, these patterns emerge consistently across both the first-order federal election and the second-order state election despite their different political and informational environments. Taken together, the findings suggest that observed wishful thinking reflects not only citizens' partisan predispositions but also the conditions under which electoral expectations are elicited. This qualifies the interpretation of wishful thinking as solely reflecting stable partisan bias. More broadly, the findings demonstrate that electoral expectations capture not only what citizens believe about election outcomes but also how they engage with the task of expressing those beliefs.

2. Electoral expectations and wishful thinking

Electoral expectations. Citizens' expectations about electoral outcomes and their varying accuracy are important to study for at least three reasons: First, they play a crucial role in psychological and economic theories of voting behavior (see, e.g., Meffert et al., 2011). Electoral expectations can affect voting behavior as demonstrated by research on strategic voting (see, e.g., Meffert et al., 2011) or bandwagon effects (see, e.g., Barnfield, 2020). When voting strategically, voters do not base their votes solely on their preferences for certain parties or candidates, but make strategic considerations based on their expectations of the election outcome (Cox, 1997). Rational voters maximize their expected utility by weighting their preferences based on their expectations. Strategic voting can thus mean defecting from voting for the preferred party, as it is not expected to pass the electoral threshold, to avoid a wasted vote. Alternatively, if the preferred party's coalition partner is close to the threshold, a rental vote for it rather than the preferred party to secure threshold insurance can also occur (Meffert and Gschwend, 2010). If citizens' expectations are inaccurate, their voting behavior may not reflect their true preferences. In some cases, individuals might even choose to abstain strategically based on misinformed beliefs. Additionally, campaign donations or volunteering can also be shaped by expectations (Searles et al., 2016). Second, expectations could shape satisfaction with democracy and perceptions of electoral integrity. If electoral outcomes diverge from citizens' expectations, this may erode trust in democratic processes (see Mongrain, 2023a, for evidence on the 2020 United States (US) election) or lead to disappointment with the outcome (Krizan and Sweeny, 2013). Third, electoral expectations can be aggregated into citizen forecasts, drawing on the wisdom of crowds. Building on seminal work in the US (Lewis-Beck and Skalaban, 1989; Lewis-Beck and Tien, 1999), citizen forecasts have since been shown to outperform not only classical vote intention polls but also quantitative models, expert judgments, and prediction markets (Graefe, 2014). The literature has continued to grow, with further contributions from the US (Murr and Lewis-Beck, 2021; Thompson-Collart et al., 2025) and many more countries, including the United Kingdom (Boon, 2012; Murr, 2011; Lewis-Beck and Stegmaier, 2011; Murr, 2016), France (Dufresne et al., 2022), Canada (Leiter et al., 2020), Mexico (Murr, 2025), Argentina (Thompson-Collart et al., 2024), and Germany (Ganser and Riordan, 2015; Lehrer et al., 2019; Leininger et al., 2026; Graefe, 2016; Murr and Lewis-Beck, 2022). Beyond simple equal-weighted aggregation, some scholars have explored weighted approaches that privilege expectations from more knowledgeable or competent individuals (Mongrain, 2021b; Murr, 2015).

Wishful thinking. Expectations are, however, often shaped by partisan preferences. As early as 1932, Hayes (1936) examined Americans' electoral expectations. Descriptive insights revealed that, among the over 8000 respondents, a large share expected their preferred candidate to win. More recent research has also identified partisan predispositions as the main driver of electoral expectations across different elections and countries (Mongrain, 2021a, 2023b). Further evidence exists from Israel (Babad, 1995), Austria, Germany (Meffert et al., 2011), Belgium (Stiers and Dassonneville, 2018), Sweden (Sjöberg, 2009), the US (Dolan and Holbrook, 2001; Granberg and Brent, 1983; Lewis-Beck and Tien, 1999; Miller et al., 2012; Lewis-Beck and Skalaban, 1989), and the United Kingdom (Morisi and Leeper, 2024).

Partisan predispositions bias voters either in the wrong direction (identifying with and overestimating the loser) or coincidentally in the correct one (if their preferred party wins) (Lewis-Beck and Skalaban, 1989). This is known as the preference–expectation link (Granberg and Brent, 1983) or wishful thinking (Babad et al., 1992). Searles et al. (2018) define wishful thinking as "a cognitive and motivational bias

in which an individual's desire for a preferred outcome increases their judgment of the likelihood of that outcome occurring" (Searles et al., 2018, 889). Wishful thinking can even persist after elections, when it comes to expectations about coalition formation (Barnfield et al., 2025). Additionally, it can occur not only in relation to one's own party or candidate, by overestimating its support, but also toward disliked parties, by underestimating theirs. Tikochinski and Babad (2023) label this wishful thinking for self and for others, contrasting two blocs: Center/Left and Right voters. Following most of the literature, the present study focuses on the preferred party by looking at deviations in expected vote shares compared to election results for the preferred party, as opposed to other parties. Additionally, to account for potential underestimations of less liked parties, degrees of preference are assessed.

3. Monetary incentives, effort and motivation

In this study, monetary incentives for accurate expectations serve as an experimental tool to examine whether the magnitude of observed wishful thinking depends on respondents' motivation to engage with the task of forming and reporting electoral expectations. Building on dual-process theory, satisficing, and motivated reasoning, I argue that incentives increase cognitive effort and promote accuracy goals, thereby improving forecasting accuracy and attenuating wishful thinking. Answering a survey item ideally involves four steps: First, comprehending the question, and second, searching for and retrieving information from memory (Tourangeau, 1984), in this case, for example, polls or past election results (Meffert and Gschwend, 2011; Irwin and van Holsteyn, 2002; Blais and Bodet, 2006; Stoetzer et al., 2024). Third, once relevant information has been retrieved, this information has to be integrated into one's judgment (Tourangeau, 1984). Previous research has shown that electoral expectations are not solely based on objective information, but that partisan wishes play an important role (Blais and Bodet, 2006). Finally, the respondent needs to select and report a response (Tourangeau, 1984), i.e., a vote share for each party.

Dual-process theory distinguishes between two modes of thinking: the automatic, intuitive, and fast System 1, which requires little effort, and the effortful and slower mode of thinking known as System 2 (Kahneman, 2011). System 1 includes the use of heuristics and biases, such as wishful thinking. When System 1 lacks an automatic, intuitive answer, System 2 and its more effortful reasoning process come into play. If people are aware of their biases, effortful thinking can be used to (potentially) overcome them. In survey research, this duality in cognitive processes and effort is reflected in satisficing versus optimizing. Respondents can execute the steps to answer an item with varying levels of effort. When they do so thoroughly, they engage in optimizing. Satisficing, by contrast, refers to executing these steps less thoroughly and choosing the first answer that seems acceptable (weak satisficing) or even skipping the steps of retrieval and judgment (strong satisficing). The likelihood of satisficing increases with higher task difficulty, lower respondent ability, and lower motivation (Krosnick et al., 1996). Eliciting vote share expectations across multiple parties is a cognitively demanding task, making satisficing a genuine concern. Monetary incentives for accurate responses aim to increase respondents' motivation and effort. They raise the salience of the task by signaling its importance and rewarding effortful processes—if they yield accurate responses. Thereby, they aim to encourage optimizing over satisficing. Crucially, the goal is not to change respondents' underlying beliefs or the information available in their memory, but to motivate them to draw on that information more thoroughly. Response latencies can serve as a proxy for information processing and respondents' degrees of elaboration, ranging from unconscious, automatic, and spontaneous information processing identified through fast response latencies to conscious, deliberative, and controlled processing, indicated by long latencies (Mayerl, 2013). Monetary incentives could not only increase internal effort to search for knowledge in one's memory, but also external effort through information search, e.g., via the internet. While the former allows the elicitation of subjective beliefs, the latter would allow for learning about belief updating (Grewenig et al., 2022) and thus about what affects electoral expectations, which is not the focus of this paper and should therefore be eliminated.

Additionally, when searching for electoral expectations, individuals may pursue different motivational goals. The concept of motivated reasoning from social psychology (Kunda, 1990) differentiates between two types of motives or goals: Individuals may be motivated to reach a directional (desired) conclusion or to arrive at an accurate conclusion. When motivated by a directional goal, individuals use beliefs and strategies intended to reach the desired conclusion. However, when motivated by an accuracy goal, individuals will seek strategies and beliefs that are appropriate and engage in more complex and elaborate reasoning (Kunda, 1990). Directional goals closely align with the concept of wishful thinking, as individuals process information in ways that align with their preferences. Partisan directional motives have been defined as "motives to support one's own political party by answering in a certain way" (Bullock and Lenz, 2019, 328). Monetary rewards for accurate responses could, however, create an accuracy goal, motivating respondents to report their genuine beliefs more faithfully. Importantly, the mechanism at work here is not that of motivated learning (Khanna and Sood, 2018): While past research has shown that recall of polls can also be prone to partisan bias (Levine, 2007) and that baseline information varies depending on which (partisan) news outlets one turns to (Searles et al., 2018, 2016), incentives are neither intended nor able to change which information individuals process, how they process it, or how they form beliefs in the first place (Khanna and Sood, 2018). Rather, the goal is to motivate respondents to report the beliefs they actually hold, counteracting motivated responding by reducing the pull of directional goals (Khanna and Sood, 2018).

Monetary stakes in electoral expectations are best known from prediction markets such as the Iowa Electronic Markets. In such markets, traders buy and sell contracts on election outcomes, and the market prices have been used successfully to forecast elections (see, e.g., Gruca and Rietz, 2025; Kou and Sobel, 2004). However, the mechanism underlying prediction markets is distinct from the incentive design used in this study in four ways. First, markets, like citizen forecasts, focus on aggregate accuracy, whereas this study asks whether incentives improve the accuracy of individuals' reported beliefs, independent of any aggregation. Second, the causal order is reversed: In markets, expectations lead people to trade, whereas here the incentive is an experimental treatment intended to increase effort and accuracy when reporting expectations. Third, traders self-select into markets and may seek out information before trading, whereas respondents in this study are randomly assigned to the incentive regardless of their interest, confidence, or expertise, and the design aims to elicit existing beliefs rather than to induce information search. Fourth, traders invest their own money and risk losing it, whereas respondents here can only gain.

Whether monetary incentives reduce partisan biases in survey responses has so far mainly been tested with factual questions. If incentives reduce such biases, the biases must at least partly reflect low effort and directional goals in responding rather than sincerely held beliefs. The evidence shows that this is often, but not always, the case. Monetary incentives improved accuracy on political knowledge items (Prior and Lupia, 2008). Incentives also reduced partisan gaps in answers to factual questions about economic conditions (Bullock et al., 2015; Prior et al., 2015), suggesting that respondents perceive such questions not only as factual questions but also as opportunities to express partisan support. Similarly, incentives improved citizens' accuracy in identifying true and false political headlines and reduced partisan bias therein (Rathje et al., 2023). Incentives do not, however, close every partisan gap: Farjam and Bravo (2024) found that they reduced partisan differences in numerical estimates about global warming, but not about race and crime, where survey responses appear to reflect sincerely held beliefs. Monetary rewards can thus shrink partisan biases in responding, but not differences in what citizens believe.

Electoral expectations are a harder test than factual knowledge questions. Factual questions have a true answer that respondents can know, retrieve from memory, and be incentivized to report truthfully. Election outcomes, by contrast, are unknown when expectations are formed. That incentives can also work for forecasts is suggested by evidence on facts whose answers lie in the future: Respondents who were incentivized to estimate figures on international events spent more time on the task and were more accurate, although all respondents were allowed to search for information (Jamieson and Weller, 2020). In the present design, by contrast, respondents were asked not to consult outside sources, so that incentives would affect how they used the information they already held. For electoral expectations and wishful thinking, the evidence is limited to two studies from Israel, and findings are mixed. Babad (1997) offered a financial incentive before the 1993 Jerusalem mayoral and city council elections and measured wishful thinking as the predicted margin between two candidates across five groups, ranging from strong supporters of one candidate to strong supporters of the other. The incentive shifted predictions significantly, but wishful thinking remained substantial. Tikochinski and Babad (2023), by contrast, found that incentives eliminated wishful thinking in the 2020 Israeli national election, measured relative to contemporaneous polls in a within-subject design. They attributed this to a shift from intuitive, fast *Type 1* thinking to slower, more deliberate *Type 2* reasoning and concluded that citizens can actually think rationally about electoral outcomes when sufficiently motivated to do so. However, they inferred this shift from changes in reported expectations alone, without a behavioral measure of effort such as response latency. Whether incentives reduce wishful thinking relative to the actual election outcome, and whether they do so through the incentive–effort–accuracy chain, thus remains an open question.

To sum up, the dual-process model differentiates between intuitive and fast thinking with low effort, which, in the survey context, is shown through satisficing, as opposed to more effortful and slower thinking. Motivated reasoning distinguishes between being motivated by directional goals, such as wishful thinking, and by an accuracy goal. Monetary rewards are intended to shift respondents toward more effortful thinking and toward being motivated by an accuracy goal, thereby increasing accuracy and decreasing wishful thinking. Based on this, three different hypotheses can be formed. First, if monetary incentives motivate individuals to prioritize accuracy and shift from intuitive to more effortful thinking, they should take longer to form their expectations:

H1 (Response Latency) *Offering a prize for the most accurate electoral expectations increases response times.*

Additionally, incentivized respondents will not only take longer, but will also have an accuracy goal that makes their expectations more accurate:

H2 (Overall Accuracy) *Offering a prize for the most accurate electoral expectations increases the accuracy of individuals' expectations.*

Finally, this increase in accuracy should be particularly pronounced for expectations about the preferred party. Directional goals, intuition, and wishful thinking should be overcome, producing more accurate outcomes through more effortful reasoning:

H3 (Wishful Thinking) *Offering a prize for the most accurate electoral expectations decreases wishful thinking.*

While H2 on individual forecasting accuracy was pre-registered, H1 testing the time mechanism and H3 on wishful thinking were not. For more details, see Appendix A.

4. The German federal election 2025 and the Saxon state election 2024

A major contribution of this paper is testing these hypotheses in two contexts that differ substantially in salience, stakes, and information environment. The 2025 German federal election is a first-order election with high media attention and extensive polling coverage, whereas the 2024 Saxon state election is a second-order election, characterized by lower turnout, weaker media coverage, and sparser polling (Reif and Schmitt, 1980). While turnout in both elections was the highest since German reunification, comparing turnout within the same electorate reveals the expected pattern: Among Saxons eligible to vote, turnout was 81.1% in the federal election, compared to 74.4% in the state election (Bundeswahlleiterin, 2025; Statistisches Landesamt des Freistaates Sachsen, 2024b). Additionally, while nearly 90 vote intention polls were published in the three months preceding the federal election (see Zicht and Cantow, 2025), only 10 were published ahead of the state election (see Zicht and Cantow, 2026). This thinner information environment and lower perceived stakes could plausibly affect respondents' baseline motivation and the effort required for the forecasting task. If the financial incentives operate through the effort- and accuracy-goal mechanisms outlined above, a similar pattern of results should emerge in both the first-order and second-order elections, despite their different political and informational environments.

While the federal election was originally scheduled for September 2025, an early election was held on February 23, 2025, following the collapse of the "traffic-light" coalition, composed of the Social Democratic Party (SPD), the Alliance 90/The Greens (Greens), and the Liberals (FDP). The election not only came early, but it was also the first time that the Bündnis Sahra Wagenknecht (BSW), founded in 2024 by former members of The Left (Die Linke), competed in a federal election. In federal elections, German voters cast two ballots under a mixed-member electoral system that combines proportional representation with single-member districts. The second vote (*Zweitstimme*) is for a party list in the voter's federal state and determines the overall allocation of seats in parliament based on proportional representation. A 5% electoral threshold is in place. With the nominal vote (*Erststimme*), voters elect a candidate in their single-member district. This study focuses solely on the *second vote*. Table 1 illustrates the results. On election day, the opposition parties increased their vote share compared to 2021. Christian Democrats (CDU/CSU) came in first, while the right-wing populist AfD ("Alternative for Germany") doubled its vote share, coming in second. The Left, which had polled near the threshold for much of the campaign, gained momentum in the final weeks before the election (starting in late January/early February). In contrast, all parties of the traffic-light coalition experienced substantial losses. Falling below the 5% electoral threshold, the FDP failed to gain representation in parliament. The newly founded BSW also narrowly missed entering parliament.

The Saxon state election took place on September 1, 2024, roughly half a year before the federal election. As at the federal level, Saxony has a mixed-member proportional electoral system, with a candidate vote (*Direktstimme*) and a party list vote (*Listenstimme*). The latter is

Table 1
Election Results.

	Germany 2025		Saxony 2024	
Party	Share	Difference to 2021 (pp)	Share	Difference to 2019 (pp)
CDU	22.6	+3.6	31.9	−0.2
CSU	6.0	+0.8	–	–
AfD	20.8	+10.4	30.6	+3.1
SPD	16.4	−9.3	7.3	−0.4
Greens	11.6	−3.1	5.1	−3.5
FDP	4.3	−7.1	0.9	−3.6
The Left	8.8	+3.9	4.5	−5.9
BSW	4.98	–	11.8	–
Others	4.6	−4.1	7.8	−1.4

Note: The table shows the vote shares in percent for the second votes (*Zweitstimmen*) for the federal election on February 23, 2025, and list votes (*Listenstimmen*) for the Saxon state election on September 1, 2024 as well as changes relative to the previous elections (2021 federal election and 2019 Saxon state election) in percentage points. The BSW did not run in the previous elections, and the CSU only stands for election in Bavaria. The Free Voters (2.3% in Saxony) are included in Others. *Sources:* The Federal Returning Officer (2025a); Statistisches Landesamt des Freistaates Sachsen (2024b).

decisive for seat allocation and is the focus of this paper. Again, a 5% threshold applies. As in 2019, the CDU received the most votes, followed by the AfD. The BSW entered parliament with 11.8%, while especially the Greens, FDP, and the Left lost substantially. The Free Voters (Freie Wähler) and the Left did not pass the electoral threshold, but gained representation in parliament through one and two direct mandates, respectively.

5. Data and methods

5.1. Data

The federal survey was fielded from January 24 until February 21, 2025, two days before election day, as part of the zweitstimme.org research project. 29,195 Germans eligible to vote were recruited through an online access panel from Bilendi. Quotas were implemented to approximate gender balance (targeting 50% men and 50% women). The subnational study was conducted ahead of the Saxon state election and was fielded from July 26 until August 30, 2024, two days before the election. A total of 5659 respondents were recruited through the Bilendi online access panel ($n = 2223$) and through social media advertisements on Instagram and Facebook ($n = 3436$). For the panel recruits, quotas were implemented to approximate gender, age, and educational balance (interlocking gender and age groups). The social media sample was recruited following the best-practice principles proposed by Neundorf and Öztürk (2023, 2021). In the Facebook Ad Manager, four conversion campaigns were set up targeting young women, young men, older women, and older men, respectively.

Neither the federal nor the Saxon sample is representative of its electorate. Respondents self-selected into the surveys. Most of them were members of the online access panel: They had signed up for the panel and, once invited to the survey, could decide whether to participate, possibly also motivated by the small compensation they receive for taking part. The Saxon social media respondents self-selected into the survey by clicking on a social media advertisement for a scientific election study. Tables B.3 and B.4 compare the samples with their respective electorates. The federal and the Saxon samples both share typical distortions of online non-probability samples. Citizens aged 60 and above are strongly underrepresented (25.4% in both samples vs. 42.6% of the federal and 45.8% of the Saxon electorate), while those aged 30 to 59 are overrepresented. Citizens aged 18 to 29 are represented well. In line with this, CDU/CSU voters are underrepresented (−7.8 percentage points in the federal and −9.9 percentage points in the Saxon sample). Compared to the German and Saxon populations, the samples are also considerably more highly educated: Those with high educational attainment are strongly overrepresented (Germany: 52.3% vs. 37.2%; Saxony: 49.1% vs. 29.8%), while those with low educational attainment are strongly underrepresented (Germany: 10.7% vs. 29.7%; Saxony: 4.9% vs. 15.6%). Those with medium educational attainment are somewhat overrepresented in the federal sample (+5.8 percentage points) but underrepresented in the Saxon sample (−8.4 percentage points). The federal sample resembles the electorate closely in gender, federal state, and turnout intention, but overrepresents voters of smaller "other" parties (+9.8 percentage points). The Saxon sample shows further distortions, which are especially pronounced among the respondents recruited via social media. Women are underrepresented overall (40.7% vs. 51.9%), driven by social media respondents (−17.8 percentage points). Additionally, a disproportionately large share of respondents reported intending to vote (90.6% vs. an actual turnout of 74.4%), concentrated among social media respondents (97.8%). Voters of the Greens are overrepresented, more strongly among social media respondents (+6.7 percentage points) than in the online access panel (+1.5 percentage points), while voters of the AfD are overrepresented only among social media respondents (+5.9 percentage points) and slightly underrepresented in the online access panel (−1.6 percentage points). These distortions must be kept in mind when generalizing beyond the samples.

The surveys included several items on respondents' political preferences, expectations, and sociodemographic characteristics. In this study, the main focus is on respondents' expectations about the parties' vote shares. Before providing their answers, respondents were shown a commitment pledge asking whether they agreed not to consult outside sources. As an experimental treatment, a subgroup of respondents was informed that those who made the most accurate predictions would receive a financial reward after the election. Subsequently, all respondents reported their expectations about the parties' vote shares. Response times for this item were collected. Respondents' political preferences were assessed using three items: their propensity to vote for specific parties (PTV), their vote intention (VI), and their party identification (PID). The detailed wording of the items can be found in Appendix Table B.1 for the federal election survey and Table B.2 for the Saxon survey.

Commitment pledge. To elicit respondents' genuine beliefs and prevent them from engaging in information-seeking behavior, a commitment pledge was included (see Tables B.1 and B.2). In it, respondents were asked whether they would commit themselves to not using outside sources when answering the expectation items. This approach has been successfully implemented when asking political knowledge questions (Clifford and Jerit, 2016). Unlike knowledge questions, the true answers to questions about future election outcomes cannot be found. Complete cheating is thus not possible. However, respondents could look for vote intention polls or past election results. Previous research has found them to be a baseline for electoral expectations (Meffert and Gschwend, 2011; Irwin and van Holsteyn, 2002; Blais and Bodet, 2006; Stoetzer et al., 2024). Over 98% of respondents committed themselves to not consulting other sources.

Monetary incentive. To motivate respondents to be more accurate and to make more effort when answering the expectation items, monetary incentives for accurate expectations were used as a treatment. Generally, all respondents from the online access panel received a small baseline compensation for participating in the survey. In addition, two-thirds of the federal participants and one-half of the Saxon respondents had the opportunity to earn a monetary reward for accurate forecasts. At the federal level, two different tournament payoffs were used, each distributing €500 among the most accurate respondents. The treatments varied in the number of top-ranked respondents rewarded, and thereby in the amount rewarded per person and the winning probabilities. While respondents were informed of the number of winners, they were not aware of the total number of participants. Specifically, one-third of the federal participants and half of the Saxon participants (the *Treatment (500)* groups) were informed that the three most accurate respondents would share a prize of €500. In the federal survey, another third (the *Treatment (5)* group) were told they would receive €5 each if they ranked among the 100 most accurate participants. Participants in the control group received no such information. Respondents were randomly assigned to one of the groups. The experimental groups are similar in age, gender, education, voting behavior, and (for the federal election) state of residence (see Tables B.3 and B.4). Because the incentives serve to increase effort and motivation rather than to study prize design, employing two different incentives at the federal level provides a check on whether the effect is robust to reward structure. Ex ante, it is not obvious which format should be more motivating. Experimental studies that hold the total prize fixed generally find that concentrating it in fewer prizes elicits higher effort than spreading it across many (Dechenaux et al., 2015), in line with a tendency to overweight small chances of large gains (Kahneman and Tversky, 1979). However, dispersing prizes more widely can raise effort under some conditions, such as when participants are risk-averse or differ substantially in ability (Dechenaux et al., 2015). Because respondents were not told how many others they were competing with, participants in neither treatment group could accurately gauge their odds of winning. Given these competing considerations, I have no

strong expectation regarding the two treatments' relative effects and treat them as complementary means of inducing accuracy motivation.

Vote share expectations. The next question asked for the expected vote shares of all parties that might have had a chance of entering parliament. In the federal election, this included CDU/CSU, SPD, AfD, Greens, FDP, Left, BSW, plus others. The survey item read as follows: *If you first think about the party list votes, which determine the distribution of seats in the federal parliament: What percentage of the votes do you expect the various parties to receive nationwide?* Respondents used sliders ranging from 0% to 50% to allocate their vote shares among the seven parties and others. A maximum of 50% on each slider prevented respondents from providing very extreme or irrational answers. Thus, the accuracy estimates can be considered conservative. The last time a party received an absolute majority in a federal election was in 1957, when the CDU/CSU won. All sliders were initially set to 0%. While adjusting the sliders, participants were continuously shown the percentage assigned to each party and the total percentage allocated so far. Participants could adjust the sliders or enter the percentages. If the total reached 100%, further increasing any slider was not possible; respondents first had to reduce another party's share. Respondents could proceed to the next page only after allocating exactly 100%. In Saxony, the item was very similar but also included the Free Voters, and voters could distribute vote shares up to 100% per party. As the items were forced-response items, I interpret assigning 100% to one party in Saxony and 50% to two parties at the federal level as satisficing. I therefore exclude 765 federal respondents and 97 Saxon respondents. This yields sample sizes of 28,430 and 5562 respondents, respectively. Because the treatment itself may affect such behavior, I test whether exclusion differs by experimental group. Treated respondents were indeed less likely to be excluded (federal: 3.1% in the control group, 2.6% in *Treatment (5)*, and 2.2% in *Treatment (500)*; Saxony: 2.2% in the control group vs. 1.2% in *Treatment (500)*; see Tables C.3 and C.4), in line with the argument that monetary incentives reduce satisficing. More highly educated respondents and those intending to vote were also less likely to be excluded, as were older respondents in the federal sample and respondents recruited via social media in Saxony. However, as only a few respondents are excluded, the composition of the samples barely changes: No share changes by more than 0.6 percentage points, overall or within any experimental group. Experimental groups remain comparable (Tables B.3 and B.4). As a robustness check, all models are nevertheless estimated without excluding these respondents.

Fig. 1 illustrates the distribution of individuals' answers per party, also indicating the actual election result and the average vote share expectation, which is equivalent to citizen forecasts. Individual expectations are widely dispersed, with standard deviations (SD) at the federal level ranging from 4.89 for the FDP to 8.55 for the AfD. In Saxony, a similar picture emerges, with standard deviations ranging from 4.19 for the FDP to 12.45 for the AfD. Tables B.5 and B.6 in the Appendix present average expected vote shares and their deviations from the actual results for each experimental group per party. Standard deviations are mostly slightly larger in the control groups than in the treatment groups.

5.2. Modeling

Response latency. If respondents in the treatment groups put more effort into reporting their electoral expectations, it is reasonable to assume that they took longer to answer the question about parties' expected vote shares (H1). To test this expectation, the time it took the respondents to move all sliders and proceed to the next page was measured. Following common practice, all respondents whose response time exceeded two standard deviations above the mean are removed (Mayerl and Urban, 2008). For the federal election, this results in the exclusion of 144 respondents who took longer than 414.4 seconds to answer the item. The distribution of response times is shown in Fig. B.1. It is reasonable to assume that these respondents were distracted when answering (Mayerl and Urban, 2008; Mayerl, 2013). Given a mean of 66.9 seconds and a standard deviation of 173.8 seconds, no response could fall more than two standard deviations below the mean. Speedy responders are thus not removed from the analyses. In Saxony, the average response time was higher, at 110 seconds, with a standard deviation of 330 seconds. 8 respondents who took longer than 770 seconds are excluded. The dependent variable is the logarithm of the raw response times, as the response times are positively skewed (see Fig. B.2 for the distributions of the raw and logged response times). To account for respondents' individual speed, a baseline speed measure is constructed as the arithmetic mean of five filler latencies from single-choice items that are not of interest to the analyses (Mayerl, 2013): eligibility to vote, year of birth, turnout intention, vote intention, and political interest (see Tables B.1 and B.2 for the item wording). Prior to aggregation, values more than two standard deviations above or below the item-specific mean are excluded. As with the expectation response time, the baseline speed measure is log-transformed. Additionally, the total survey duration (in minutes) was measured. To test H1, I regress the logged response latency on treatment status, controlling for logged baseline response speed and total survey duration, and including survey date fixed effects, which capture the lead time, i.e., the number of days before the election. For better substantive interpretation, I rely on the approach of Rittmann et al. (2025) to transform the logged outcomes into seconds. For this, I

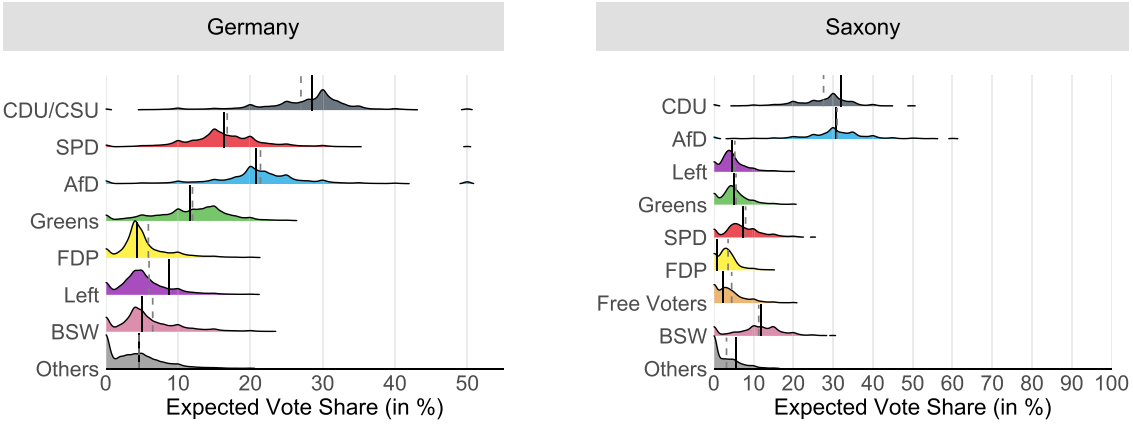


Fig. 1. Distribution of individuals' party vote share expectations. *Note:* The figure illustrates the respondents' expectations on the parties' vote shares. Respondents used sliders ranging from 0% to 50% (Germany) or 100% (Saxony), and all vote shares had to sum to 100%. The black solid lines indicate the parties' actual vote shares and the gray dashed ones the average across individuals' expectations. Values can be found in Tables B.5 and B.6.

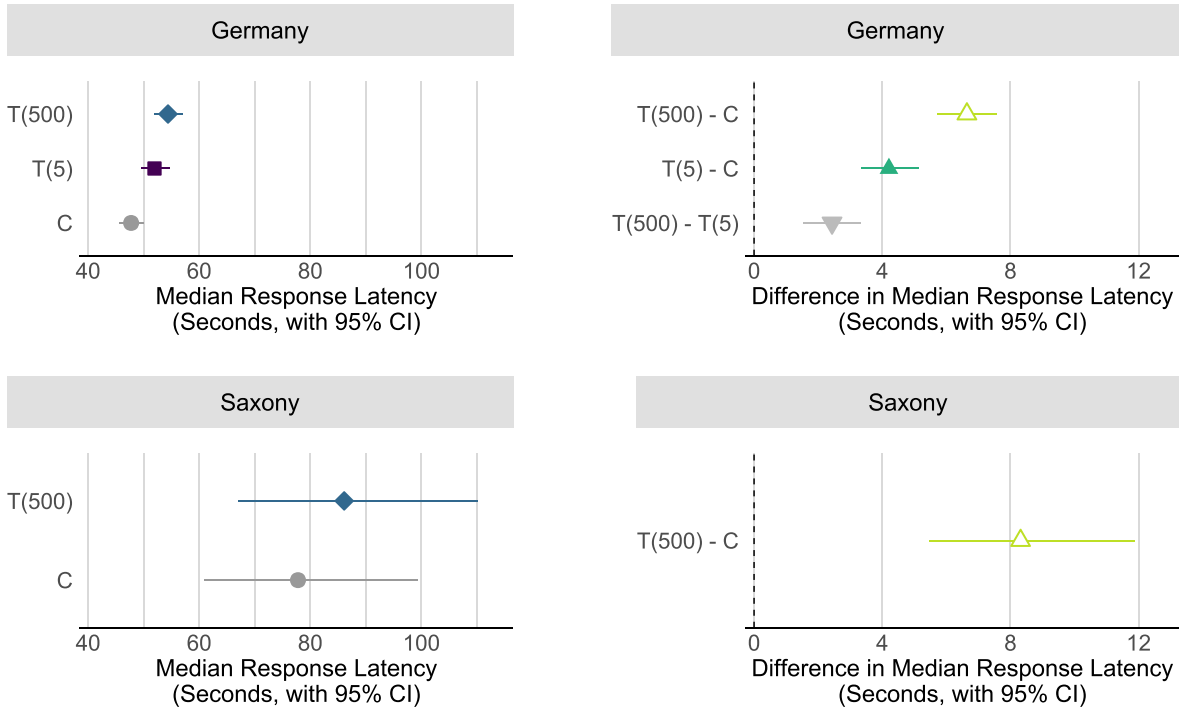


Fig. 2. Effects of treatment on median response latency. *Note:* The left panels display the conditional median response latencies (in seconds) for the expectation item by group. The right panels show the first differences between these groups. The top row reports results from the federal election, and the bottom row from Saxony. Estimates are based on model 1, and transformed to seconds (Rittmann et al., 2025) for the average-case scenario described in Section 5.2. Lines correspond to 95% confidence intervals. Treated respondents took longer, on average, than respondents in the control condition. Additionally, for the federal election, the *Treatment (500)* condition yielded, on average, longer response times than the *Treatment (5)* condition. As shown in the right panels, all group differences are statistically different from zero.

use an average-case scenario, setting the logged baseline speed and the total duration to their means (2.1 and 22.5 minutes, respectively), and the lead time to 15 days. For the Saxon case, this corresponds to a logged baseline speed of 2 and a duration of 21.8 minutes, and a lead time of 20 days. I present the conditional median response latency in seconds for the different groups, along with the first differences, i.e., the differences between two groups, conditional on this scenario. The conditional mean response latency is reported in the Appendix.

Overall accuracy. To assess individuals' accuracy in their vote share expectations across all parties and a residual "others" category with regard to the eventual election result, the mean absolute error (MAE) is used. It is defined as

$$MAE = \frac{1}{P}\sum_{p=1}^{P} |\hat{e}_p - v_p|$$

where $p = 1, \ldots, P$ indexes the parties and "others". The sum of the absolute differences between each party's expected vote share $\hat{e}_p$ and its electoral result v_p is divided by the number of parties P. In the federal case, $P = 8$ (seven parties plus others), while Saxony also included the Free Voters ($P = 9$). Accuracy levels are broadly comparable across the two samples, with Saxon respondents showing a marginally higher average MAE (4.77 vs. 4.46) and marginally greater dispersion ($SD = 3.15$ vs. $SD = 2.91$). Yet, in both elections, individual spread was quite broad, ranging from 0.43 to 20.48 (federal) and from 0.50 to 21.50 (Saxony). To test H2, I regress the MAE on the treatment status with survey date fixed effects. I present predicted values based on the observed-value approach: Each respondent is set to each treatment condition while all other variables are kept at their observed values, and predictions are then averaged over the sample (Hanmer and Ozan Kalkan, 2013). As an alternative accuracy measure, the root mean squared error (RMSE) serves as a robustness check.

Wishful thinking. The phenomenon of wishful thinking, defined as deviations in electoral expectations from the election result based on partisan preferences, is at the heart of this paper. To assess wishful thinking across partisan preferences and to test the effect of the motivational incentive, I create a stacked dataset representing respondent-party dyads, pairing each respondent's electoral expectation with their party preference. The residual category "Others" is excluded because it pools a heterogeneous set of smaller parties, and preferences are only measured for major parties. I create a dummy variable indicating whether a party was the respondent's preferred party. I operationalize the preferred party as the party for which respondents have the highest *propensity to vote*. The PTV item asked respondents about their likelihood of ever voting for each of seven parties: CDU/CSU, SPD, AfD, Greens, FDP, the Left, and BSW (see Table B.1), plus Free Voters in Saxony (see Table B.2). Answer options ranged from 1 ("would never vote for this party") to 11 ("would most likely vote for this party"). I identify respondents with a clearly preferred party as those with a single party having the highest PTV score. This party is coded as 1, while parties with lower scores are coded as less preferred parties (0). At the federal level, for 5354 respondents, more than one party received the highest score, indicating no clear preference. Additionally, 20 respondents did not answer any PTV item. This leaves 23,056 respondents for whom a clearly preferred party p^* can be identified based on the PTV measure. In Saxony, 8 respondents did not answer any propensity to vote item, while 889 had more than one preferred party based on this operationalization. For 4665 Saxons, one clearly preferred party can be identified.

The dependent variable measures individuals' forecast error per party, i.e., the deviation between a respondent's expected vote share $\hat{e}_p$ and the party's actual vote share v_p, giving a percentage point under- or overestimation: $\hat{e}_p - v_p$. For the preferred party, most respondents overestimated the party's vote share, though underestimations also occurred (see Figs. B.5 and B.6). This underestimation is especially pronounced

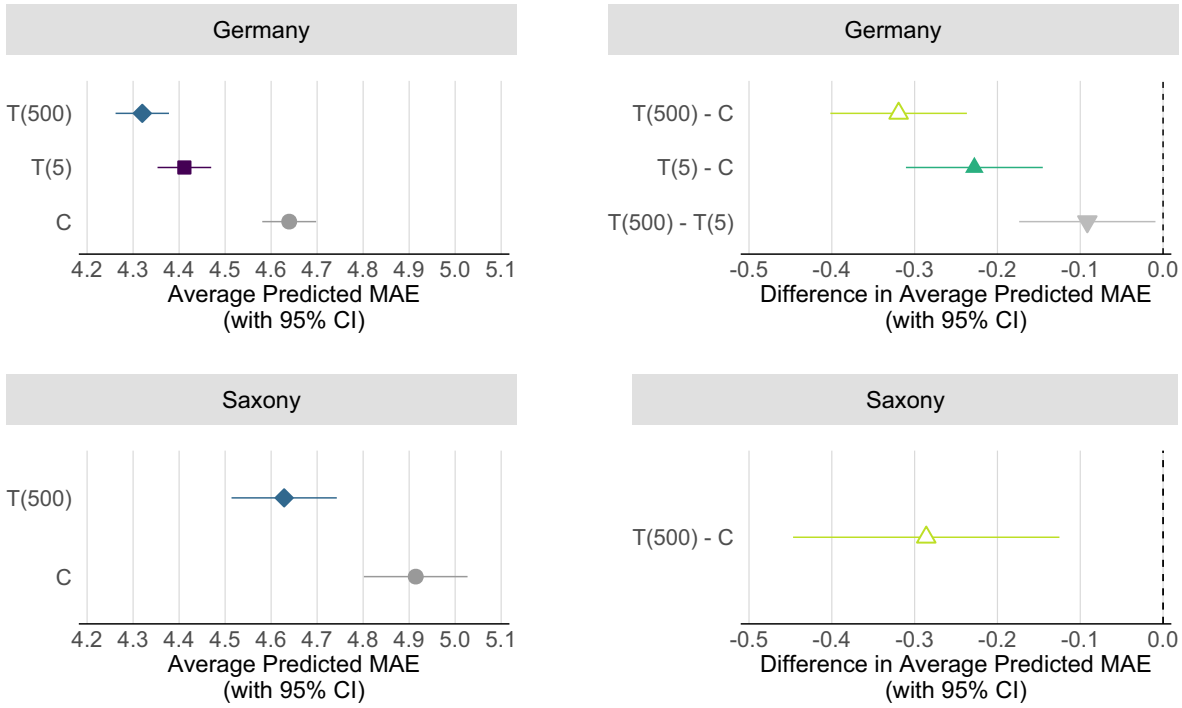


Fig. 3. Effects of treatment on MAE. *Note:* The dots in the left panels show the average predicted MAE of respondents' expected vote shares by experimental group based on model 2a, using an observed-value approach. The right panels show the differences between groups. Lines correspond to 95% confidence intervals. The top row reports the federal election, the bottom row reports Saxony. In the federal election, respondents in the *Treatment (500)* condition had the lowest error, i.e., were most accurate, followed by *Treatment (5)*, while the control group had the highest error. In Saxony, respondents in the *Treatment (500)* condition were likewise more accurate than the control group. As the right panels show, all differences between the groups are statistically distinguishable from zero.

for the Left, which gained momentum in the last weeks of the federal campaign, and for the CDU in Saxony.

To assess wishful thinking, I first analyze how the forecast errors for a given party differ between individuals who prefer that party and those who do not. This is done by using separate linear regression models by party, each regressing a respondent's forecast error for that party on a dummy variable indicating whether the respondent prefers that party or not. To test H3, I rely on the stacked dataset of respondent-party dyads and regress the respondent's forecast error for a party on treatment status, interacted with a dummy for whether the respondent prefers that party, as well as survey date and party fixed effects. Because each respondent contributes expectations for multiple parties, observations are not independent within respondents. I therefore report respondent-clustered standard errors. Some respondents did not provide PTV scores for all parties. The corresponding respondent-party dyads are excluded, meaning the number of dyads per respondent is not always identical across the sample.

As robustness checks, I use three alternative operationalizations of the preferred party. First, based on the PTV measure, I allow respondents to have more than one preferred party, coding all parties ranked highest as 1, instead of excluding respondents with more than one preferred party. Second, I use the party a respondent *intends to vote for*, restricting the analysis to respondents intending to vote for one of the parties for which expectations were elicited. Vote intention may not perfectly capture respondents' preferred party, as it could reflect strategic defection from a sincere first preference. The propensity to vote measure is more general, asking about the likelihood of ever voting for a party, but this too could be affected by strategic considerations. Third, I therefore use *the party respondents identify with*. For the federal election, this item was fielded in a later wave (February 13–21), which should not be of concern given the relative stability of party identification, though it is available for only a third of the respondents. Among respondents for whom all three measures are available, the measures yield the same party for 78.7% at the federal level and for 84.5% in Saxony. Figs. B.3 and B.4 compare the measures in detail. Additionally, I use the PTV scores instead of dichotomizing them and interact them with treatment status to account for degrees of preference.

6. Results

Thinking more slowly. Turning to the time component first, respondents who are treated are expected to take longer to answer the expectation question than those in the control group (H1). Respondents in the treatment groups indeed took longer to respond to the expectation question on average (see model 1 in Tables C.1 and C.2), suggesting more effortful responding. Fig. 2 displays the conditional median response latencies in seconds and the corresponding first differences. Respondents in the control group were substantially quicker to report their expectations than respondents who could receive a reward for accurate expectations. At the subnational level, response latencies were considerably higher than those observed for the federal election: The conditional median in the Saxon control group was, on average, 77.8 seconds (with the 95% confidence interval ranging from 60.8 to 99.3 seconds) compared to 47.7 [45.6; 50.1] seconds at the federal level. This likely reflects multiple factors. Saxon respondents had to provide vote share estimates for one additional party, the Free Voters, but the substantially longer times suggest further reasons as well. In particular, party vote shares may be less salient at the subnational level, given that state elections receive less polling coverage, leaving respondents with less information to draw on. The treatments increased response times in all cases. In Saxony, the median response time among the treated was, on average, 8.3 seconds higher [5.5; 11.9] than in the control group. At the federal level, the median response time was, on average, 6.6 [5.7; 7.6] seconds higher than in the control group in the *Treatment (500)* condition, and 4.2 [3.4; 5.1] seconds higher in the *Treatment (5)* condition. This is consistent with H1 and the expectation that the monetary incentive motivates respondents

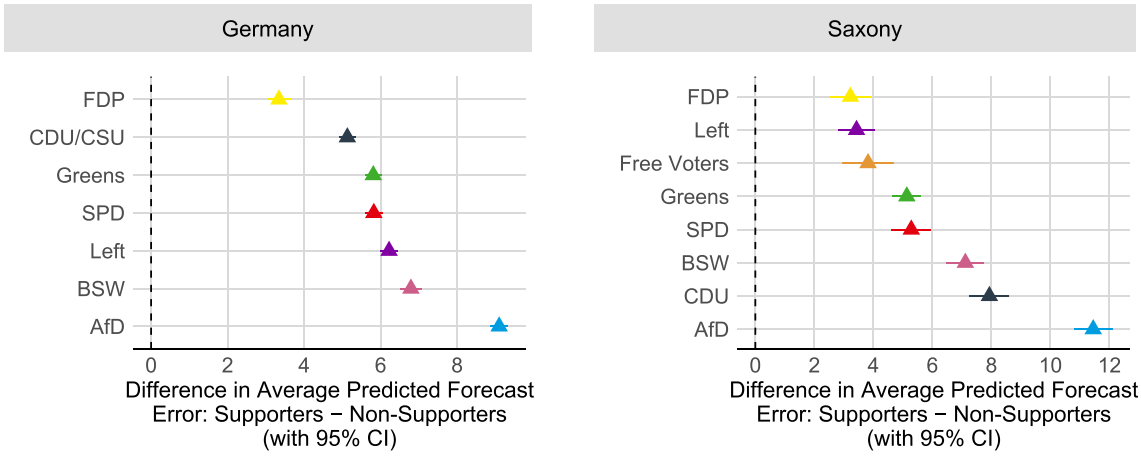


Fig. 4. Wishful thinking in electoral expectations. *Note:* The figures show the difference in average predicted forecast error (expected minus realized vote share) between supporters and non-supporters for each party, for the federal election (left) and the Saxon state election (right). Lines correspond to 95% confidence intervals. Across all parties, federal supporters overestimate their preferred party by between 3.3 [3.0; 3.7] and 9.1 [8.9; 9.3] percentage points on average compared to non-supporters. In Saxony, the difference ranges from 3.2 [2.5; 3.9] to 11.4 [10.8; 12.1] percentage points. For the predicted forecast errors by supporters and non-supporters, refer to Fig. B.7.

to spend more effort on the task. It is not known how treated respondents used the additional time. They could have used the additional time to think more about the task or to seek external information, such as polls. Yet, the vast majority of respondents committed themselves to not using outside sources. Additionally, it is noteworthy that the *Treatment (500)* respondents took on average 2.4 [1.5; 3.3] seconds longer than those in the *Treatment (5)* condition.

Overall accuracy. Moving from the time respondents took to answer to their actual expectations, it was expected that treated respondents would be more accurate overall (H2), as indicated by a lower MAE. This expectation is confirmed: Monetary incentives, on average, lead to more accurate expectations, i.e., lower MAEs (see model 2a in Tables C.1 and C.2), consistent with incentivized respondents exerting greater effort on the task. Fig. 3 shows the average predicted MAEs by group and their differences. Individuals in the control group had the largest error: The average predicted MAE was 4.64 [4.58; 4.70] in the federal election and 4.91 [4.80; 5.03] in Saxony. On average, the treatments reduced this error by 0.32 [0.24; 0.40] (*Treatment (500)*), 0.23 [0.15; 0.31] (*Treatment (5)*), and 0.29 [0.13; 0.45] (Saxony) percentage points, respectively. Again, at the federal level, *Treatment (500)* had a slightly larger effect than *Treatment (5)*. The findings thus support Hypothesis 2.

Exploratory analyses by educational attainment show, as one would expect, that more highly educated respondents were more accurate, i.e., had a lower MAE (see Fig. C.9), than those with lower educational degrees. Turning to heterogeneous treatment effects, the incentives improve accuracy among respondents with high and medium levels of educational attainment in the federal election. Among those with low attainment, only the effect of *Treatment (500)* is significant. In Saxony, the treatment has an effect only among the highly educated, but the sample sizes are also considerably smaller (e.g., $n = 267$ respondents with low attainment). Political interest shows a similarly clear accuracy pattern: more self-reported interest is associated with a lower MAE (see Fig. C.10), but no clear heterogeneous treatment effects can be identified.

Wishful thinking. Moving from overall accuracy to individuals' forecast errors by party preference, Fig. 4 shows that wishful thinking is evident across all parties in both elections: Those who prefer a party expect it to perform better than those who do not, as indicated by positive differences in average predicted forecast errors between supporters and non-supporters.

Having shown that wishful thinking exists, I test whether the treatment dampens it (H3; model 3a in Tables C.1 and C.2). Fig. 5 shows the average predicted forecast errors by experimental group and by whether the party is one's preferred party or not, as well as the differences between groups. It yields three insights. First, as shown in Fig. 4, wishful thinking is evident for preferred parties. While for non-preferred parties, respondents deviate little on average (a slight underestimation of less than one percentage point across groups and elections), the effect for the preferred party is pronounced: Respondents in the control group overestimate their preferred party by, on average, 5.19 [5.00; 5.38] percentage points in the federal election and by 6.29 [5.81; 6.78] percentage points in Saxony. Second, monetary incentives seem to reduce the wishful thinking distortion, though the effect is modest relative to overall wishful thinking. Regression coefficients are significant at the 5%-level. Turning to the treatment effects (Fig. 5) in the federal election, *Treatment (500)* reduced the forecast error by 0.33 [0.07; 0.59] percentage points. For *Treatment (5)*, the reduction in predicted forecast error slightly misses statistical significance (0.25 [−0.01; 0.51]). Differences between treatments are not significant. In Saxony, the average reduction was larger: 1.50 [0.86; 2.14] percentage points. Third, the accuracy gain through the monetary incentive is not entirely attributable to the preferred party. A small but consistent effect also emerges for non-preferred parties. In Saxony, the treatment attenuated the negative forecast error by 0.19 [0.10; 0.28] percentage points on average, drawing expectations closer to the actual result. Nationally, *Treatment (500)* shifted the forecast error 0.09 [0.04; 0.14] percentage points closer to the election result, and *Treatment (5)* by 0.07 [0.02; 0.12] percentage points. Nevertheless, the effect is more pronounced for the preferred party, particularly in Saxony, confirming H3. Taken together, these results indicate that when respondents are motivated to pursue accuracy—here induced experimentally through monetary incentives—expectations for the preferred party move closer to the actual outcome. The magnitude of observed wishful thinking thus depends in part on respondents' engagement with the forecasting task rather than on partisan predispositions alone.

Previous research has found that the extent of wishful thinking depends on citizens' educational attainment, with more educated citizens overestimating to a lesser extent than less educated citizens (Stiers and Dassonneville, 2018; Babad, 1997). Exploratory analyses confirm this pattern among the German and Saxon respondents (see Fig. C.11).

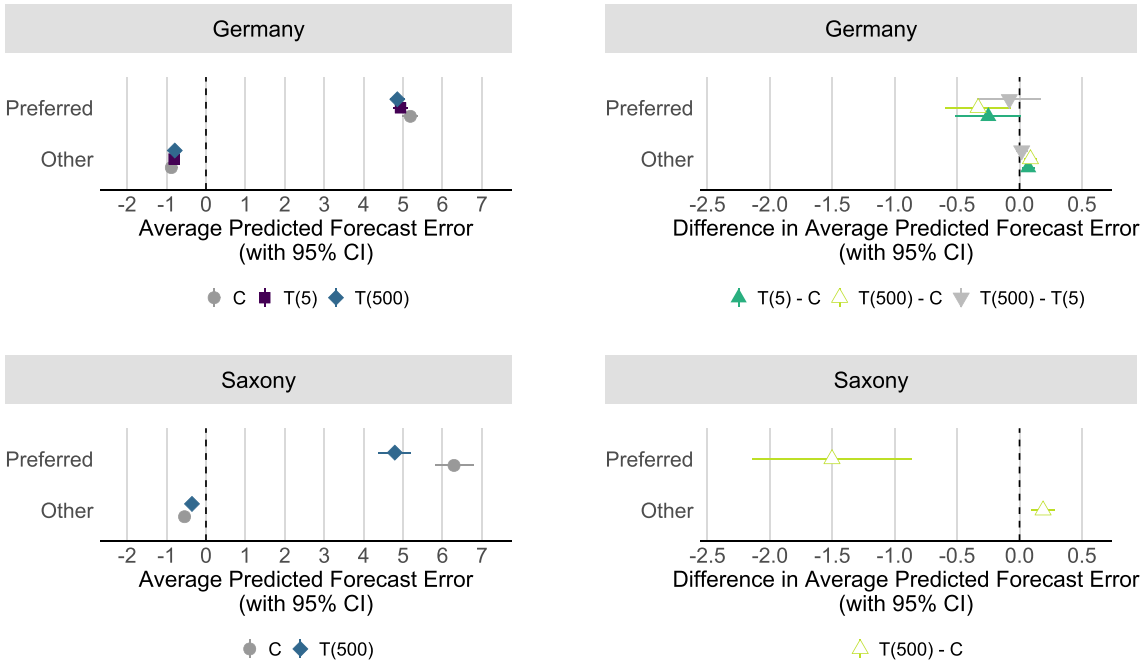


Fig. 5. Effects of treatment on forecast error by party preference (identified through PTV). *Note:* The dots in the left panels show the average predicted forecast error by treatment status for the respondent's preferred party (identified through the highest PTV score) and for the other parties (Other), based on model 3a and an observed-value approach. The right panels show the differences between experimental groups. Lines correspond to 95% confidence intervals. The top row reports the federal election, the bottom row reports Saxony. Estimates are reported in the text. Results are based on 23,056 respondents (federal election) and 4665 respondents (Saxony).

Additionally, in Saxony, the monetary incentive dampens wishful thinking among respondents with medium and high educational attainment (while the sample size is again small among low-educated respondents). At the federal level, the treatments reduce wishful thinking among low- and medium-educated respondents (for the former only *Treatment (5)*), but not among highly educated respondents. Based on reported political interest, no clear patterns emerge (see Fig. C.12).

Robustness checks. For H1, the conditional mean response time yields the same pattern as the median (see Fig. C.1). Findings on individuals' accuracy (H2) operationalized through the MAE are also robust to using the RMSE as an alternative specification of the dependent variable (see model 2b in Tables C.1 and C.2 and Fig. C.2 in the Appendix). For wishful thinking and H3, the findings are also robust to measuring the preferred party through respondents' vote intention (see Fig. C.4 and model 3c) or party identification (model 3d, Fig. C.5) instead of the propensity to vote. When including those respondents who gave their highest PTV score to more than one party, the coefficients are statistically significant at the 5%-level (see model 3b), but the confidence intervals of the treatment effect slightly include zero at the federal level (see Fig. C.3), suggesting that effects are stronger for clearly preferred parties. Being measured on an 11-point scale from "would never vote for this party" (1) to "would most likely vote for this party" (11), PTV allows for assessing not only dichotomized preference but also degrees of wishful thinking. Respondents, on average, underestimate parties for which they have a low propensity to vote, whereas this pattern reverses into overestimation around a score of 4 in Saxony and 5 at the federal level, indicating a clear wishful thinking pattern (see Fig. C.6 and model 3e in Tables C.1 and C.2). Treatment effects are small but consistent in direction: Monetary incentives attenuate both underestimation for low-PTV parties and overestimation for high-PTV parties toward zero forecast error, i.e., the election result. Among federal respondents, at a PTV score of 1, the control group underestimates by, on average, 2.21 [2.14; 2.28] percentage points, attenuated by 0.21 [0.11; 0.30] percentage points through *Treatment (5)* and by 0.26 [0.17; 0.36] percentage points through *Treatment (500)*. At a very high propensity to vote (PTV = 11), control respondents overestimate by 4.80 [4.64; 4.95] percentage points, with incentives reducing this by 0.34 [0.14; 0.55] and 0.40 [0.20; 0.61] percentage points, respectively. Saxon respondents show a similar pattern with notably larger treatment effects: Underestimation of 1.80 [1.67; 1.93] percentage points is dampened by 0.41 [0.24; 0.58] percentage points, while overestimation of 4.49 [4.17; 4.80] percentage points is reduced by 0.92 [0.52; 1.33] percentage points. Overall, in both elections, the incentive treatments reduce wishful thinking by pulling expectations closer to the actual election result, i.e., a forecast error of zero.

The previous analyses exclude respondents who allocated 50% to two parties at the federal level (765, 2.6%) or 100% to one party in Saxony (97, 1.7%), which I interpret as satisficing. While members of the control group were more likely to be excluded, re-estimating all models on the full samples without excluding satisficers leads to the same conclusions (Tables C.5 and C.6, Fig. C.7). However, all treatment effects are larger, since the excluded respondents come disproportionately from the control groups. Excluding them thus yields conservative estimates. In Saxony, the respondents recruited via social media deviate more strongly from the electorate than those from the online access panel (see Table B.4). Estimating the models separately by recruitment channel shows that all treatment effects point in the same direction and are statistically significant in both subsamples, while the effects on accuracy and wishful thinking are larger in the online access panel (see Table C.7 and Fig. C.8). The findings are therefore not driven by the self-selected social media respondents. The smaller effects among them may reflect that they are already highly motivated, yet even among them, the incentive improves accuracy and reduces wishful thinking.

To sum up, monetary incentives lead respondents, on average, to take more time when stating their expectations about party vote shares, consistent with the assumption that incentives increase their engagement with the forecasting task (H1). This slower answering behavior is

accompanied by overall more accurate expectations (H2), directly supporting the incentive–effort–accuracy chain that this paper set out to test. Wishful thinking biases remain evident even among incentivized respondents, confirming that partisan predispositions continue to shape electoral expectations. However, the treatment attenuates these biases to a meaningful extent (H3), indicating that the magnitude of wishful thinking is not fixed but responds to the conditions under which expectations are elicited. Importantly, this pattern is not limited to a single electoral context: the evidence from the high-salience federal election is corroborated by consistent findings from the lower-salience Saxon state election, strengthening confidence that the relationship between motivation, accuracy, and wishful thinking generalizes across different informational and political environments.

7. Discussion and conclusion

Citizens' expectations about election outcomes play an important role in strategic voting, election forecasting, and democratic politics more broadly. Yet decades of research have shown that these expectations are systematically distorted by wishful thinking. While previous research has primarily documented the existence of this bias and how it varies across citizens, e.g., by educational attainment, this paper asks a different question: whether the observed magnitude of wishful thinking reflects only citizens' partisan predispositions or also depends on the conditions under which electoral expectations are elicited. To address this question, monetary incentives were employed as an experimental tool to examine whether a stronger motivation to answer accurately can serve as an accuracy goal and motivate respondents to be more effortful and thereby change how respondents approach the task of forming and reporting electoral expectations.

Based on two original pre-election survey experiments conducted before the 2025 German federal election and the 2024 Saxon state election, the analyses provide consistent evidence that monetary incentives increase respondents' engagement with the forecasting task: Respondents in the treatment conditions spent more time answering the expectation item, suggesting greater cognitive effort and less satisficing. These longer response times were accompanied by modest yet systematic improvements in forecasting accuracy, reflected in lower mean absolute errors. At the same time, electoral expectations exhibited clear wishful thinking: Respondents systematically overestimated the electoral prospects of parties they preferred, while slightly underestimating other parties. Importantly, these partisan distortions became smaller, especially among preferred parties, when respondents were incentivized to provide accurate forecasts. While wishful thinking remained evident, stronger accuracy motivations can attenuate, though not eliminate, this bias.

The findings contribute to the literature in four respects. First, they qualify the interpretation of wishful thinking as solely reflecting stable partisan predispositions. Rather than viewing wishful thinking exclusively as an enduring characteristic of citizens' political beliefs, the results suggest that citizens' underlying predispositions and the conditions under which expectations are elicited jointly shape the wishful thinking observed.

Second, by evaluating respondents' expectations and their wishful thinking against realized election outcomes rather than contemporaneous polls, this study assesses genuine forecasting accuracy rather than agreement with available polling information. Polls provide respondents with an important informational benchmark, but citizens may reasonably depart from them if they anticipate polling errors or late campaign dynamics. Therefore, comparing expectations to realized outcomes offers a more direct assessment of citizens' predictive ability.

Third, the response time analyses provide behavioral evidence consistent with the theoretical argument that monetary incentives increase cognitive effort and reduce satisficing. Previous research has inferred this mechanism indirectly from changes in reported expectations alone. While response times cannot conclusively identify the cognitive processes underlying respondents' answers, they allow for a direct behavioral test of the assumed incentive–effort–accuracy chain.

Fourth, these findings emerge across two different electoral settings. The German federal election represents a highly salient first-order election characterized by extensive media attention and a dense polling environment, whereas the Saxon state election constitutes a lower-salience second-order election with considerably less polling information available to voters. Nevertheless, the effects of monetary incentives on effort, forecasting accuracy, and wishful thinking are consistent across both elections. The effects on effort and accuracy are of similar size, while the reduction of wishful thinking is considerably stronger in the subnational election. This consistency increases confidence that the observed relationship between motivation, forecasting accuracy, and wishful thinking is not confined to a particular electoral setting or campaign environment.

The present findings also help reconcile previous work on monetary incentives and electoral expectations. Babad (1997) reported only small reductions in wishful thinking, insufficient to meaningfully reduce partisan distortions, whereas Tikochinski and Babad (2023) concluded that financial incentives largely eliminate wishful thinking by moving respondents' expectations closer to contemporaneous polling. The present study points to an intermediate conclusion: Monetary incentives neither eliminate wishful thinking nor leave it unaffected. Instead, they reduce partisan distortions while simultaneously increasing response effort and improving forecasting accuracy. Taken together, the accumulating evidence suggests that stronger accuracy goals can meaningfully attenuate observed wishful thinking, even though partisan predispositions continue to shape electoral expectations.

The findings also speak to the literature on citizen forecasting. Electoral expectations are being used to generate aggregate election forecasts based on the wisdom of crowds, as individual errors cancel out. A growing body of work has moved beyond this classical democratic view toward a wisdom of the knowledgeable, or technocratic view, by weighting more expert forecasters more heavily to improve aggregate accuracy (Mongrain, 2021b; Murr, 2015). The present study shows that motivating respondents to engage carefully with the forecasting task improves the quality of individual expectations. Whether these individual-level gains translate into systematically more accurate aggregate forecasts remains, however, an open question: Aggregating the treated respondents' expectations improved the citizen forecast slightly in the case of the large-sample national election (MAE of 1.08 in *Treatment (500)*, 1.09 in *Treatment (5)*, compared to 1.14 in the control group; see Table B.5), whereas in Saxony the control group's aggregated forecast was more accurate (1.57 vs. 1.67, see Table B.6). In both samples, however, aggregation reduced the error considerably, in line with the wisdom of crowds. The average individual MAE was 4.46 (federal) and 4.77 (Saxony). How, and under which conditions, motivational gains at the individual level can be leveraged for aggregate forecasting is thus an important question for future research.

This study is subject to several limitations, which, at the same time, point to promising avenues for future research. First, the samples are not representative of the respective electorates (see Tables B.3 and B.4): Citizens aged 30 to 59 and highly educated citizens are overrepresented, while citizens with lower levels of education, older citizens, CDU/CSU voters, and, in Saxony, women are underrepresented. Saxon respondents additionally reported a very high turnout intention. This most likely also reflects the fact that respondents self-selected into the online surveys, especially in Saxony, where part of the sample was recruited through social media advertisements. The data must therefore not be read as a representative cross-section of the German or Saxon electorate, and self-selection may also operate through unobserved characteristics. A comparison of recruitment channels in Saxony and analyses by educational attainment show that the effects are not confined to the most self-selected respondents and do not consistently concentrate among the highly educated, but they cannot show whether the effects hold to the same extent among citizens who did not participate. Thus, while the experimental design ensures high internal validity, generalizations beyond the samples studied

here warrant caution. Similarly, the data come from a single country. Extending this design to additional countries and electoral contexts, ideally with representative samples also covering older citizens and those with lower educational attainment, would allow for assessing the generalizability of accuracy-motivated expectation formation and differences in elicited expectations.

Second, although response times provide behavioral evidence consistent with greater cognitive effort, they remain an indirect measure of the underlying cognitive processes. The design cannot distinguish whether incentives primarily affect how respondents retrieve relevant information, integrate it into a judgment, or translate their beliefs into a survey response. Moreover, while treated respondents took, on average, longer to express their expectations than untreated ones, the study design does not allow for disentangling what citizens used this additional time for. Respondents may have engaged in more careful reasoning, but they may also have consulted external information such as polls. Although a commitment pledge asked respondents not to consult outside sources, and over 98% agreed to it, actual compliance cannot be verified. To better identify the underlying mechanisms, future research could replicate the experiment in a more controlled laboratory setting to restrict information search behavior. A related but distinct question is what citizens actually draw on when forming their expectations. Rather than restricting information search, future studies could also deliberately allow it and explicitly track this behavior and its content, building, for example, on the work of Meffert and Gschwend (2011) and Stoetzer et al. (2024), to advance our understanding of the baseline knowledge citizens rely on and how they translate it into electoral expectations.

Third, while this study demonstrates that monetary incentives increase response effort and improve electoral expectations, it remains unclear whether comparable effects could be achieved through less costly interventions. Monetary incentives were introduced as an experimental tool to motivate respondents to spend more effort on the task and to be more accurate. However, the design cannot fully separate whether effects are based on the prospect of financial gain or the heightened task salience that mentioning a reward may itself create. The federal experiment also included two different monetary incentives. Although I had no prior expectation as to whether concentrating the prize on a few respondents or spreading it across many would be more effective, respondents in the *Treatment (500)* condition took, on average, slightly longer and were slightly more accurate than those in the *Treatment (5)* condition, while the two treatments did not differ significantly in their effect on wishful thinking. As the two conditions differ not only in prize size but also in the number of rewarded respondents (three people share €500, or 100 receive €5 each), and respondents were not aware of the total sample size, the role of incentive size, perceived winning probability, and salience cannot be fully disentangled within this design. However, based on this study's results, concentrating prizes on fewer respondents seems more effective. A related alternative to monetary incentives is to appeal to respondents to be objective or accurate, which would eliminate both the costs of the rewards and the administrative burden of identifying winners and administering payouts (Prior et al., 2015). Existing evidence on such appeals is mixed: Objectivity instructions yielded only marginal effects on wishful thinking among sports fans and in electoral expectations (Babad and Katz, 1991; Babad et al., 1992), whereas accuracy appeals reduced partisan bias in reports of economic perceptions as effectively as monetary incentives (Prior et al., 2015). An important avenue for future research is therefore to examine which other accuracy motivations could be applied to activate accuracy goals.

Finally, voters are unlikely to be exposed to monetary incentives in the real world when forming their views on the political world and making voting decisions. Rather, the incentives served as an experimental tool to induce accuracy goals and thereby identify the role of motivation in shaping reported electoral expectations. In this sense, the contribution of this study extends beyond demonstrating that financial rewards improve survey responses: it shows that the wishful thinking observed in surveys is not a fixed quantity, its magnitude also depends on respondents' motivation to engage with the forecasting task. Electoral expectations thus capture not only what citizens believe about election outcomes but also how they engage with the task of expressing those beliefs, a point that matters for research on wishful thinking as much as for the use of electoral expectations in studies of strategic voting and citizen forecasting. Encouragingly, the results show that respondents know more about upcoming elections than their unincentivized survey responses suggest: when motivated to be accurate, they can and do provide more accurate and less biased forecasts.

Declaration of Generative AI and AI-assisted technologies in the writing process

Statement: During the preparation of this work, the author used ChatGPT and Claude in order to get assistance with coding and to copy-edit written text. After using these tools/services, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

I thank Stine Bartram, William Bruce, Marietta Sujin Hüttemann, and Lisa-Marie Zikesch for excellent research assistance. I am also grateful to Florian Foos, Thomas Gschwend, Michael S. Lewis-Beck, Frauke Riebe, Oliver Rittmann, and Lukas F. Stoetzer for feedback on earlier versions of this manuscript. Earlier versions of the paper were presented at the MZES Mittelbaukolloquium, the Colloquium of the Chair of Political Science, Quantitative Methods in the Social Sciences, at the University of Mannheim, at the 2025 EPSA panel "When do campaigns work?", the 2025 Zweitstimme.org Workshop on Elections, Public Opinion & Elite Behavior, and the 49th Annual Conference of the International Association for the Study of German Politics. I thank all participants and discussants for their helpful feedback. This work was supported by the University of Mannheim's Graduate School of Economic and Social Sciences and funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation; 529544178).

Appendix A. Preregistration

This study was partially pre-registered on OSF on July 24, 2024 (Saxon election; https://osf.io/aytwj), and January 24, 2025 (federal election; https://osf.io/4xtkw), respectively. The preregistrations specified three hypotheses concerning the effect of monetary incentives on forecasting behavior, covering the effect of monetary incentives on individual-level accuracy, and on the accuracy of aggregated citizen forecasts.

> ***H1 (Individual Forecasting Accuracy): Offering a prize for the most accurate electoral expectations increases accuracy of individuals' expectations.***
> *H2 (Citizen Forecasts' Certainty): Offering a prize for the most accurate electoral expectations decreases the variance of the electoral expectations.*
> *H3 (Citizen Forecasts' Accuracy) Offering a prize for the most accurate electoral expectations increases the accuracy of the groups' expectations.*

Given the scope of the present paper, I focus exclusively on the preregistered hypothesis concerning individual forecasting accuracy (H1 in the preregistration, H2 in the main body of the paper), and on how this could come about.

Citizen forecasting is left for future research, as it speaks to a different literature and also to different mechanisms, namely, how aggregation behaves with (potentially) more accurate individual expectations.

Analyses for the hypothesis on individual forecasting accuracy were pre-registered as:

> *For each categorical item I will regress the Correct Forecast measure on a treatment dummy (logistic regression).* ***For the vote share items, I will regress MAE and RMSE respectively on a treatment dummy (linear regression).*** *Moreover, I will run separate models for each party's vote shares by regressing the absolute deviations on the treatment dummy. Here I focus on the main parties currently in parliament (AfD, BSW, CDU, FDP, Grüne, Linke, SPD).*

In line with the preregistration, individual-level forecasting accuracy for vote share expectations is assessed using MAE and RMSE, and regressed on a treatment indicator. However, while the survey included additional items measuring expectations for other outcomes (such as the chancellor, the coalition, or the first vote), this paper focuses on expectations for party vote shares. This focus reflects considerations of scope and analytical clarity. Importantly, expectations about party vote shares were measured immediately after the treatment, ensuring that the outcome captures respondents' direct responses to the experimental manipulation. Additionally, while the preregistration also specified party-level analyses based on absolute deviations, the paper instead examines directional deviations between the expected and realized vote share for each respondent's preferred party (H3). This hypothesis and analysis were not pre-registered. H1 and analyses of response times were likewise not pre-registered and are presented as exploratory evidence regarding a potential mechanism.

Appendix B. Items and descriptives

B.1. Items

B.1.1. Federal

Table B.1
Item wording in English and German (federal study).

Item	English Wording (Translation)	German Wording (Original)
Eligibility to Vote	The federal election in Germany will take place on 23 February. Are you eligible to vote in the federal election? *[Single-choice]* • Yes • No • Don't know	Am 23. Februar ist die Bundestagswahl in Deutschland. Sind Sie bei der Bundestagswahl wahlberechtigt? *[Einfachauswahl]* • Ja • Nein • Weiß nicht
Year of Birth	Please select the year you were born. *[Dropdown]* 1925–2007	Wählen Sie bitte aus, in welchem Jahr Sie geboren sind. *[Dropdown]* 1925–2007
Turnout Intention	When elections take place, many people cast their votes, while others are unable to vote or do not participate in the election for other reasons. Please indicate how likely you are to vote in the general election on 23 February 2025. I will…*[Single-choice]* "1 = definitely not vote" to "10 = definitely vote"	Wenn Wahlen stattfinden, geben viele Leute ihre Stimme ab, andere kommen nicht dazu, ihre Stimme abzugeben, oder nehmen aus anderen Gründen nicht an der Wahl teil. Geben Sie bitte an, wie wahrscheinlich es ist, dass Sie am 23. Februar 2025 zur Bundestagswahl gehen. Ich werde … *[Einfachauswahl]* "1 = sicher nicht zur Wahl gehen" bis "10 = sicher zur Wahl gehen"
Political Interest	Generally speaking, how interested are you in politics? *[Single-choice]* • Very interested • Quite interested • Moderately interested • Not very interested • Not interested at all	Einmal ganz allgemein gesprochen: Wie stark interessieren Sie sich für Politik? *[Einfachauswahl]* • Sehr stark • Stark • Mittelmäßig • Weniger stark • Überhaupt nicht
Vote Intention	In the federal election, you can cast two votes: the first vote for a candidate in your constituency and the second vote for a party list in your federal state. And now to the second vote: Which party will you vote for/would you vote for/did you vote for? *[Single-choice]* • CDU *(CSU in Bavaria)* • SPD • AfD • Greens • Liberals • Left • BSW • *Schleswig-Holstein only:* SSW • Another party • Invalid vote • Don't know	Bei der Bundestagswahl können Sie zwei Stimmen vergeben: Die Erststimme für eine Kandidatin oder einen Kandidaten in Ihrem Wahlkreis und die Zweitstimme für eine Parteiliste in Ihrem Bundesland. Und nun zur Zweitstimme: Welche Partei werden Sie wählen/würden Sie wählen/haben Sie gewählt? *[Einfachauswahl]* • CDU *(CSU in Bayern)* • SPD • AfD • B'90/Die Grünen • FDP • Die Linke • BSW • *nur Schleswig-Holstein:* SSW • Sonstige • Ungültige Stimme • Weiß ich nicht
Propensity to Vote	Now we are interested in your general party preferences. There are a number of political parties in Germany. Each of them would like to get your vote. Please indicate for each of the following parties, using this scale, how likely you are to ever vote for that party. *[Matrix question]* Scale 1 ("would never vote for this party") to 11 ("would most likely vote for this party"). • CDU (CSU in Bavaria) • SPD • AfD • Greens • Liberals • Left • BSW	Nun sind wir an Ihren generellen Parteipräferenzen interessiert. Es gibt eine Reihe von politischen Parteien in Deutschland. Jede davon würde gerne Ihre Stimme bekommen. Geben Sie bitte für jede der folgenden Parteien anhand dieser Skala an, wie wahrscheinlich es ist, dass Sie diese Partei jemals wählen würden. *[Maxtrixfrage]* Skala 1 ("Würde diese Partei niemals wählen") bis 11 ("Würde diese Partei sehr wahrscheinlich wählen"). • CDU (CSU in Bayern) • SPD • AfD • B'90/Die Grünen • FDP • Die Linke • BSW
Party Identification	In Germany, many people lean toward a particular political party over a longer period of time, even though they occasionally vote for a different party. What about you: Generally speaking, do you lean toward a particular party? And if so, which one? • CDU/CSU • SPD • AfD • Greens • Liberals • Left • BSW • Another party • No party	In Deutschland neigen viele Leute längere Zeit einer bestimmten politischen Partei zu, obwohl sie auch ab und zu eine andere Partei wählen. Wie ist das bei Ihnen: Neigen Sie – ganz allgemein gesprochen – einer bestimmten Partei zu? Und wenn ja, welcher? • CDU/CSU • SPD • AfD • B'90/Die Grünen • FDP • Die Linke • BSW • Anderer Partei • Keiner Partei

(*continued on next page*)

Table B.1 *(continued)*

Item	English Wording (Translation)	German Wording (Original)
Commitment Pledge	Below, we will ask you a few questions about your expectations regarding the outcome of the federal election. We kindly ask that you refrain from consulting other sources (e.g., other websites). We are interested in your genuine perceptions. Are you willing to refrain from consulting other sources when answering the following questions? *[Single-choice]* • Yes, I am willing • No, I am not willing	Im Folgenden stellen wir Ihnen einige Fragen zu Ihren Erwartungen über den Ausgang der Bundestagswahl. Dafür möchten wir Sie bitten, keine anderen Quellen heranzuziehen (z. B. andere Internetseiten). Wir sind an Ihren unverfälschten Einschätzungen interessiert. Sind Sie bereit, für die Beantwortung der nächsten Fragen keine anderen Quellen heranzuziehen? *[Einfachauswahl]* • Ja, ich bin bereit • Nein, ich bin nicht bereit
Treatment (500)	If your expectations for the outcome of the federal election are close to the actual result, you have the chance to win a prize. The top 3 participants will receive prizes worth a total of €500 after the election.	Wenn Ihre Erwartungen zum Ausgang der Bundestagswahl nah am tatsächlichen Ergebnis liegen, haben Sie die Chance, einen Preis zu gewinnen. Die besten 3 Teilnehmenden erhalten nach der Wahl Preise im Gesamtwert von 500 Euro.
Treatment (5)	If your expectations for the outcome of the federal election are close to the actual result, you have the chance to win a prize. The top 100 participants will receive a prize of €5 each after the vote.	Wenn Ihre Erwartungen zum Ausgang der Bundestagswahl nah am tatsächlichen Ergebnis liegen, haben Sie die Chance, einen Preis zu gewinnen. Die besten 100 Teilnehmenden erhalten nach der Wahl einen Preis in Höhe von je 5 Euro.
Vote Expectation	If you first think about the party list votes, which determine the distribution of seats in the federal parliament: What percentage of the votes do you expect the various parties to receive nationwide? Please use the following input mask to enter your expectations. The total number of votes should add up to exactly 100%. *[Sliders from 0% to 50%]* • CDU/CSU • SPD • AfD • Greens • Liberals • Left • BSW • Others	Wenn Sie zunächst an die Zweitstimmen denken, die über die Verteilung der Mandate im Bundestag entscheiden: Wie viel Prozent der Stimmen erwarten Sie, werden die verschiedenen Parteien bundesweit erhalten? Bitte nutzen Sie die folgende Eingabemaske, um Ihre Erwartungen einzugeben. Die Summe der Stimmen soll genau 100% ergeben. *[Slider von 0% bis 50%]* • CDU/CSU • SPD • AfD • B'90/Die Grünen • FDP • Die Linke • BSW • Sonstige

B.1.2. Saxony

Table B.2
Item wording in English and German (Saxony study).

Item	English Wording (Translation)	German Wording (Original)
Year of Birth	Please enter the year you were born.	Tragen Sie bitte ein, in welchem Jahr Sie geboren sind.
Gender	Please indicate your gender. *[Single-choice]* • Male • Female • Diverse	Geben Sie bitte Ihr Geschlecht an. *[Einfachauswahl]* • Männlich • Weiblich • Divers
Turnout Intention	Now some questions about the upcoming state election. When elections take place, many people cast their votes, while others are unable to vote or do not participate for other reasons. Please indicate how likely you are to vote in the state election on 1 September 2024. I will… *[Single-choice]* • Definitely vote • Probably vote • Maybe vote • Probably not vote • Definitely not vote • I have already voted by postal ballot • Don't know	Nun einige Fragen zur anstehenden Landtagswahl. Wenn Wahlen stattfinden, geben viele Leute ihre Stimme ab, andere kommen nicht dazu, ihre Stimme abzugeben, oder nehmen aus anderen Gründen nicht an der Wahl teil. Geben Sie bitte an, wie wahrscheinlich es ist, dass Sie am 1. September 2024 zur Landtagswahl gehen. Ich werde … *[Einfachauswahl]* • bestimmt zur Wahl gehen • wahrscheinlich zur Wahl gehen • vielleicht zur Wahl gehen • wahrscheinlich nicht zur Wahl gehen • bestimmt nicht zur Wahl gehen • ich habe bereits Briefwahl gemacht • weiß ich noch nicht
Political Interest	Generally speaking, how interested are you in politics? *[Single-choice]* • Very interested • Quite interested • Moderately interested • Not very interested • Not interested at all	Einmal ganz allgemein gesprochen: Wie stark interessieren Sie sich für Politik? *[Einfachauswahl]* • sehr stark • stark • mittelmäßig • weniger stark • überhaupt nicht

(*continued on next page*)

Table B.2 *(continued)*

Item	English Wording (Translation)	German Wording (Original)
Vote Intention	And now to the party list vote: Which party will you vote for/would you vote for/have you voted for? *[Single-choice]* • CDU • AfD • Left • Greens • SPD • FDP • Freie Wähler • BSW • Another party • Invalid vote	Und nun zur Listenstimme: Welche Partei werden Sie wählen/haben Sie gewählt/würden Sie wählen? *[Einfachauswahl]* • CDU • AfD • Die Linke • B'90/Die Grünen • SPD • FDP • Freie Wähler • Bündnis Sahra Wagenknecht (BSW) • Eine andere Partei • Ich werde ungültig wählen.
Propensity to Vote	Now we are interested in your general party preferences. There are a number of political parties in Germany. Each of them would like to get your vote. Please indicate for each of the following parties, using this scale, how likely you are to ever vote for that party. *[Matrix question]* Scale 1 ("would never vote for this party") to 11 ("would most likely vote for this party"). • CDU • AfD • Left • Greens • SPD • FDP • Freie Wähler • BSW	Nun sind wir an Ihren generellen Parteipräferenzen interessiert. Es gibt eine Reihe von politischen Parteien in Deutschland. Jede davon würde gerne Ihre Stimme bekommen. Geben Sie bitte für jede der folgenden Parteien anhand dieser Skala an, wie wahrscheinlich es ist, dass Sie diese Partei jemals wählen würden. *[Matrixfrage]* Skala 1 ("Würde diese Partei niemals wählen") bis 11 ("Würde diese Partei sehr wahrscheinlich wählen"). • CDU • AfD • Die Linke • B'90/Die Grünen • SPD • FDP • Freie Wähler • Bündnis Sahra Wagenknecht (BSW)
Party Identification	In Germany, many people lean toward a particular political party over a longer period of time, even though they occasionally vote for a different party. What about you: Generally speaking, do you lean toward a particular party? And if so, which one? • CDU • AfD • Left • Greens • SPD • FDP • BSW • Another party • No party	In Deutschland neigen viele Leute längere Zeit einer bestimmten politischen Partei zu, obwohl sie auch ab und zu eine andere Partei wählen. Wie ist das bei Ihnen: Neigen Sie – ganz allgemein gesprochen – einer bestimmten Partei zu? Und wenn ja, welcher? • CDU • AfD • Die Linke • B'90/Die Grünen • SPD • FDP • Bündnis Sahra Wagenknecht (BSW) • Andere Partei, und zwar: • keiner Partei
Commitment Pledge	Below, we will ask you a few questions about your expectations regarding the outcome of the state election. We kindly ask that you refrain from consulting other sources (e.g., other websites). We are interested in your genuine perceptions. Are you willing to refrain from consulting other sources when answering the following questions? *[Single-choice]* • Yes, I am willing. • No, I am not willing.	Im Folgenden stellen wir Ihnen einige Fragen zu Ihren Erwartungen über den Ausgang der Landtagswahl. Dafür möchten wir Sie bitten, keine anderen Quellen heranzuziehen (z. B. andere Internetseiten). Wir sind an Ihren unverfälschten Einschätzungen interessiert. Sind Sie bereit, für die Beantwortung der nächsten Fragen keine anderen Quellen heranzuziehen? *[Einfachauswahl]* • Ja, ich bin bereit. • Nein, ich bin nicht dazu bereit.
Treatment (500)	If your expectations for the outcome of the state election are close to the actual result, you have the chance to win a prize. The top 3 participants will receive prizes worth a total of €500 after the election.	Wenn Ihre Erwartungen zum Ausgang der Landtagswahl nah am tatsächlichen Ergebnis liegen, haben Sie die Chance, einen Preis zu gewinnen. Die besten 3 Teilnehmenden erhalten nach der Wahl Preise im Gesamtwert von 500 Euro.
Vote Expectation	If you first think about the party list votes ("second votes"), which determine the distribution of seats in the state parliament: What percentage of the votes do you expect the various parties to receive statewide? Please use the following input mask to enter your expectations. The total should add up to exactly 100%. *[Sliders from 0% to 100%]* • CDU • AfD • Left • Greens • SPD • FDP • Freie Wähler • BSW • Others	Wenn Sie zunächst an die Listenstimmen („Zweitstimmen") denken, die über die Verteilung der Mandate im Landtag entscheiden: Wie viel Prozent der Stimmen erwarten Sie, werden die verschiedenen Parteien landesweit erhalten? Bitte nutzen Sie die folgende Eingabemaske, um Ihre Erwartungen einzugeben. Die Summe der Stimmen soll genau 100% ergeben. *[Slider von 0% bis 100%]* • CDU • AfD • Die Linke • B'90/Die Grünen • SPD • FDP • Freie Wähler • Bündnis Sahra Wagenknecht (BSW) • Sonstige

B.2. Sample composition

B.2.1. Federal

Table B.3
Demographic composition of the federal sample (before and after exclusions).

		Before exclusions (difference to benchmark)				Main sample (difference to before exclusions)			
	Bench-mark	All n = 29,195	C n = 9697	T(5) n = 9728	T(500) n = 9770	All n = 28,430 (−765)	C n = 9400 (−297)	T(5) n = 9475 (−253)	T(500) n = 9555 (−215)
Age Group									
18–29	13.0	13.1 (+ 0.1)	13.1 (+ 0.1)	13.2 (+ 0.3)	13.0 (0.0)	13.0 (− 0.1)	13.0 (0.0)	13.2 (− 0.1)	12.9 (− 0.1)
30–39	13.8	19.9 (+ 6.1)	20.0 (+ 6.2)	20.2 (+ 6.4)	19.5 (+ 5.7)	19.7 (− 0.2)	19.8 (− 0.2)	20.0 (− 0.2)	19.4 (− 0.1)
40–49	14.2	20.3 (+ 6.1)	20.5 (+ 6.4)	20.4 (+ 6.3)	19.9 (+ 5.7)	20.1 (− 0.1)	20.4 (− 0.1)	20.3 (− 0.1)	19.7 (− 0.2)
50–59	16.5	21.3 (+ 4.8)	20.7 (+ 4.2)	21.5 (+ 5.0)	21.6 (+ 5.2)	21.4 (+ 0.1)	20.7 (+ 0.1)	21.6 (+ 0.1)	21.8 (+ 0.1)
60+	42.6	25.4 (−17.2)	25.7 (−16.9)	24.6 (−18.0)	25.9 (−16.7)	25.7 (+ 0.3)	26.0 (+ 0.3)	24.9 (+ 0.3)	26.2 (+ 0.3)
Gender									
Men*	48.6	49.8 (+ 1.2)	49.9 (+ 1.4)	49.8 (+ 1.2)	49.6 (+ 1.0)	49.7 (0.0)	49.8 (− 0.1)	49.8 (0.0)	49.6 (0.0)
Women	51.4	50.2 (− 1.2)	50.1 (− 1.4)	50.2 (− 1.2)	50.4 (− 1.0)	50.3 (0.0)	50.2 (+ 0.1)	50.2 (0.0)	50.4 (0.0)
Education									
Low	29.7	10.7 (−19.0)	10.6 (−19.1)	10.6 (−19.1)	11.1 (−18.7)	10.4 (− 0.4)	10.2 (− 0.4)	10.2 (− 0.4)	10.7 (− 0.3)
Medium	29.4	35.3 (+ 5.8)	35.3 (+ 5.9)	34.9 (+ 5.5)	35.6 (+ 6.1)	35.2 (0.0)	35.3 (0.0)	34.9 (0.0)	35.5 (− 0.1)
High	37.2	52.3 (+15.0)	52.6 (+15.4)	52.6 (+15.3)	51.6 (+14.4)	52.7 (+ 0.4)	53.1 (+ 0.5)	53.0 (+ 0.5)	52.0 (+ 0.4)
Still in School	3.6	0.6 (− 3.0)	0.4 (− 3.1)	0.7 (− 2.8)	0.7 (− 2.9)	0.6 (0.0)	0.4 (0.0)	0.7 (0.0)	0.7 (0.0)
Turnout Intention									
Yes	82.5	81.1 (− 1.4)	80.7 (− 1.8)	80.9 (− 1.6)	81.8 (− 0.7)	81.6 (+ 0.4)	81.3 (+ 0.6)	81.4 (+ 0.5)	82.1 (+ 0.2)
Vote Intention									
AfD	20.8	20.6 (− 0.2)	21.1 (+ 0.3)	19.8 (− 1.0)	21.0 (+ 0.2)	20.2 (− 0.4)	20.6 (− 0.5)	19.4 (− 0.4)	20.7 (− 0.3)
BSW	5.0	6.2 (+ 1.2)	5.9 (+ 0.9)	6.0 (+ 1.1)	6.5 (+ 1.5)	6.1 (0.0)	5.9 (0.0)	6.0 (− 0.1)	6.5 (0.0)
CDU	22.6	16.9 (− 5.7)	16.9 (− 5.7)	16.8 (− 5.8)	17.0 (− 5.5)	17.0 (+ 0.1)	17.0 (+ 0.1)	16.9 (+ 0.1)	17.0 (0.0)
CSU	6.0	3.9 (− 2.1)	3.8 (− 2.2)	4.0 (− 2.0)	3.8 (− 2.1)	3.9 (0.0)	3.8 (0.0)	4.0 (+ 0.1)	3.9 (0.0)
FDP	4.3	3.9 (− 0.4)	3.9 (− 0.4)	3.8 (− 0.6)	4.1 (− 0.2)	4.0 (+ 0.1)	4.0 (+ 0.1)	3.8 (0.0)	4.2 (+ 0.1)
Greens	11.6	12.2 (+ 0.6)	12.3 (+ 0.7)	12.8 (+ 1.2)	11.5 (− 0.1)	12.4 (+ 0.2)	12.5 (+ 0.3)	12.9 (+ 0.2)	11.6 (+ 0.2)
Left	8.8	7.5 (− 1.2)	7.6 (− 1.2)	7.7 (− 1.0)	7.3 (− 1.5)	7.6 (0.0)	7.7 (0.0)	7.8 (+ 0.1)	7.3 (0.0)
SPD	16.4	14.2 (− 2.2)	14.0 (− 2.4)	14.2 (− 2.2)	14.5 (− 1.9)	14.3 (+ 0.1)	14.0 (0.0)	14.2 (0.0)	14.6 (+ 0.1)
Other	4.6	14.4 (+ 9.8)	14.3 (+ 9.7)	14.8 (+10.2)	14.0 (+ 9.4)	14.3 (0.0)	14.3 (0.0)	14.8 (0.0)	13.9 (0.0)
State									
Baden-Württemberg	12.6	10.8 (− 1.9)	10.7 (− 1.9)	11.1 (− 1.6)	10.5 (− 2.2)	10.8 (0.0)	10.7 (0.0)	11.1 (0.0)	10.5 (0.0)
Bavaria	15.7	14.8 (− 0.9)	14.3 (− 1.4)	15.8 (+ 0.1)	14.3 (− 1.3)	14.8 (0.0)	14.3 (0.0)	15.9 (+ 0.1)	14.3 (0.0)
Berlin	4.0	5.7 (+ 1.6)	6.0 (+ 2.0)	5.2 (+ 1.1)	5.9 (+ 1.8)	5.7 (0.0)	6.0 (− 0.1)	5.1 (0.0)	5.9 (0.0)
Brandenburg	3.4	3.3 (− 0.1)	3.1 (− 0.3)	3.4 (0.0)	3.3 (0.0)	3.3 (0.0)	3.1 (0.0)	3.4 (0.0)	3.3 (0.0)
Bremen	0.7	0.8 (+ 0.1)	0.8 (+ 0.1)	0.8 (+ 0.1)	0.9 (+ 0.2)	0.8 (0.0)	0.8 (0.0)	0.8 (0.0)	0.9 (0.0)
Hamburg	2.1	3.0 (+ 0.8)	3.2 (+ 1.0)	2.8 (+ 0.6)	3.0 (+ 0.9)	3.0 (0.0)	3.2 (0.0)	2.8 (0.0)	3.0 (0.0)
Hesse	7.2	7.7 (+ 0.5)	7.6 (+ 0.4)	7.8 (+ 0.6)	7.7 (+ 0.5)	7.7 (0.0)	7.6 (0.0)	7.7 (0.0)	7.7 (0.0)
Lower Saxony	10.0	9.0 (− 1.0)	8.8 (− 1.2)	9.1 (− 0.9)	9.0 (− 0.9)	9.0 (0.0)	8.8 (0.0)	9.1 (0.0)	9.1 (0.0)
Mecklenburg-Western Pomerania	2.1	2.0 (− 0.2)	2.2 (0.0)	1.7 (− 0.4)	2.0 (− 0.1)	1.9 (0.0)	2.1 (0.0)	1.7 (− 0.1)	2.0 (0.0)
North Rhine-Westphalia	21.3	20.5 (− 0.8)	21.1 (− 0.2)	20.1 (− 1.2)	20.4 (− 0.9)	20.5 (0.0)	21.1 (0.0)	20.1 (0.0)	20.4 (0.0)
Rhineland-Palatinate	5.0	4.8 (− 0.2)	4.9 (− 0.1)	4.5 (− 0.4)	4.9 (− 0.1)	4.8 (0.0)	4.9 (0.0)	4.5 (0.0)	4.9 (0.0)
Saarland	1.2	1.3 (+ 0.1)	1.3 (0.0)	1.3 (+ 0.1)	1.4 (+ 0.1)	1.3 (0.0)	1.3 (0.0)	1.3 (0.0)	1.4 (0.0)
Saxony	5.3	6.6 (+ 1.3)	6.6 (+ 1.3)	6.2 (+ 0.9)	7.0 (+ 1.7)	6.6 (0.0)	6.5 (0.0)	6.3 (+ 0.1)	7.0 (0.0)
Saxony-Anhalt	2.9	3.1 (+ 0.2)	2.9 (+ 0.1)	3.4 (+ 0.6)	2.9 (+ 0.1)	3.1 (0.0)	2.9 (0.0)	3.4 (0.0)	2.9 (0.0)
Schleswig-Holstein	3.7	3.8 (+ 0.1)	3.7 (− 0.1)	3.8 (+ 0.1)	3.9 (+ 0.2)	3.8 (0.0)	3.7 (0.0)	3.9 (0.0)	3.9 (0.0)
Thuringia	2.7	3.0 (+ 0.2)	2.9 (+ 0.2)	3.1 (+ 0.3)	2.9 (+ 0.1)	3.0 (0.0)	3.0 (+ 0.1)	3.1 (0.0)	2.8 (0.0)

Note: Cells show the share of respondents (%) before exclusions (left) and in the main sample (right), overall and by experimental group. In the main sample, 765 respondents who answered 50% twice are excluded. For the overall sample, the difference from a benchmark is shown in percentage points in parentheses. For the main sample, the difference from the overall sample before exclusions is shown. Darker shading indicates larger absolute differences (scale up to 20 percentage points on the left and up to 5 percentage points on the right). Benchmarks are the eligible voting population for age, gender, and federal state (The Federal Returning Officer, 2025b), the German population for education (Statistisches Bundesamt (Destatis), 2026a), election results for vote intention, and actual turnout for turnout intention (The Federal Returning Officer, 2025a). Men* includes men and diverse respondents.

B.2.2. Saxony

Table B.4
Demographic composition of the Saxon sample (before and after exclusions).

		Before exclusions (difference to benchmark)					Main sample (difference to before exclusions)				
	Bench-mark	All n = 5659	OAP n = 2223	SM n = 3436	C n = 2889	T(500) n = 2770	All n = 5562 (−97)	OAP n = 2161 (−62)	SM n = 3401 (−35)	C n = 2824 (−65)	T(500) n = 2738 (−32)
Age Group											
18–29	10.9	11.3 (+ 0.4)	15.2 (+ 4.3)	8.8 (− 2.1)	11.3 (+ 0.4)	11.3 (+ 0.4)	11.3 (0.0)	15.1 (0.0)	8.9 (0.0)	11.3 (0.0)	11.3 (− 0.1)
30–39	12.4	17.6 (+ 5.2)	21.1 (+ 8.6)	15.4 (+ 3.0)	18.0 (+ 5.5)	17.3 (+ 4.9)	17.6 (0.0)	21.0 (− 0.1)	15.5 (+ 0.1)	18.0 (0.0)	17.3 (0.0)
40–49	14.8	22.4 (+ 7.6)	19.0 (+ 4.1)	24.7 (+ 9.8)	22.6 (+ 7.7)	22.3 (+ 7.5)	22.5 (0.0)	18.9 (− 0.1)	24.8 (+ 0.1)	22.5 (− 0.1)	22.5 (+ 0.2)
50–59	16.1	23.2 (+ 7.1)	22.4 (+ 6.3)	23.7 (+ 7.7)	23.2 (+ 7.2)	23.2 (+ 7.1)	23.2 (0.0)	22.4 (0.0)	23.7 (0.0)	23.3 (0.0)	23.1 (− 0.1)
60 +	45.8	25.4 (−20.4)	22.4 (−23.3)	27.3 (−18.5)	24.9 (−20.8)	25.9 (−19.9)	25.4 (0.0)	22.7 (+ 0.2)	27.1 (− 0.2)	24.9 (0.0)	25.9 (0.0)
Gender											
Men*	48.1	59.3 (+11.2)	49.2 (+ 1.1)	65.9 (+17.8)	59.7 (+11.6)	58.9 (+10.8)	59.4 (+ 0.1)	49.3 (+ 0.1)	65.9 (0.0)	59.7 (0.0)	59.1 (+ 0.2)
Women	51.9	40.7 (−11.2)	50.8 (− 1.1)	34.1 (−17.8)	40.3 (−11.6)	41.1 (−10.8)	40.6 (− 0.1)	50.7 (− 0.1)	34.1 (0.0)	40.3 (0.0)	40.9 (− 0.2)
Education											
Low	15.6	4.9 (−10.7)	8.6 (− 6.9)	2.5 (−13.1)	5.1 (−10.5)	4.8 (−10.8)	4.8 (− 0.1)	8.4 (− 0.2)	2.5 (0.0)	4.8 (− 0.2)	4.8 (0.0)
Medium	51.4	42.9 (− 8.4)	53.4 (+ 2.0)	36.2 (−15.2)	41.4 (−10.0)	44.6 (− 6.8)	42.7 (− 0.2)	53.2 (− 0.2)	36.0 (− 0.2)	41.1 (− 0.2)	44.3 (− 0.3)
High	29.8	49.1 (+19.2)	33.9 (+ 4.1)	58.9 (+29.0)	50.2 (+20.4)	47.9 (+18.0)	49.4 (+ 0.3)	34.2 (+ 0.3)	59.1 (+ 0.2)	50.6 (+ 0.4)	48.1 (+ 0.3)
Still in School	3.2	1.3 (− 2.0)	2.2 (− 1.0)	0.6 (− 2.6)	1.4 (− 1.8)	1.1 (− 2.1)	1.3 (0.0)	2.3 (+ 0.1)	0.6 (0.0)	1.5 (0.0)	1.1 (0.0)
Turnout Intention											
Yes	74.4	90.6 (+16.2)	79.5 (+ 5.1)	97.8 (+23.4)	90.7 (+16.3)	90.6 (+16.2)	90.9 (+ 0.3)	79.9 (+ 0.4)	97.9 (+ 0.1)	91.0 (+ 0.3)	90.8 (+ 0.3)
Vote Intention											
AfD	30.6	33.6 (+ 3.0)	29.1 (− 1.6)	36.6 (+ 5.9)	34.0 (+ 3.4)	33.2 (+ 2.6)	33.3 (− 0.4)	28.6 (− 0.5)	36.2 (− 0.4)	33.4 (− 0.6)	33.2 (− 0.1)
Greens	5.1	9.8 (+ 4.7)	6.7 (+ 1.5)	11.8 (+ 6.7)	9.7 (+ 4.6)	9.9 (+ 4.8)	9.9 (+ 0.1)	6.8 (+ 0.1)	11.9 (+ 0.1)	9.9 (+ 0.2)	9.9 (0.0)
BSW	11.8	11.1 (− 0.7)	13.4 (+ 1.6)	9.7 (− 2.2)	10.7 (− 1.1)	11.6 (− 0.2)	11.2 (+ 0.1)	13.6 (+ 0.2)	9.7 (0.0)	10.8 (+ 0.1)	11.7 (+ 0.1)
CDU	31.9	22.0 (− 9.9)	25.3 (− 6.6)	19.9 (−12.0)	22.0 (− 9.9)	22.0 (− 9.9)	22.2 (+ 0.2)	25.6 (+ 0.3)	20.1 (+ 0.2)	22.4 (+ 0.3)	22.1 (+ 0.1)
FDP	0.9	2.1 (+ 1.2)	2.8 (+ 1.9)	1.7 (+ 0.8)	2.0 (+ 1.1)	2.3 (+ 1.4)	2.2 (0.0)	2.8 (0.0)	1.7 (0.0)	2.1 (0.0)	2.3 (0.0)
Left	4.5	4.7 (+ 0.3)	5.0 (+ 0.5)	4.6 (+ 0.1)	4.8 (+ 0.4)	4.6 (+ 0.2)	4.8 (0.0)	5.0 (+ 0.1)	4.6 (0.0)	4.9 (0.0)	4.6 (0.0)
SPD	7.3	8.4 (+ 1.1)	7.3 (0.0)	9.1 (+ 1.8)	8.9 (+ 1.6)	7.9 (+ 0.6)	8.5 (+ 0.1)	7.5 (+ 0.1)	9.2 (+ 0.1)	9.0 (+ 0.1)	8.0 (+ 0.1)
Free Voters	2.3	2.7 (+ 0.4)	3.2 (+ 0.9)	2.3 (+ 0.1)	2.6 (+ 0.4)	2.7 (+ 0.4)	2.7 (0.0)	3.2 (0.0)	2.3 (0.0)	2.7 (0.0)	2.7 (0.0)
Other	5.6	5.1 (− 0.4)	6.9 (+ 1.3)	4.0 (− 1.6)	4.9 (− 0.6)	5.3 (− 0.2)	5.0 (− 0.2)	6.5 (− 0.4)	4.0 (0.0)	4.7 (− 0.2)	5.2 (− 0.1)

Note: Cells show the share of respondents (%) before exclusions (left) and in the main sample (right), overall, by recruitment channel (OAP = online access panel, SM = social media), and by experimental group. In the main sample, 97 respondents who allocated 100% to one party are excluded. For the overall sample, the difference to a benchmark is shown in percentage points in parentheses. For the main sample, the difference to the overall sample before exclusions is shown. Darker shading indicates larger absolute differences (scale up to 20 percentage points on the left and up to 5 percentage points on the right). Benchmarks are the eligible voting population for age and gender (Statistisches Landesamt des Freistaates Sachsen, 2024a), the overall population for education (Statistisches Bundesamt (Destatis), 2026b), election results for vote intention, and actual turnout for turnout intention (Statistisches Landesamt des Freistaates Sachsen, 2024b). Men* includes men and diverse respondents.

B.3. Descriptives

B.3.1. Response time

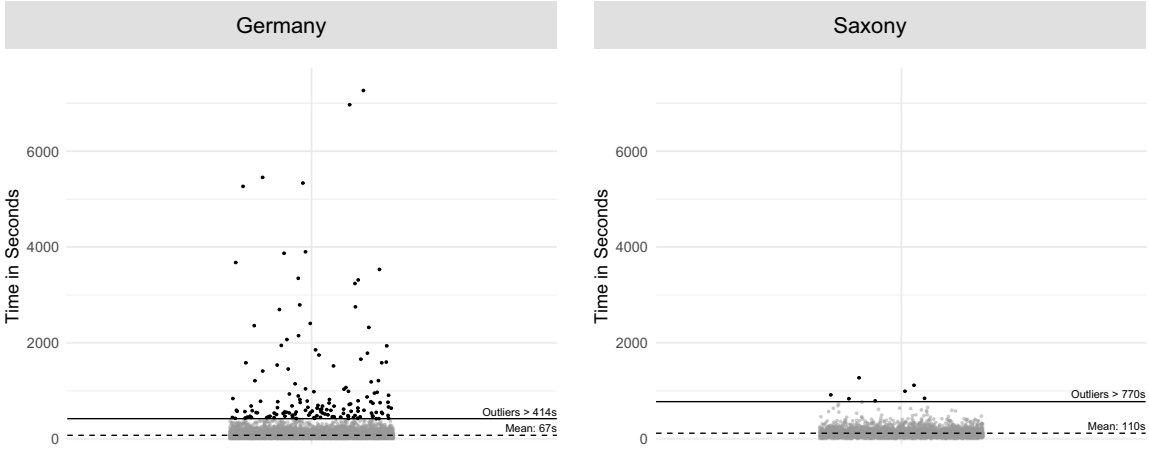


Fig. B.1. Distribution of response times. *Note:* The figure shows the time respondents took to answer the vote expectation item. Points indicate individual response times. Jittering along the x-axis is used solely for visualization purposes and has no substantive interpretation. The dashed line marks the mean. Outliers are defined as those lying more than two standard deviations above or below the mean. The solid line marks the upper cutoff. In both cases, no extremely low outliers are present. High outliers (black points) are removed from the analyses, leaving the respondents with response times depicted in gray. For the federal election (left), response times ranged from 2.7 to 20,496.6 seconds, with a mean of 66.9 seconds (SD = 173.8). 144 respondents with times exceeding 414.4 seconds are removed, leaving 28,286 respondents. For Saxony (right), response times ranged from 3.8 to 23,989 seconds, with a mean of 110 seconds (SD = 330). 8 respondents with times exceeding 770 seconds are removed, leaving 5554 respondents.

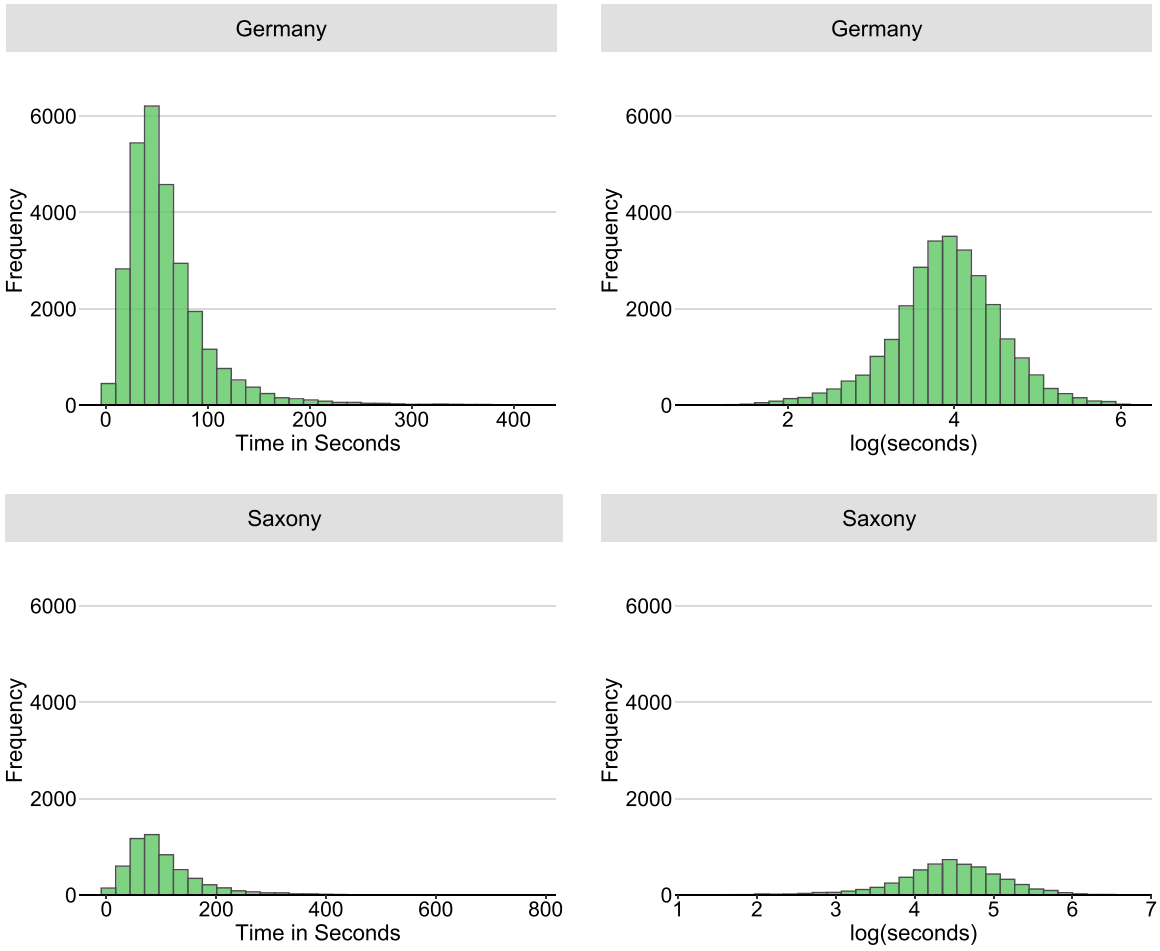


Fig. B.2. Response times. *Note:* The figures show histograms of the response times with outliers removed. The top row shows the federal election (n = 28,286), the bottom row shows Saxony (n = 5554). The left panels show the positively skewed distribution of raw response times in seconds, while the right panels show the logged response times used as the dependent variable to test H1.

B.3.2. Expectations across groups

Table B.5
Forecasting accuracy by party and treatment group (federal).

		Full Sample			Control			Treatment (5)			Treatment (500)		
Party	Result	Mean	Dev.[1]	SD	Avg.	Dev.	SD	Avg.	Dev.	SD	Avg.	Dev.	SD
AfD	20.80	21.35	0.55	8.55	21.29	0.48	8.95	21.34	0.54	8.48	21.42	0.62	8.21
BSW	4.98	6.48	1.50	5.76	6.62	1.63	6.05	6.37	1.39	5.63	6.44	1.46	5.60
CDU/CSU	28.52	26.95	-1.57	7.83	26.74	-1.78	8.02	26.93	-1.59	7.78	27.18	-1.34	7.69
FDP	4.33	5.85	1.53	4.89	6.01	1.68	5.21	5.87	1.54	4.80	5.69	1.36	4.66
Greens	11.61	12.00	0.39	6.19	11.90	0.29	6.31	12.10	0.49	6.20	12.00	0.40	6.04
Left	8.77	5.98	-2.80	5.38	5.99	-2.78	5.36	6.05	-2.73	5.55	5.89	-2.88	5.21
SPD	16.41	16.79	0.38	6.88	16.69	0.28	7.04	16.80	0.39	6.78	16.88	0.46	6.82
Others	4.57	4.60	0.03	5.34	4.77	0.20	5.71	4.54	-0.03	5.14	4.49	-0.08	5.14
MAE			1.09			1.14			1.09			1.08	
RMSE			1.39			1.45			1.37			1.36	

[1] Dev. = Mean - Result. The mean of the individual expectations equals the citizen forecast. The 'MAE' / 'RMSE' rows give the accuracy of these group-level vote share forecasts.

Table B.6
Forecasting accuracy by party and treatment group (Saxony).

		Full sample			Control			Treatment (500)		
Party	Result	Mean	Dev.[1]	SD	Avg.	Dev.	SD	Avg.	Dev.	SD
AfD	30.63	31.04	0.41	12.45	31.14	0.50	12.88	30.94	0.31	11.99
BSW	11.82	11.28	-0.54	7.45	11.23	-0.59	7.82	11.33	-0.49	7.04
CDU	31.90	27.56	-4.34	10.54	27.91	-4.00	11.06	27.21	-4.70	9.97
FDP	0.89	3.57	2.67	4.19	3.56	2.66	4.62	3.58	2.69	3.70
Greens	5.11	5.59	0.48	5.68	5.60	0.49	6.23	5.57	0.46	5.04
Left	4.47	5.28	0.82	5.55	5.20	0.73	5.64	5.37	0.90	5.45
SPD	7.33	8.02	0.69	6.67	7.90	0.58	6.73	8.13	0.80	6.61
Free Voters	2.26	4.46	2.20	5.30	4.33	2.07	5.24	4.60	2.34	5.36
Others	5.58	3.20	-2.39	4.58	3.13	-2.46	4.77	3.27	-2.31	4.38
MAE			1.62			1.57			1.67	
RMSE			2.07			1.98			2.16	

[1] Dev. = Mean - Result. The mean of the individual expectations equals the citizen forecast. The 'MAE' / 'RMSE' rows give the accuracy of these group-level vote share forecasts.

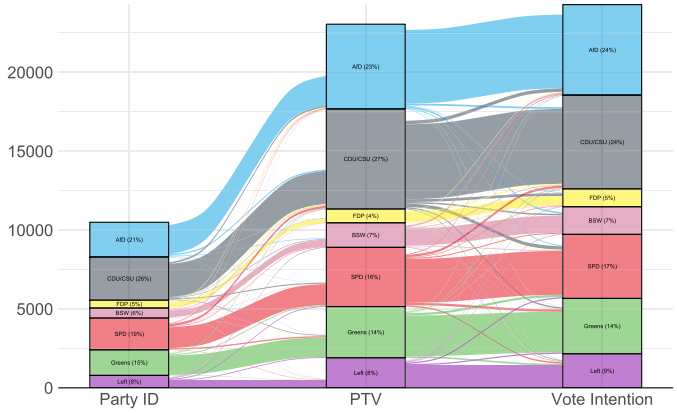


Fig. B.3. Comparison of preferred party measures for the federal election. *Note:* The figure compares three measures for the preferred party: the highest PTV score, vote intention, and party identification. The plot is based on 26,967 respondents who have a value for at least one measure. Based on the PTV, 85.5% have a preferred party (n = 23,056), and for vote intention, 90.1% (n = 24,296). Party identification was only measured in a follow-up survey wave and is available for 10,499 re-interviewed respondents (38.9%). Among respondents with both PTV scores and vote intentions available (n = 20,598), 86.8% preferred the same party on both measures. For PTV and party identification (n = 8971), agreement was 85.4%, and for vote intention and party identification (n = 9806), 77.4%. Among the 8491 respondents with all three measures, 78.7% have the same preferred party on all three.

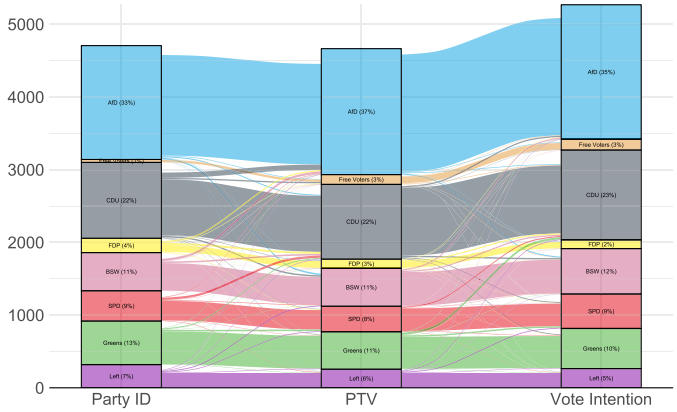


Fig. B.4. Comparison of preferred party measures for the Saxon election. *Note:* The figure compares three measures for the preferred party: the highest PTV score, vote intention, and party identification. The plot is based on 5481 respondents who have a value for at least one measure. Based on the PTV, 85.1% have a preferred party (n = 4665), and for vote intention, 96.1% (n = 5269). Party identification is available for 4707 respondents (85.9%). Among respondents with both PTV scores and vote intentions available (n = 4470), 92.6% prefer the same party on both measures. For PTV and party identification (n = 4039), agreement was 88.6%, and for vote intention and party identification (n = 4627), 82.3%. Among the 3976 respondents with all three measures, 84.5% have the same preferred party on all three.

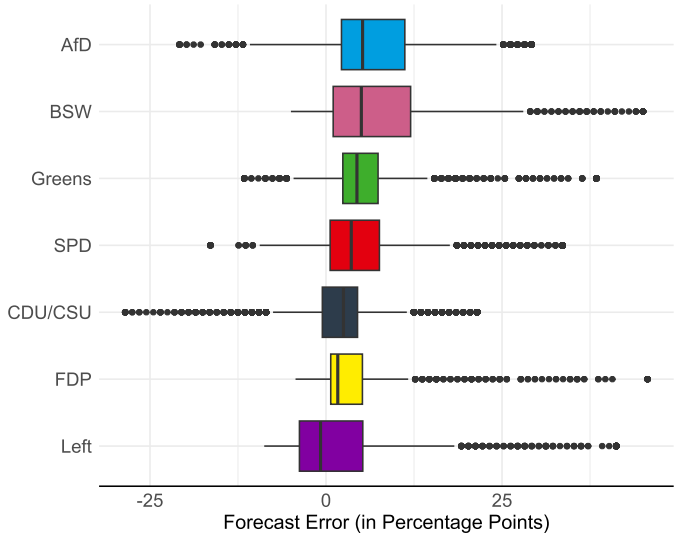


Fig. B.5. Forecast error for preferred party (Federal). *Note:* Boxplots illustrate the deviations between respondents' expected vote shares and actual vote shares (forecast error) for respondents' preferred party (identified through the highest PTV score). Parties are ordered by median forecast error. Of the 23,056 respondents, 27.5% prefer the CDU/CSU, 23.3% the AfD, 16.3% the SPD, 14.1% the Greens, 8.3% the Left, 6.7% the BSW, and 3.8% the FDP. For 40.33% of them the preferred party had a PTV score of 11, for 22.16% a score of 10, for 16.31% a score of 9, for 11.35% a score of 8, for 4.80% a score of 7, and for 1165 respondents (5.05%) the score was below 7. While most respondents overestimated their preferred party, 22% underestimated it, most notably for the Left, where more than half expected it to perform worse than it actually did, likely because data collection began 30 days before the election, when the party was still polling at around 4% to 5%, and gained momentum over the field period. As the measure is defined as $\hat{e}_{p^*} - v_{p^*}$, possible minimum and maximum values differ across parties: although expected vote shares could range from 0% to 50% for each party, actual results differed, so that larger parties such as the CDU/CSU could be underestimated to a greater extent, whereas smaller parties like the FDP, BSW, and the Left could be more strongly overestimated (specifically, values could range from −20.8 to 29.2 for the AfD, −5 to 45 for the BSW, −11.6 to 38.4 for the Greens, −16.4 to 33.6 for the SPD, −28.5 to 21.5 for CDU/CSU, −4.3 to 45.7 for the FDP, and −8.8 to 41.2 for the Left); see Table 1 for the election results.

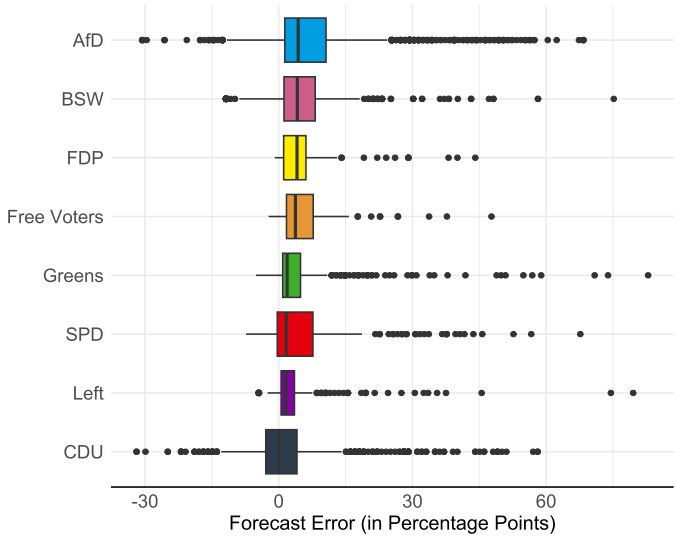


Fig. B.6. Forecast error for preferred party (Saxony). *Note:* Boxplots illustrate the deviations between respondents' expected vote shares and actual vote shares (forecast error) for respondents' preferred party (identified through the highest PTV score). Parties are ordered by median forecast error. Of the 4665 respondents, 37.1% prefer the AfD, 22.0% the CDU, 11.1% the BSW, 11.0% the Greens, 7.5% the SPD, 5.5% the Left, 2.9% the Free Voters, and 2.8% the FDP. For 62.92% of them, the preferred party had a PTV score of 11, for 18.18% a score of 10, and for 9.90% the highest score was a 9. For 5.42%, the score was 8, and for 167 respondents, the highest PTV score was below 8. While most respondents overestimated their preferred party, 25.4% underestimated it, most notably for the CDU, where nearly half (44.7%) expected it to perform worse than it actually did. As the measure is defined as $\hat{e}_{p^*} - v_{p^*}$, possible minimum and maximum values differ across parties: although expected vote shares could range from 0% to 100% for each party, actual results differed, so that larger parties such as the CDU and AfD could be underestimated to a greater extent, whereas smaller parties like the FDP, the Free Voters, and the Left could be more strongly overestimated (specifically, values could range from −30.6 to 69.4 for the AfD, −11.8 to 88.2 for the BSW, −31.9 to 68.1 for the CDU, −0.9 to 99.1 for the FDP, −2.3 to 97.7 for the Free Voters, −5.1 to 94.9 for the Greens, −4.5 to 95.5 for the Left, and −7.3 to 92.7 for the SPD); see Table 1 for the election results.

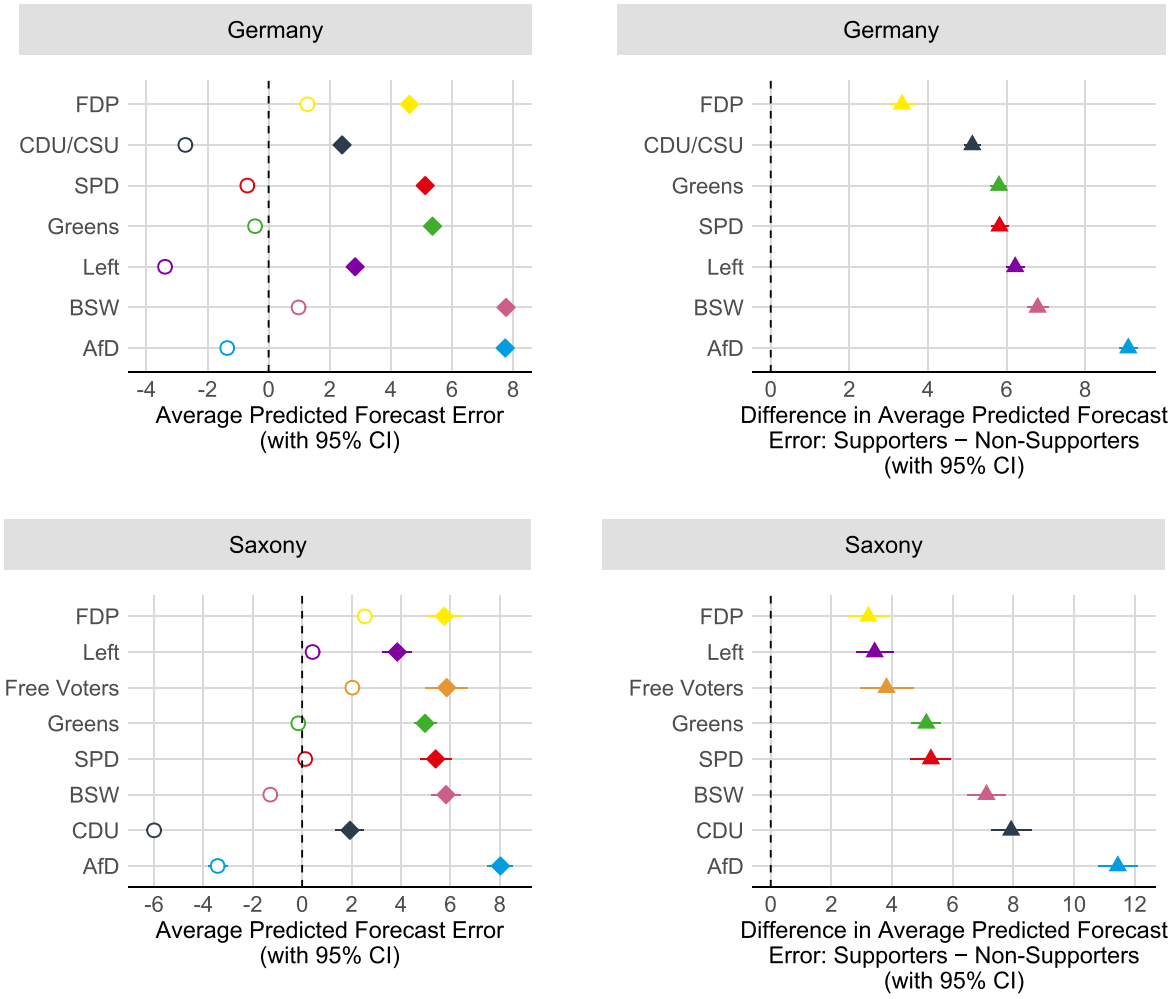


Fig. B.7. Wishful thinking in federal and Saxon electoral expectations. *Note:* The left panels show the average predicted forecast error (difference between the expected vote share and the realized vote share) for each party, differentiating between those who prefer the respective party (diamonds) and those who do not (circles). The top row refers to the federal election, the bottom row to the Saxon state election. In both elections, the forecast error for those who prefer a party (diamonds) is always positive; supporters overestimate their party on average. The average forecast error among those who do not prefer the specific party (circles) is always smaller. The right panels show the differences in forecast errors between supporters and non-supporters: At the federal level, supporters overestimate their preferred party by 3.3 percentage points [3.0; 3.7] to 9.1 percentage points [8.9; 9.3] on average compared to non-supporters. In Saxony, the difference ranges from 3.2 percentage points [2.5; 3.9] to 11.4 percentage points [10.8; 12.1]. The right plots are also part of the main paper (see Fig. 4). Results are based on the 23,056 federal and 4665 Saxon respondents with a clearly preferred party (PTV).

Appendix C. Results

C.1. Regression models (main specification)

Regression models for the federal election (see Table C.1) and Saxony (see Table C.2).

Table C.1
Linear regression models for the federal election.

	H1	H2		H3				
	Time	MAE	RMSE	PTV Dummy	PTV Dummy (incl. Ties)	VI Dummy	PID Dummy	PTV Score
	(1)	(2a)	(2b)	(3a)	(3b)	(3c)	(3d)	(3e)
Intercept	2.99***	3.93***	4.88***	−0.68***	−0.64***	−0.86***	−0.41***	−2.61***
	(0.08)	(0.37)	(0.46)	(0.09)	(0.11)	(0.09)	(0.10)	(0.14)
T(5) (ref: C)	0.08***	−0.23***	−0.27***	0.07**	0.07**	0.08***	0.11**	0.26***
	(0.01)	(0.04)	(0.05)	(0.03)	(0.02)	(0.02)	(0.04)	(0.06)
T(500)	0.13***	−0.32***	−0.39***	0.09***	0.09***	0.12***	0.15***	0.33***
	(0.01)	(0.04)	(0.05)	(0.02)	(0.02)	(0.02)	(0.04)	(0.06)
Baseline Speed	0.36***							
	(0.01)							
Survey Duration	0.00***							
	(0.00)							
Preferred Party				6.62***	5.24***	6.62***	5.82***	
				(0.12)	(0.10)	(0.11)	(0.16)	
T(5) x Preferred Party				−0.32*	−0.26*	−0.36*	−0.50*	
				(0.15)	(0.13)	(0.15)	(0.22)	
T(500) x Preferred Party				−0.42**	−0.26*	−0.59***	−0.63**	
				(0.15)	(0.13)	(0.15)	(0.22)	
PTV Score								0.77***
								(0.01)
T(5) x PTV Score								−0.05***
								(0.01)
T(500) x PTV Score								−0.07***
								(0.01)
Date Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓
Party Fixed Effects	–	–	–	✓	✓	✓	✓	✓
Clustered SE	–	–	–	✓	✓	✓	✓	✓
R^2	0.09	0.01	0.01	0.16	0.13	0.15	0.15	0.17
Adj. R^2	0.09	0.01	0.01	0.16	0.13	0.15	0.15	0.17
Num. obs.	28,286	28,430	28,430	160,290	198,870	170,072	73,493	197,547
Num. respondents	28,286	28,430	28,430	23,056	28,410	24,296	10,499	28,410

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Different linear regression models have been estimated. All include survey date fixed effects. The first *Time* model (1) has response latency (log(s)) as the dependent variable to test H1. Models (2a) and (2b) are concerned with overall accuracy, measured as *MAE* and, as a robustness check, *RMSE*, to test H2. For H3, the dependent variable is the forecast error (difference between respondents' expected vote share for a party and the actual result). Models include party fixed effects and standard errors are clustered by respondent. In Model (3a), the *preferred party* is the party to which respondents assign the highest PTV; respondents who give the highest PTV to more than one party are not included. As robustness checks, the preferred party is operationalized as the party with the highest PTV including ties (Model (3b)), the party respondents intend to vote for (Model (3c)) or the party respondents identify with (Model (3d)). Model (3e) assesses degrees of party preference through the propensity to vote (measured from 1 to 11). For Models (3a)–(3e), *Num. obs.* counts respondent × party observations and *Num. respondents* counts the respondents.

Table C.2
Linear regression models for Saxony.

	H1	H2		H3				
	Time	MAE	RMSE	PTV Dummy	PTV Dummy (incl. Ties)	VI Dummy	PID Dummy	PTV Score
	(1)	(2a)	(2b)	(3a)	(3b)	(3c)	(3d)	(3e)
Intercept	2.69***	5.69***	7.53***	−2.02***	−1.94***	−2.36***	−2.12***	−3.56***
	(0.07)	(0.30)	(0.43)	(0.18)	(0.18)	(0.18)	(0.19)	(0.22)
T(500) (ref: C)	0.10***	−0.29***	−0.44***	0.19***	0.19***	0.19***	0.17***	0.55***
	(0.02)	(0.08)	(0.12)	(0.05)	(0.05)	(0.05)	(0.05)	(0.12)
Baseline Speed	0.75***							
	(0.02)							
Survey Duration	0.00***							
	(0.00)							
Preferred Party				8.51***	6.84***	8.75***	8.27***	
				(0.30)	(0.24)	(0.28)	(0.30)	
T(500) x Preferred Party				−1.69***	−1.25***	−1.75***	−1.48***	
				(0.37)	(0.31)	(0.36)	(0.39)	
PTV Score								0.76***
								(0.02)
T(500) x PTV Score								−0.13***
								(0.03)
Date Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓
Party Fixed Effects	–	–	–	✓	✓	✓	✓	✓
Clustered SE	–	–	–	✓	✓	✓	✓	✓
R^2	0.25	0.07	0.06	0.16	0.14	0.16	0.15	0.15
Adj. R^2	0.24	0.06	0.06	0.16	0.14	0.16	0.15	0.15
Num. obs.	5554	5562	5562	37,187	44,432	42,152	37,656	44,274
Num. respondents	5554	5562	5562	4665	5554	5269	4707	5554

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Different linear regression models have been estimated. All include survey date fixed effects. The first *Time* model (1) has response latency (log(s)) as the dependent variable to test H1. Models (2a) and (2b) are concerned with overall accuracy, measured as *MAE* and, as a robustness check, *RMSE*, to test H2. For H3, the dependent variable is the forecast error (difference between respondents' expected vote share for a party and the actual result). Models include party fixed effects and standard errors are clustered by respondent. In Model (3a), the *preferred party* is the party to which respondents assign the highest PTV; respondents who give the highest PTV to more than one party are not included. As robustness checks, the preferred party is operationalized as the party with the highest PTV including ties (Model (3b)), the party respondents intend to vote for (Model (3c)) or the party respondents identify with (Model (3d)). Model (3e) assesses degrees of party preference through the propensity to vote (measured from 1 to 11). For Models (3a)–(3e), *Num. obs.* counts respondent × party observations and *Num. respondents* the respondents.

C.2. Robustness checks: different operationalizations

C.2.1. H1 – response latency

Rittmann et al. (2025) recommend reporting the median and mean response time. As insights are very similar, the main body only reports the median (see Fig. 2). This section also illustrates the mean (see Fig. C.1). While the mean provides overall higher estimates, the patterns are the same.

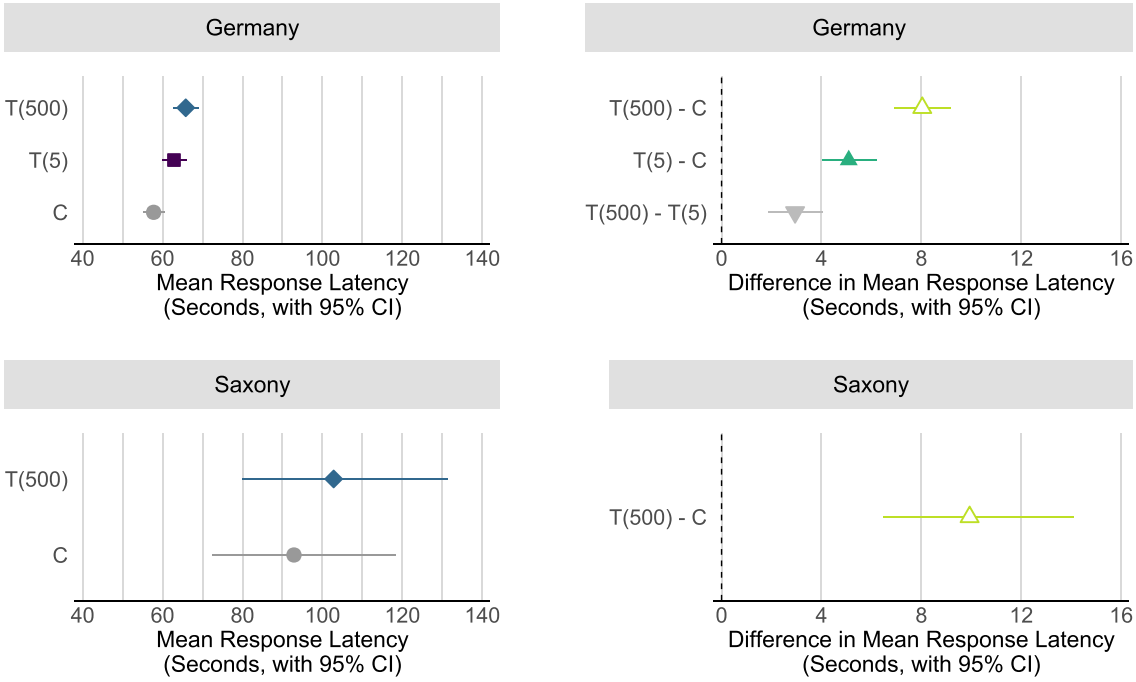


Fig. C.1. Effects of treatment on mean response latency. *Note:* The left panels display the conditional mean response latencies (in seconds) for the expectation item by group. The median is also displayed in Fig. 2 in the main body. The right panels show the first differences between these groups. The top row reports results from the federal election, the bottom row from Saxony. Estimates are based on model 1, transformed to seconds (Rittmann et al., 2025) for the average-case scenario described in Section 5.2. Lines correspond to 95% confidence intervals. While the conditional mean is always higher than the median, treated respondents took longer, on average, than respondents in the control condition; respondents in the control group were the fastest. Additionally, for the federal election, the *Treatment (500)* yielded, on average, longer response times than the *Treatment (5)* condition. As shown in the right panels, all group differences are statistically different from zero.

C.2.2. H2 – Overall accuracy

As an alternative accuracy measure to the MAE, the RMSE is used:

$$RMSE = \sqrt{\frac{1}{P}\sum_{p=1}^{P}(\hat{e}_p - v_p)^2}.$$

The RMSE ranges from 0.48 to 24.43 (mean = 5.53, SD = 3.63) for the federal election, and from 0.60 to 35.79 (mean = 6.24, SD = 4.51) in Saxony. While model 2b in Tables C.1 and C.2 reports the regression results, Fig. C.2 shows the average predicted RMSEs and their differences between groups for the federal election and the Saxon state election, indicating similar patterns to the MAE (see Fig. 3 in the main paper).

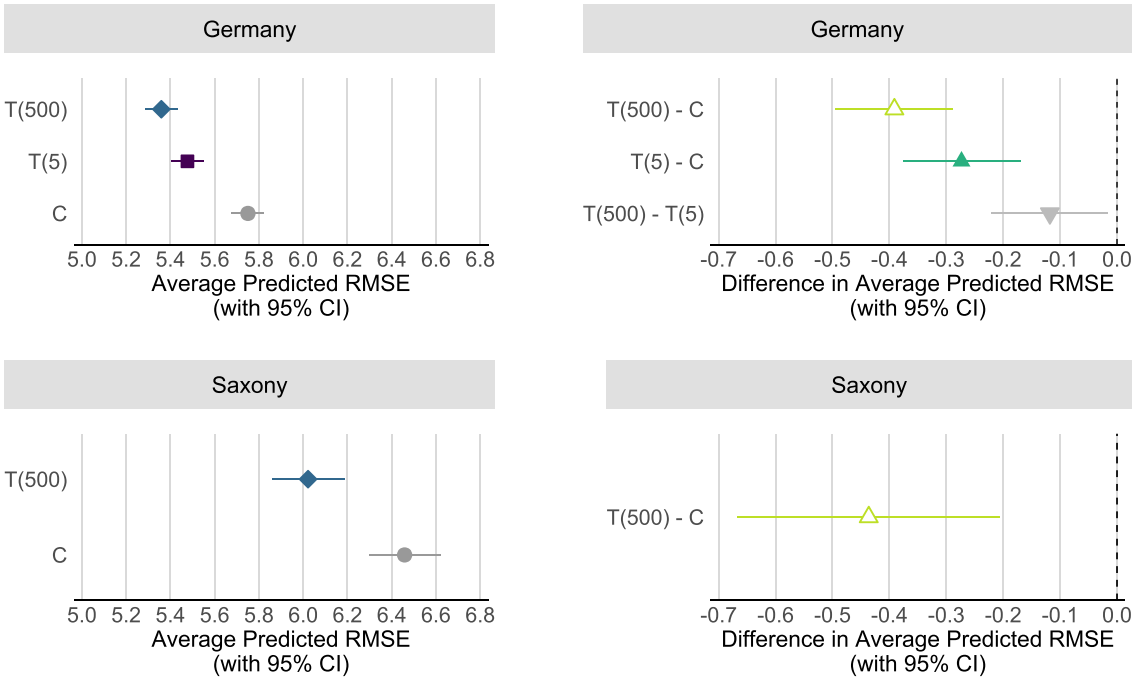


Fig. C.2. Effects of treatment on RMSE. *Note:* The dots in the left panels show the average predicted RMSE of respondents' expected vote shares by experimental group based on model 2b, using an observed-value approach. The right panels show the differences between groups. Lines correspond to 95% confidence intervals. The top row reports the federal election, the bottom row reports Saxony. In the federal election, respondents in the *Treatment (500)* condition had the lowest error, i.e., were most accurate, followed by *Treatment (5)*, while the control group had the highest error. In Saxony, respondents in the *Treatment (500)* condition were likewise more accurate than the control group. As the right panels show, all differences between the groups are statistically different from zero. On average, the treatments changed the RMSE in the federal election by −0.27 [−0.38; −0.17] (*Treatment (5)* – C), −0.39 [−0.49; −0.29] (*Treatment (500)* – C), and −0.12 [−0.22; −0.02] (*Treatment (500)* – *Treatment (5)*) percentage points, and in Saxony by −0.44 [−0.67; −0.21] (*Treatment (500)* – C) percentage points. Results are based on responses from 28,430 respondents (for the federal election) and 5562 (for Saxony), respectively.

C.2.3. H3 – Wishful thinking

The following figures show the effects of the treatment on forecasting error by party preference analogous to Fig. 5 based on three different operationalizations of the preferred party: based on highest PTV score including tied respondents, based on vote intention and based on party identification.

H3 – Wishful thinking (PTV including ties) – model 3b.

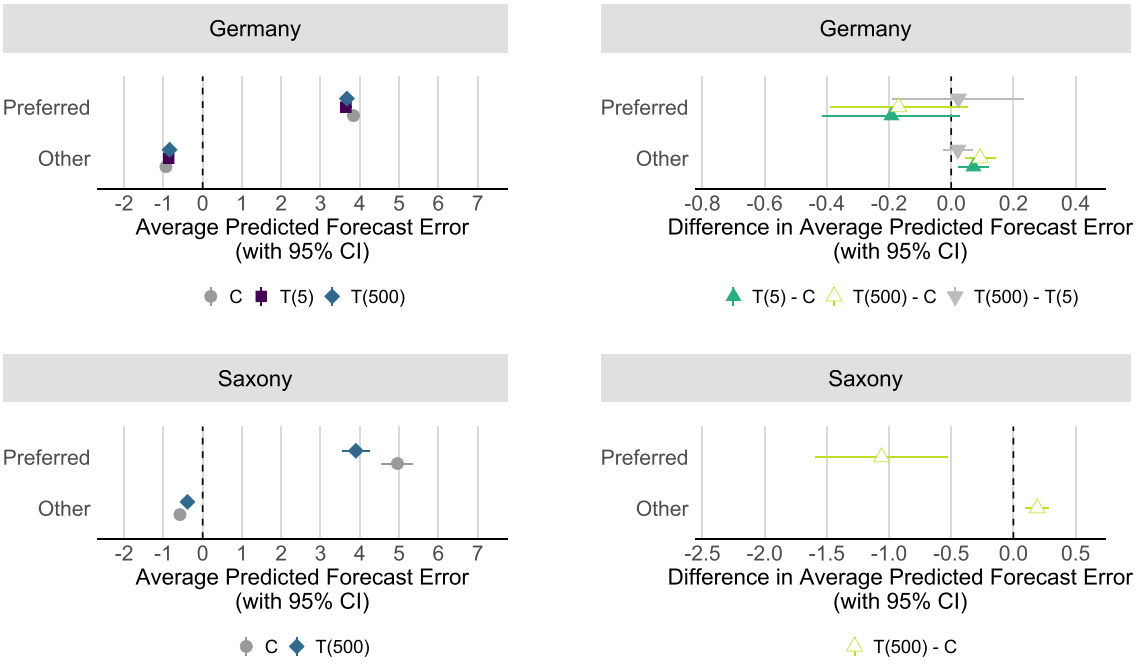


Fig. C.3. Effects of treatment on forecast error by party preference (PTV including ties). *Note:* The dots in the left panels show the average predicted forecast error depending on whether a party was the respondent's preferred party or not (Other) and on treatment status based on model 3b, using an observed-value approach. The preferred party is operationalized as the party with the highest propensity to vote (PTV) score. In this operationalization, respondents with multiple parties sharing the highest score are included, with each tied party counted as a preferred party. Lines correspond to 95% confidence intervals. The top row reports the federal election (n = 28,410), and the bottom row Saxony (n = 5554). The right panels show pairwise differences between experimental groups. In both cases, respondents deviate more positively for their preferred party than for less-preferred parties. For the preferred party, the treatment effects are −0.19 [−0.41; 0.03] (*Treatment (5)*) and −0.17 [−0.39; 0.05] (*Treatment (500)*) percentage points in the federal election (both are not significant), and −1.06 [−1.59; −0.52] percentage points in Saxony. For non-preferred parties, the treatment effects are 0.07 [0.02; 0.12] (*Treatment (5)*) and 0.09 [0.04; 0.14] (*Treatment (500)*) percentage points in the federal election, and 0.19 [0.10; 0.28] percentage points in Saxony.

H3 – Wishful thinking (vote intention) – model 3c.

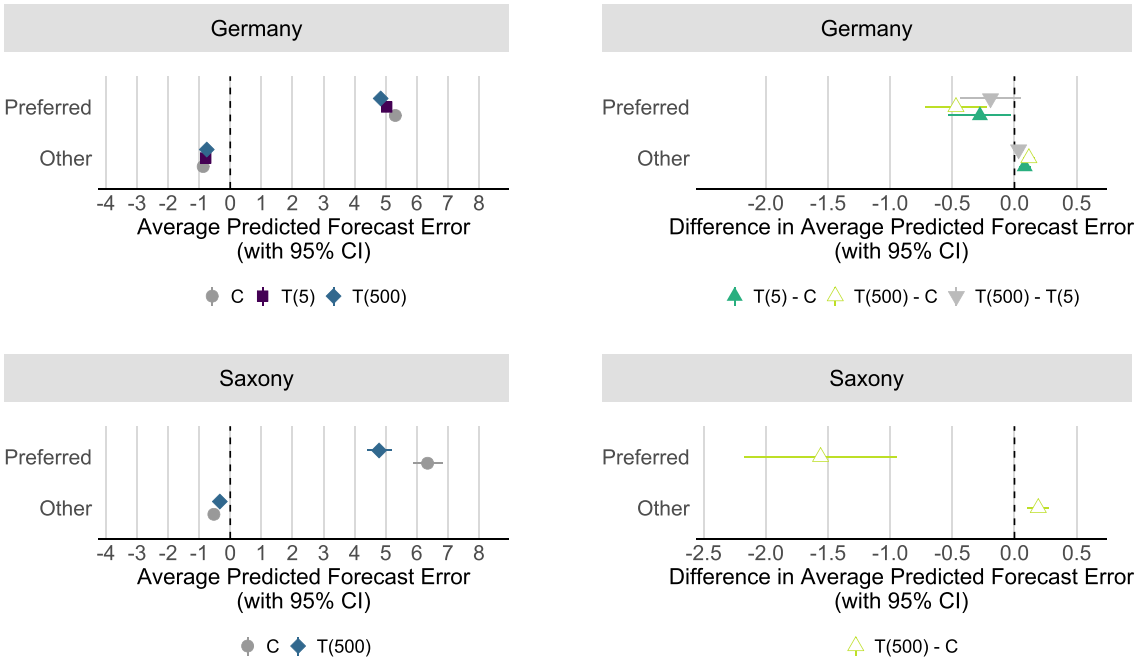


Fig. C.4. Effects of treatment on forecast error by party preference (vote intention). *Note:* The dots in the left panels show the average predicted forecast error depending on whether a party was the respondent's preferred party or not (Other) and on treatment status based on model 3c, using an observed-value approach. The preferred party is operationalized as the party respondents intend to vote for. Lines correspond to 95% confidence intervals. The top row reports the federal election (n = 24,296), the bottom row reports Saxony (n = 5269). The right panels show pairwise differences between experimental groups. In both cases, respondents deviate more positively for their preferred party than for less-preferred parties, and monetary incentives reduce this overestimation relative to the control group. For the preferred party, the treatment effects are −0.28 [−0.53; −0.03] (*Treatment (5)*) and −0.47 [−0.72; −0.22] (*Treatment (500)*) percentage points in the federal election, and −1.56 [−2.17; −0.95] percentage points in Saxony. For non-preferred parties, the treatment effects are 0.08 [0.04; 0.13] (*Treatment (5)*) and 0.12 [0.07; 0.16] (*Treatment (500)*) percentage points in the federal election, and 0.19 [0.10; 0.28] percentage points in Saxony.

H3 – Wishful thinking (party identification) – model 3d.

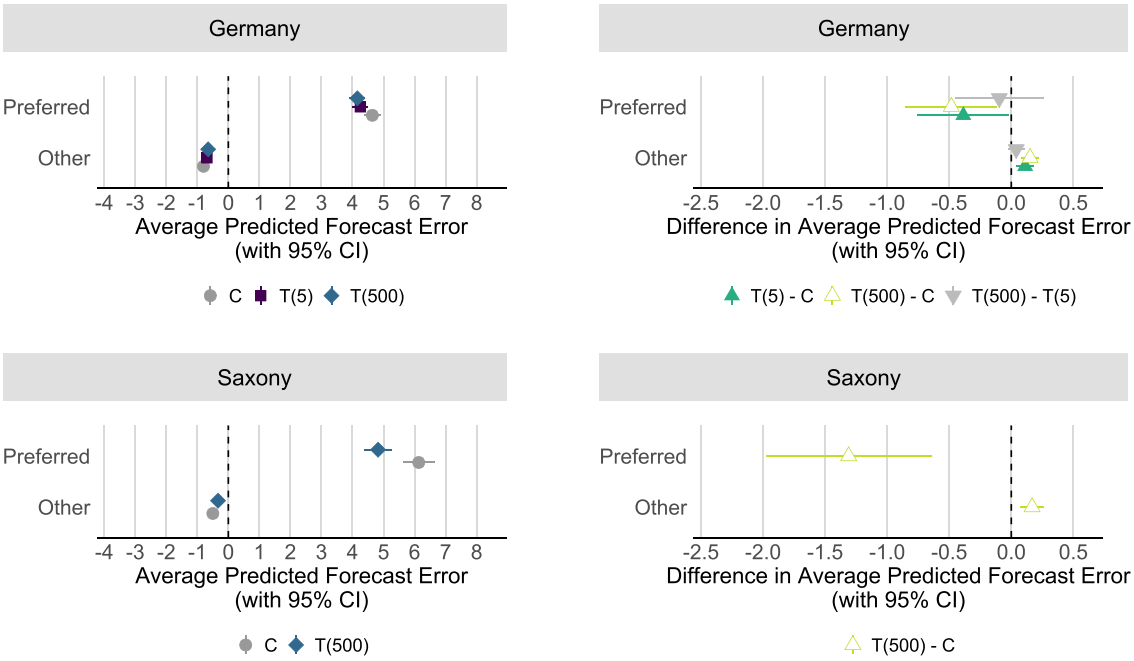


Fig. C.5. Effects of treatment on forecast error by party preference (party identification). *Note:* The dots in the left panels show the average predicted forecast error depending on whether a party was the respondent's preferred party or not (Other) and on treatment status based on model 3d, using an observed-value approach. The preferred party is operationalized as the party respondents indicate they identify with. Lines correspond to 95% confidence intervals. The top row reports the federal election (n = 10,499), the bottom row Saxony (n = 4707). The right panels show pairwise differences between experimental groups. In both cases, respondents deviate more positively for their preferred party than for less-preferred parties, and monetary incentives reduce this overestimation relative to the control group. For the preferred party, the treatment effects are −0.39 [−0.75; −0.02] (*Treatment (5)*) and −0.48 [−0.85; −0.11] (*Treatment (500)*) percentage points in the federal election, and −1.31 [−1.98; −0.64] percentage points in Saxony. For non-preferred parties, the treatment effects are 0.11 [0.04; 0.18] (*Treatment (5)*) and 0.15 [0.08; 0.22] (*Treatment (500)*) percentage points in the federal election, and 0.17 [0.07; 0.26] percentage points in Saxony.

H3 – Wishful thinking (PTV score) – model 3e.

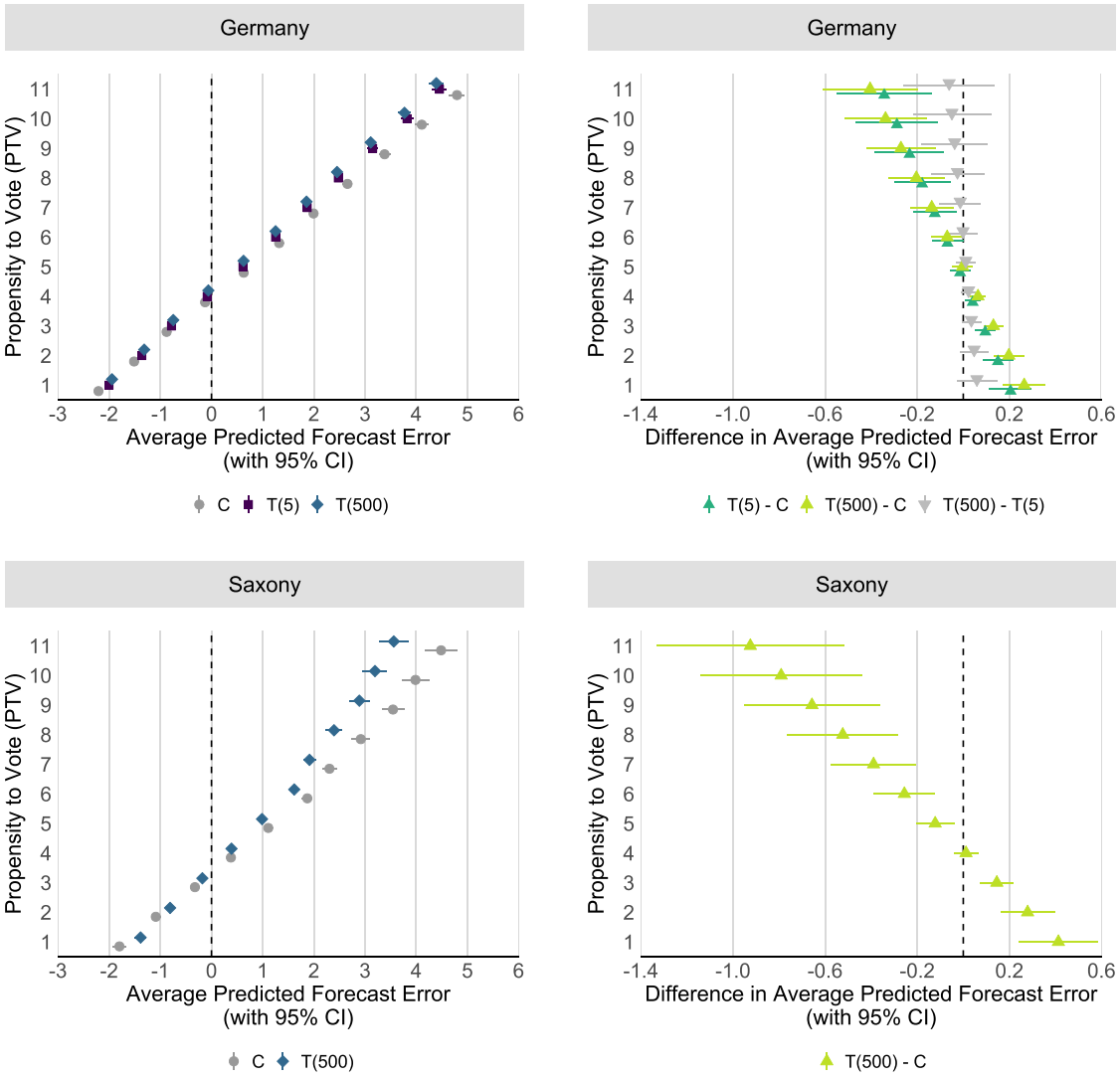


Fig. C.6. Effects of treatment on forecast error by propensity to vote. *Note:* The dots in the left panels show the average predicted forecast error depending on the propensity to vote that respondents attach to a party and on treatment status based on model 3e, using an observed-value approach. Lines correspond to 95% confidence intervals. The top row reports the federal election, whereas the bottom row reports Saxony. In both cases, respondents deviate more negatively for parties with a lower PTV score, and more positively for parties with a higher PTV score (> 3 in Saxony and > 4 at the federal level). The treatment can decrease this under- and overestimation to some extent. The right panels show experimental group differences. The treatments pull the forecast error, on average, closer to zero relative to the control group. No significant differences are observed between the two federal treatment groups. Results are based on 28,410 respondents (federal election) and 5554 respondents (Saxony).

C.3. Robustness check: models without excluding satisficers

Table C.3
Logistic regression of being excluded (answered 50% twice) on treatment and respondent characteristics (federal election).

	Excluded
Intercept	−1.77***
	(0.18)
T(5)	−0.18*
	(0.09)
T(500)	−0.34***
	(0.09)
Age 30–39	0.08
	(0.12)
Age 40–49	0.03
	(0.12)
Age 50–59	−0.50***
	(0.13)
Age 60+	−0.95***
	(0.14)
Gender: Women	−0.08
	(0.08)
Education: Medium	−0.81***
	(0.10)
Education: High	−1.44***
	(0.10)
Education: Still in School	−0.96*
	(0.43)
Intends to Vote	−0.64***
	(0.08)
State Fixed Effects	✓
Log Likelihood	−3202.63
Num. obs.	28,618

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Logistic regression of being excluded (answered 50% for two parties) on treatment and respondent characteristics. Reference categories: Control group, age 18-29, gender Men (incl. diverse), education Low, no turnout intention, state Baden-Württemberg. The model includes fixed effects for the 16 federal states, whose coefficients are not shown.

Table C.4
Logistic Regression of being excluded (allocated 100% to one party) on treatment and respondent characteristics (Saxony).

	Excluded
Intercept	−2.28***
	(0.44)
T(500)	−0.71**
	(0.22)
Age 30–39	0.07
	(0.40)
Age 40–49	0.09
	(0.39)
Age 50–59	0.08
	(0.38)
Age 60 +	0.21
	(0.38)
Gender: Women	0.10
	(0.21)
Education: Medium	−0.37
	(0.34)
Education: High	−0.95**
	(0.37)
Education: Still in School	−14.24
	(460.29)
Intends to Vote	−0.83**
	(0.26)
Recruited via Social Media	−0.70**
	(0.24)
Log Likelihood	−454.79
Num. obs.	5557

Note: * p < 0.05, ** p < 0.01, *** p < 0.001. Logistic regression of being excluded (allocated 100% to one party) on treatment and respondent characteristics. Reference categories: Control group, age 18-29, gender Men (incl. diverse), education Low, no turnout intention, recruited via online access panel. The large standard error of Still in School results from the absence of excluded respondents in this category.

Table C.5
Linear regression models for the federal election without exclusions.

	H1	H2		H3				
	Time	MAE	RMSE	PTV Dummy	PTV Dummy (incl. Ties)	VI Dummy	PID Dummy	PTV Score
	(1)	(2a)	(2b)	(3a)	(3b)	(3c)	(3d)	(3e)
Intercept	2.91***	4.63***	5.75***	−0.96***	−0.88***	−1.04***	−0.48***	−3.14***
	(0.08)	(0.43)	(0.53)	(0.18)	(0.17)	(0.14)	(0.11)	(0.21)
T(5) (ref: C)	0.09***	−0.28***	−0.34***	0.09**	0.09**	0.11***	0.15***	0.33***
	(0.01)	(0.05)	(0.06)	(0.03)	(0.03)	(0.03)	(0.04)	(0.07)
T(500)	0.14***	−0.43***	−0.52***	0.12***	0.14***	0.16***	0.20***	0.46***
	(0.01)	(0.05)	(0.06)	(0.03)	(0.03)	(0.03)	(0.04)	(0.07)
Baseline Speed	0.37***							
	(0.01)							
Survey Duration	0.00***							
	(0.00)							
Preferred Party				7.34***	5.87***	7.53***	6.71***	
				(0.13)	(0.11)	(0.12)	(0.18)	
T(5) x Preferred Party				−0.45**	−0.42**	−0.55***	−0.76**	
				(0.17)	(0.15)	(0.16)	(0.24)	
T(500) x Preferred Party				−0.65***	−0.50***	−0.90***	−0.97***	
				(0.17)	(0.15)	(0.16)	(0.24)	
PTV Score								0.87***
								(0.01)
T(5) x PTV Score								−0.07***
								(0.02)
T(500) x PTV Score								−0.10***
								(0.02)
Date Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓
Party Fixed Effects	–	–	–	✓	✓	✓	✓	✓
Clustered SE	–	–	–	✓	✓	✓	✓	✓
R^2	0.09	0.01	0.01	0.15	0.12	0.15	0.14	0.16
Adj. R^2	0.09	0.01	0.01	0.15	0.12	0.14	0.14	0.16
Num. obs.	29,049	29,195	29,195	163,754	204,071	174,538	75,299	202,580
Num. respondents	29,049	29,195	29,195	23,570	29,153	24,934	10,757	29,153

Note: * p < 0.05, ** p < 0.01, *** p < 0.001. Same models as Table C.1, but including those respondents who answered 50% twice.

Table C.6
Linear regression models for Saxony without exclusions.

	H1	H2		H3				
	Time	MAE	RMSE	PTV Dummy	PTV Dummy (incl. Ties)	VI Dummy	PID Dummy	PTV Score
	(1)	(2a)	(2b)	(3a)	(3b)	(3c)	(3d)	(3e)
Intercept	2.65***	5.76***	7.65***	−2.09***	−1.83***	−2.37***	−2.10***	−3.62***
	(0.08)	(0.34)	(0.52)	(0.20)	(0.20)	(0.19)	(0.20)	(0.24)
T(500) (ref: C)	0.12***	−0.41***	−0.66***	0.30***	0.28***	0.27***	0.24***	0.73***
	(0.02)	(0.09)	(0.14)	(0.06)	(0.06)	(0.05)	(0.06)	(0.13)
Baseline Speed	0.76***							
	(0.02)							
Survey Duration	0.00***							
	(0.00)							
Preferred Party				9.98***	7.74***	10.09***	9.58***	
				(0.37)	(0.29)	(0.34)	(0.36)	
T(500) x Preferred Party				−2.51***	−1.83***	−2.36***	−2.06***	
				(0.45)	(0.37)	(0.43)	(0.46)	
PTV Score								0.85***
								(0.03)
T(500) x PTV Score								−0.18***
								(0.03)
Date Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓
Party Fixed Effects	–	–	–	✓	✓	✓	✓	✓
Clustered SE	–	–	–	✓	✓	✓	✓	✓
R^2	0.24	0.06	0.05	0.16	0.13	0.16	0.15	0.14
Adj. R^2	0.23	0.05	0.04	0.16	0.13	0.16	0.15	0.14
Num. obs.	5650	5659	5659	37746	45208	42808	38248	45009
Num. respondents	5650	5659	5659	4740	5651	5351	4781	5651

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Same models as Table C.2, but including those respondents who allocated 100% to one party.

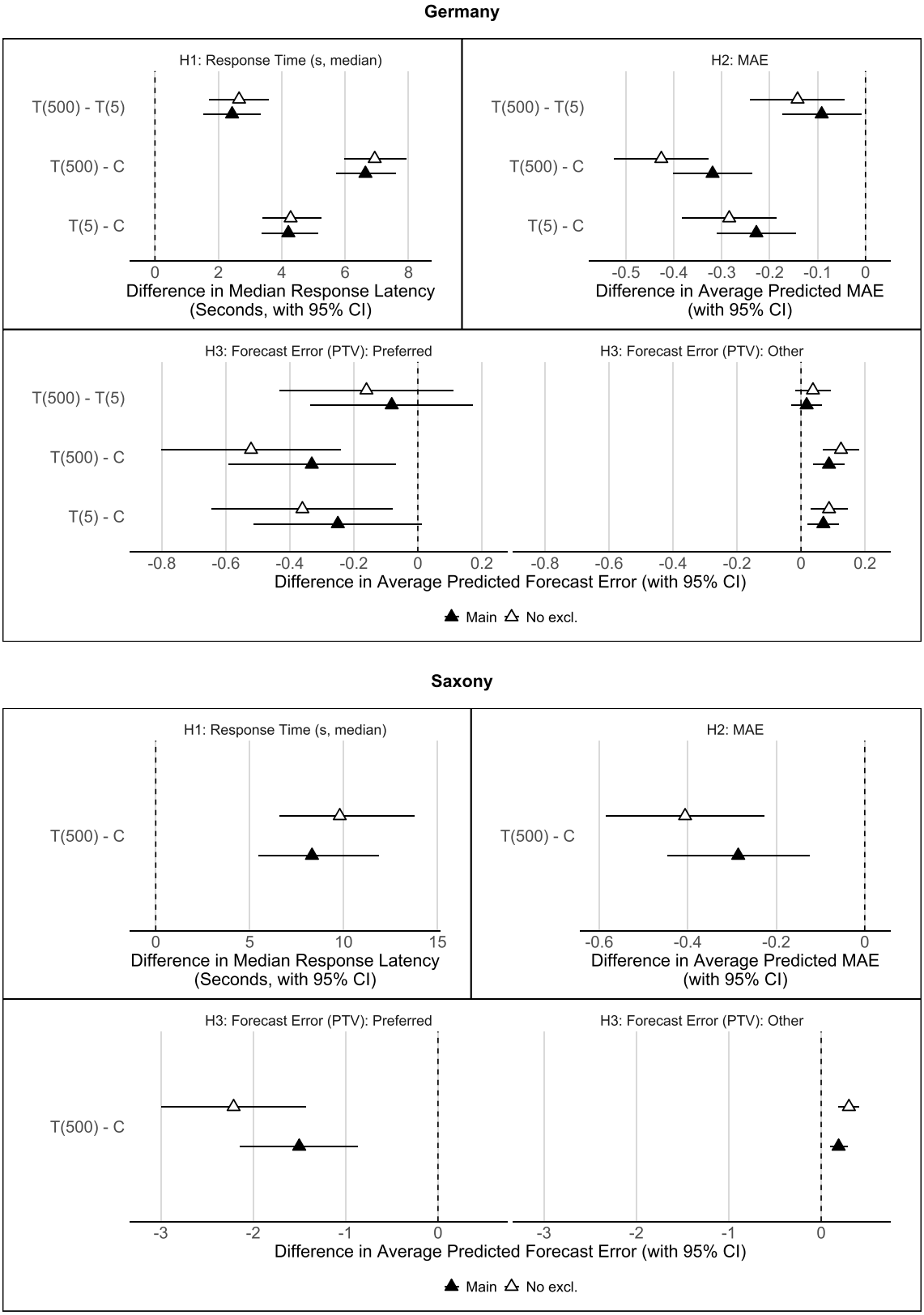


Fig. C.7. Effects of treatment with and without excluding satisficers. *Note:* The panels show the differences in predicted outcomes between experimental groups, for response time (H1), MAE (H2), and the forecast error by dichotomized PTV for preferred and other parties (H3). Filled triangles show the main sample, hollow ones show the full sample without excluding satisficers. Lines correspond to 95% confidence intervals. The top panel reports the federal election, the bottom panel shows Saxony.

C.4. Robustness check: Saxon models by recruitment channel

Table C.7
Linear regression models for Saxony by recruitment channel.

	H1		H2		H3	
	OAP	SM	OAP	SM	OAP	SM
	(1)	(1)	(2a)	(2a)	(3a)	(3a)
Intercept	2.73***	3.19***	5.82***	3.48***	−5.05***	−0.19
	(0.10)	(0.09)	(0.35)	(0.35)	(0.38)	(0.17)
T(500) (ref: C)	0.13***	0.09***	−0.52***	−0.19*	0.39***	0.11*
	(0.03)	(0.02)	(0.15)	(0.08)	(0.10)	(0.05)
Baseline Speed	0.71***	0.62***				
	(0.03)	(0.03)				
Survey Duration	0.00***	0.00***				
	(0.00)	(0.00)				
Preferred Party					12.32***	6.48***
					(0.67)	(0.25)
T(500) x Preferred Party					−3.23***	−1.10**
					(0.81)	(0.34)
Date Fixed Effects	✓	✓	✓	✓	✓	✓
Party Fixed Effects	–	–	–	–	✓	✓
Clustered SE	–	–	–	–	✓	✓
R^2	0.27	0.18	0.10	0.02	0.18	0.20
Adj. R^2	0.26	0.17	0.08	0.01	0.18	0.20
Num. obs.	2156	3398	2161	3401	13,766	23,421
Num. respondents	2156	3398	2161	3401	1723	2942

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Models (1), (2a) and (3a) from Table C.2, are estimated separately for respondents recruited through the online access panel (OAP) and through social media (SM). Standard errors for (3a) are clustered by respondent.

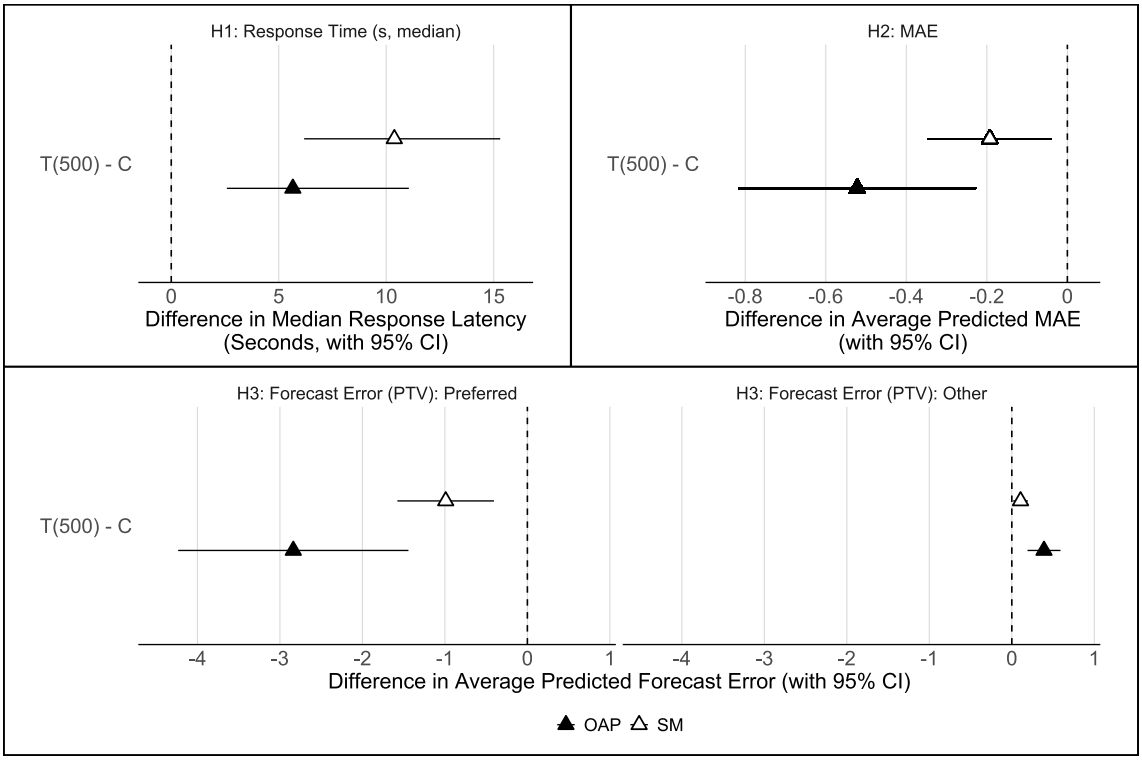


Fig. C.8. Effects of treatment among Saxons recruited via social media and the online access panel. *Note:* The panels show the differences in predicted outcomes between experimental groups, for response time (H1), MAE (H2) and the forecast error by PTV for preferred and other parties (H3). Filled triangles show the online access panel, hollow ones show the social media sample. Lines correspond to 95% confidence intervals. Response time differences are in seconds. As social media respondents took longer overall (median in the control group: 107 seconds in the social media sample vs. 42 seconds in the online access panel), their larger difference in seconds corresponds to a smaller relative effect (+9% vs. +14% in the online access panel; see Table C.7). Pooled models in which treatment is interacted with the recruitment channel show that the effects on MAE and on the forecast error for both preferred and other parties are significantly smaller among social media respondents, while the effect on response time does not differ significantly between the channels.

C.5. Exploratory heterogeneity analyses

C.5.1. H2 – Heterogeneity in overall accuracy (MAE)

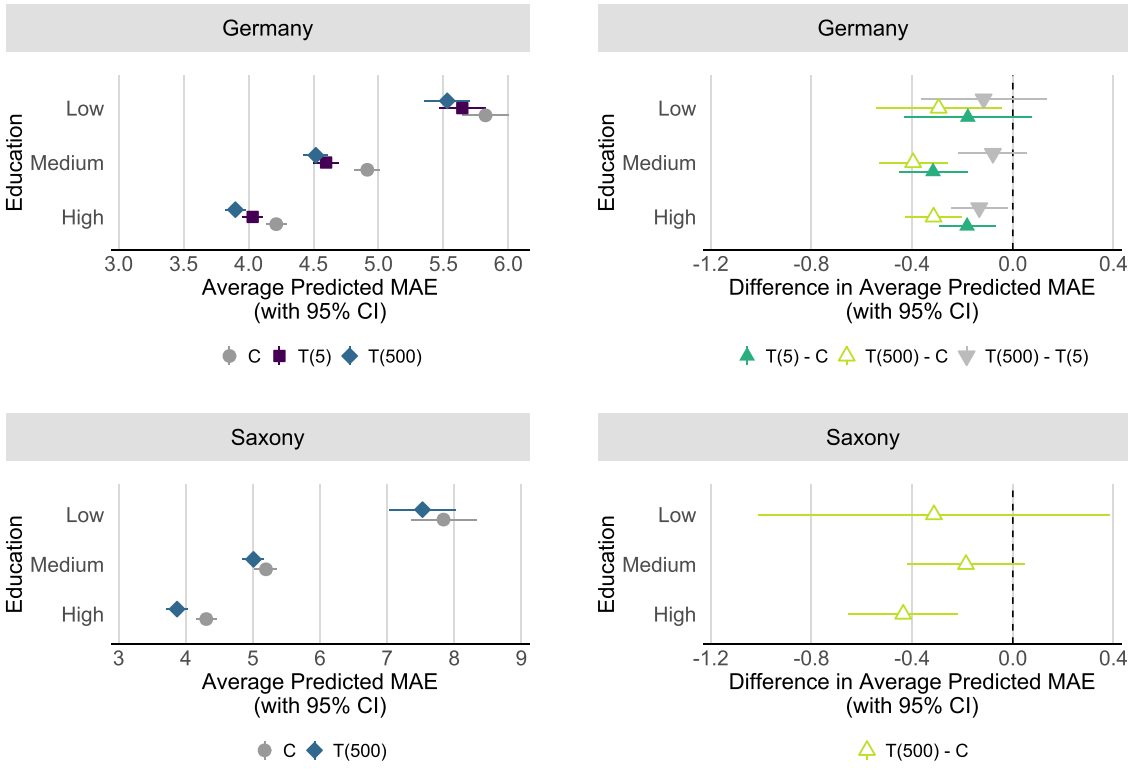


Fig. C.9. Effects of treatment on MAE by education. *Note:* The figure is based on OLS regression models, regressing respondents' MAE on treatment status interacted with their educational attainment while including survey date fixed effects. The dots in the left panels show the average predicted MAE of respondents' expected vote shares by experimental group and education level, based on an observed-value approach. The right panels show the differences between groups. Lines correspond to 95% confidence intervals. The top row reports the federal election, and the bottom row reports Saxony. In the federal election, the treatment effects of *Treatment (5)* and *Treatment (500)* relative to the control group are −0.18 [−0.29; −0.07] and −0.31 [−0.43; −0.20] (high education), −0.32 [−0.45; −0.18] and −0.40 [−0.53; −0.26] (medium), and −0.18 [−0.43; 0.08] and −0.29 [−0.54; −0.04] (low) percentage points. In Saxony, the treatment effects of *Treatment (500)* are −0.44 [−0.65; −0.22] (high), −0.19 [−0.42; 0.05] (medium), and −0.31 [−1.01; 0.38] (low) percentage points. Results are based on responses from 27,952 respondents (for the federal election) and 5390 (for Saxony), respectively.

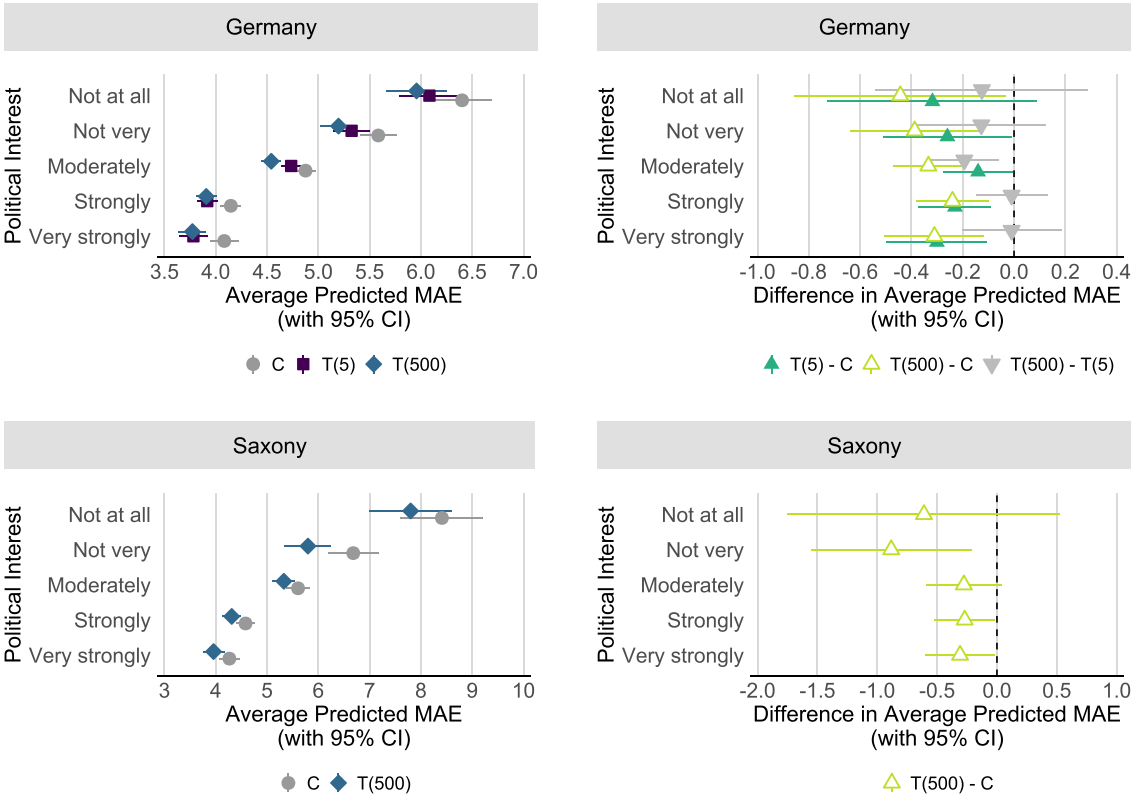


Fig. C.10. Effects of treatment on MAE by political interest. *Note:* The figure is based on OLS regression models, regressing respondents' MAE on treatment status interacted with their political interest while including survey date fixed effects. The dots in the left panels show the average predicted MAE for respondents' expected vote shares by experimental group and political interest, based on an observed-value approach. The right panels show the differences between groups. Lines correspond to 95% confidence intervals. The top row reports on the federal election, the bottom row reports on Saxony. Results are based on responses from 28,413 respondents (for the federal election) and 5559 (for Saxony), respectively.

C.5.2. H3 – Heterogeneity in wishful thinking

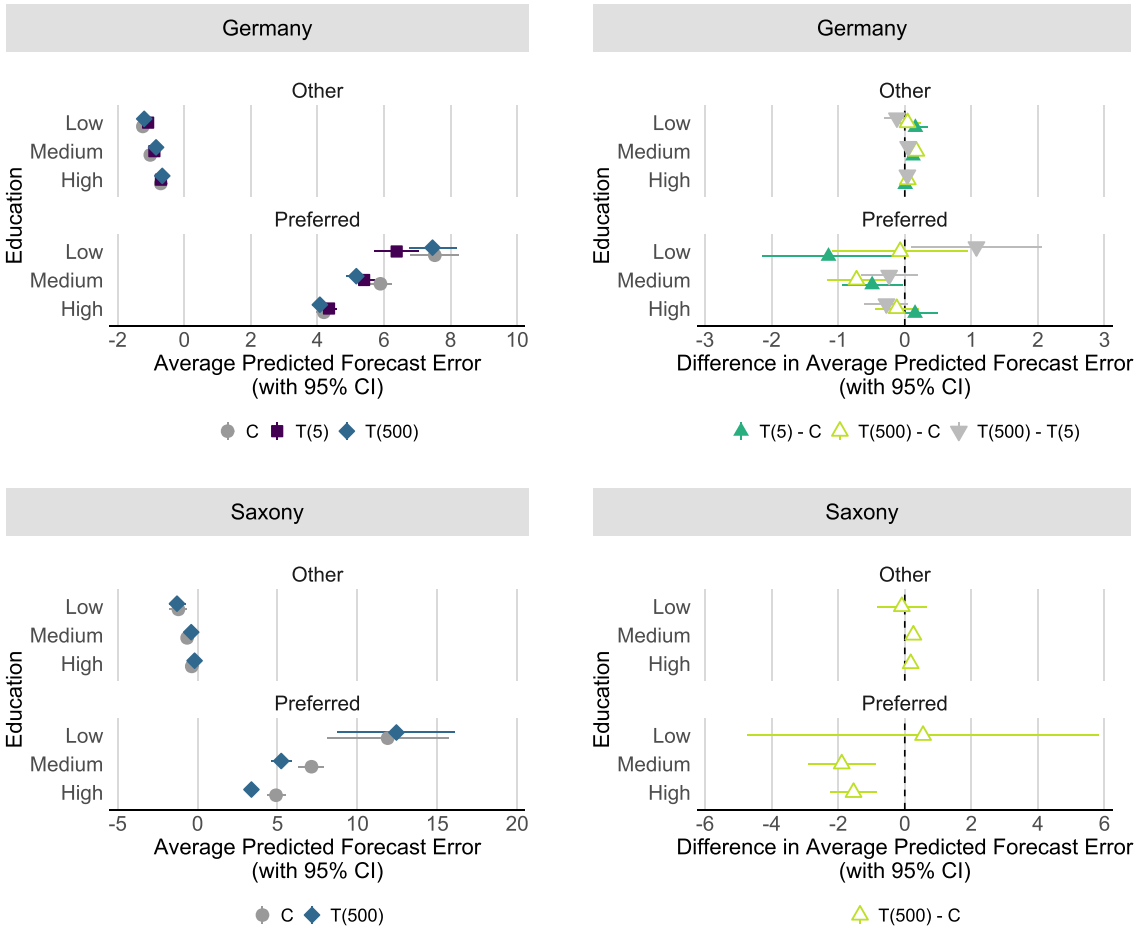


Fig. C.11. Effects of treatment on wishful thinking by education. *Note:* The figure is based on OLS regression models, that regress respondents' forecast error on an interaction of a preferred-party dummy (highest PTV score), treatment status, and educational attainment, and include survey-date fixed effects. The dots in the left panels show the average predicted forecast error by preferred-party status, experimental group, and education level, based on an observed-value approach. The right panels show the differences between groups. Lines correspond to 95% confidence intervals. The top row reports the federal election; the bottom row reports Saxony. For the preferred party, the treatment effects in the federal election are 0.15 [−0.18; 0.49] (*Treatment (5)*) and −0.12 [−0.45; 0.21] (*Treatment (500)*) percentage points for high education, −0.49 [−0.94; −0.04] and −0.72 [−1.17; −0.28] for medium education, and −1.15 [−2.14; −0.15] and −0.07 [−1.08; 0.95] for low education. In Saxony, the treatment effects of *Treatment (500)* are −1.54 [−2.24; −0.84] (high), −1.89 [−2.91; −0.88] (medium), and 0.55 [−4.74; 5.84] (low) percentage points. Results are based on responses from 22,661 respondents (for the federal election) and 4525 (for Saxony), respectively.

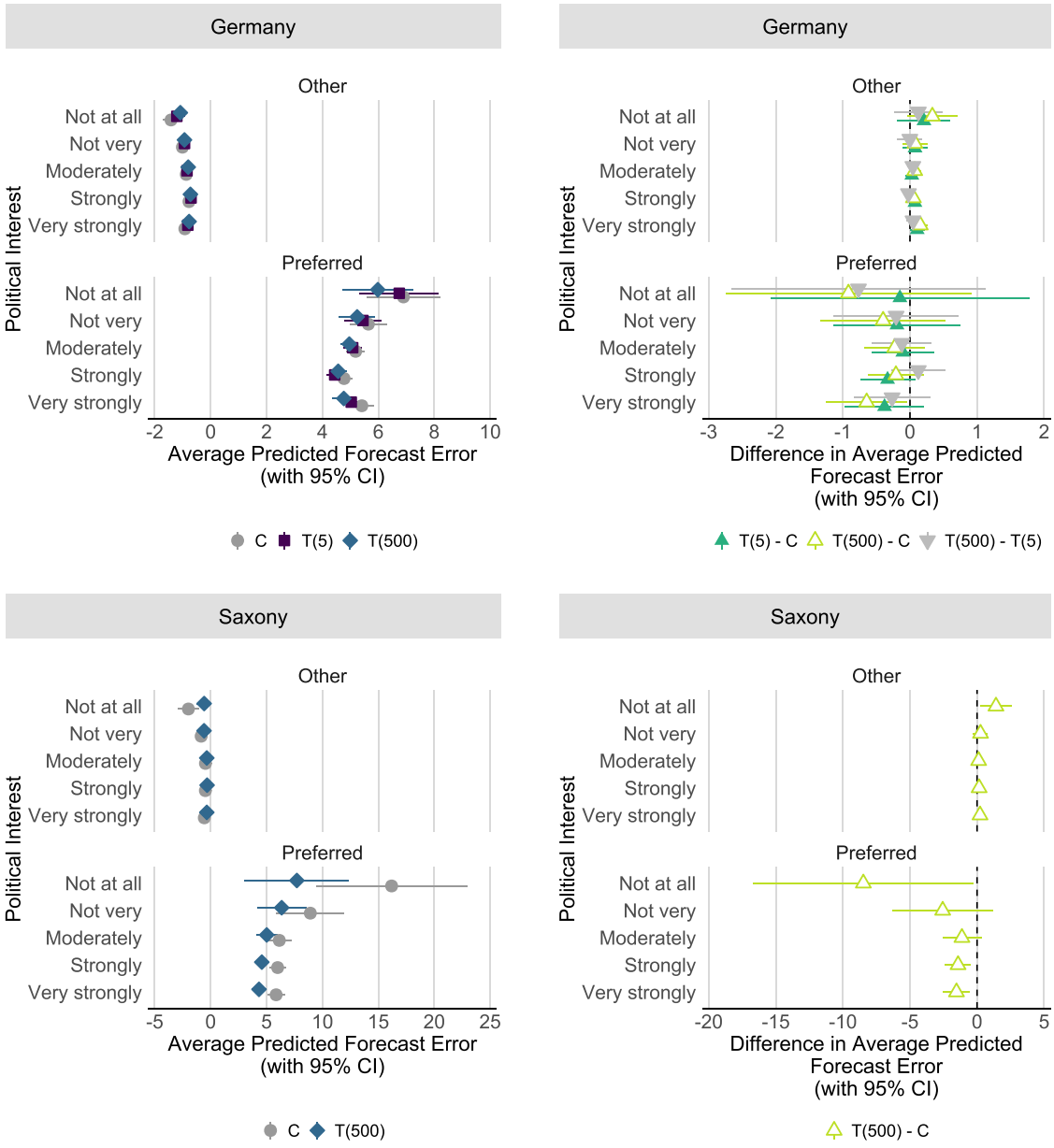


Fig. C.12. Effects of treatment on wishful thinking by political interest. *Note:* The figure is based on OLS regression models, regressing respondents' forecast error on an interaction of a preferred-party dummy (highest PTV score), treatment status, and political interest, including survey date fixed effects. The dots in the left panels show the average predicted forecast error by preferred-party status, experimental group, and political interest, based on an observed-value approach. The right panels show the differences between groups. Lines correspond to 95% confidence intervals. The top row reports the federal election, the bottom row reports Saxony. For the preferred party, the treatment effects in the federal election are −0.38 [−0.97; 0.21] (*Treatment (5)*) and −0.65 [−1.25; −0.04] (*Treatment (500)*) percentage points among the very strongly interested, −0.33 [−0.74; 0.07] and −0.21 [−0.62; 0.21] (strongly), −0.11 [−0.57; 0.36] and −0.23 [−0.68; 0.22] (moderately), −0.19 [−1.13; 0.75] and −0.40 [−1.33; 0.53] (not very), and −0.15 [−2.08; 1.78] and −0.92 [−2.74; 0.91] (not at all). In Saxony, the treatment effects of *Treatment (500)* are −1.53 [−2.52; −0.54] (very strongly), −1.42 [−2.37; −0.46] (strongly), −1.13 [−2.56; 0.31] (moderately), −2.56 [−6.31; 1.19] (not very), and −8.47 [−16.67; −0.28] (not at all) percentage points. Results are based on responses from 23,044 respondents (for the federal election) and 4662 (for Saxony), respectively.

Data availability

The data and code are available at https://github.com/hannahrajski/accuracy-goals-and-wishful-thinking-in-electoral-expectations.

References

Babad, E., 1995. Can accurate knowledge reduce wishful thinking in voters' predictions of election outcomes? J. Psychol. 129 (3), 285–300. https://doi.org/10.1080/00223980.1995.9914966

Babad, E., 1997. Wishful thinking among voters: motivational and cognitive influences. Int. J. Public Opin. Res. 9 (2), 105–125. https://doi.org/10.1093/ijpor/9.2.105

Babad, E., Katz, Y., 1991. Wishful thinking—against all odds. J. Appl. Soc. Psychol. 21 (23), 1921–1938. https://doi.org/10.1111/j.1559-1816.1991.tb00514.x

Babad, E., Hills, M., O'Driscoll, M., 1992. Factors influencing wishful thinking and predictions of election outcomes. Basic Appl. Soc. Psychol. 13 (4), 461–476. https://doi.org/10.1207/s15324834basp1304_6

Barnfield, M., 2020. Think twice before jumping on the bandwagon: clarifying concepts in research on the bandwagon effect. Polit. Stud. Rev. 18 (4), 553–574. https://doi.org/10.1177/1478929919870691

Barnfield, M., Phillips, J., Stoeckel, F., Mérola, V., Stöckli, S., Lyons, B., Thompson, J., Szewach, P., Reifler, J., 2025. Wishful thinking in response to events: evidence from the 2021 German federal election. Elect. Stud. 95, 102940. https://doi.org/10.1016/j.electstud.2025.102940

Blais, A., Bodet, M.A., 2006. How do voters form expectations about the parties' chances of winning the election? Soc. Sci. Q. 87 (3), 477–493. https://doi.org/10.1111/j.1540-6237.2006.00392.x
Boon, M., 2012. Predicting elections: a 'wisdom of crowds' approach. Int. J. Mark. Res. 54 (4), 465–483. https://doi.org/10.2501/IJMR-54-4-465-483
Bullock, J.G., Lenz, G., 2019. Partisan bias in surveys. Annu. Rev. Polit. Sci. 22 (1), 325–342. https://doi.org/10.1146/annurev-polisci-051117-050904
Bullock, J.G., Gerber, A.S., Hill, S.J., Huber, G.A., 2015. Partisan bias in factual beliefs about politics. Q. J. Polit. Sci. 10 (4), 519–578. https://doi.org/10.1561/100.00014074
Clifford, S., Jerit, J., 2016. Cheating on political knowledge questions in online surveys: an assessment of the problem and solutions. Public Opin. Q. 80 (4), 858–887. https://doi.org/10.1093/poq/nfw030
Cox, G.W., 1997. Making Votes Count: Strategic Coordination in the World's Electoral Systems. Cambridge University Press, Cambridge. https://doi.org/10.1017/CBO9781139174954
Dechenaux, E., Kovenock, D., Sheremeta, R.M., 2015. A survey of experimental research on contests, all-pay auctions and tournaments. Exp. Econ. 18 (4), 609–669. https://doi.org/10.1007/s10683-014-9421-0
Die Bundeswahlleiterin, 2025. Ergebnisse früherer Bundestagswahlen. Wiesbaden.
Dolan, K.A., Holbrook, T.M., 2001. Knowing versus caring: the role of affect and cognition in political perceptions. Polit. Psychol. 22 (1), 27–44. https://doi.org/10.1111/0162-895X.00224
Dufresne, Y., Jérôme, B., Lewis-Beck, M.S., Murr, A.E., Savoie, J., 2022. Citizen forecasting: the 2022 French presidential election. PS: Polit. Sci. & Polit. 55 (4), 730–734. https://doi.org/10.1017/S1049096522000567
Farjam, M., Bravo, G., 2024. Do you really believe that? The effect of economic incentives on the acceptance of real-world data in a polarized context. R. Soc. Open Sci. 11 (4), 240252. https://doi.org/10.1098/rsos.240252
Ganser, C., Riordan, P., 2015. Vote expectations at the next level. Trying to predict vote shares in the 2013 German federal election by polling expectations. Elect. Stud. 40, 115–126. https://doi.org/10.1016/j.electstud.2015.08.001
Graefe, A., 2014. Accuracy of vote expectation surveys in forecasting elections. Public Opin. Q. 78 (S1), 204–232. https://doi.org/10.1093/poq/nfu008
Graefe, A., 2016. Forecasting proportional representation elections from non-representative expectation surveys. Elect. Stud. 42, 222–228. https://doi.org/10.1016/j.electstud.2016.03.001
Granberg, D., Brent, E., 1983. When prophecy bends: the preference–expectation link in U.S. presidential elections, 1952–1980. J. Personal. Soc. Psychol. 45 (3), 477–491. https://doi.org/10.1037/0022-3514.45.3.477
Grewenig, E., Lergetporer, P., Werner, K., Woessmann, L., 2022. Incentives, search engines, and the elicitation of subjective beliefs: evidence from representative online survey experiments. J. Econom. 231 (1), 304–326. https://doi.org/10.1016/j.jeconom.2020.03.022
Gruca, T.S., Rietz, T.A., 2025. Iowa electronic markets: forecasting the 2024 US presidential election. PS: Polit. Sci. & Polit. 58 (2), 258–266. https://doi.org/10.1017/S1049096524000921
Hanmer, M.J., Ozan Kalkan, K., 2013. Behind the curve: clarifying the best approach to calculating predicted probabilities and marginal effects from limited dependent variable models. Am. J. Polit. Sci. 57 (1), 263–277. https://doi.org/10.1111/j.1540-5907.2012.00602.x
Hayes, S.P., 1936. The predictive ability of voters. J. Soc. Psychol. 7 (2), 183–191. https://doi.org/10.1080/00224545.1936.9921660
Irwin, G.A., van Holsteyn, J.J.M., 2002. According to the polls: the influence of opinion polls on expectations. Public Opin. Q. 66 (1), 92–104. https://doi.org/10.1086/338632
Jamieson, T., Weller, N., 2020. The effects of certain and uncertain incentives on effort and knowledge accuracy. J. Exp. Polit. Sci. 7 (3), 218–231. https://doi.org/10.1017/XPS.2019.27
Kahneman, D., 2011. Thinking, Fast and Slow, first ed. Farrar, Straus and Giroux, New York.
Kahneman, D., Tversky, A., 1979. Prospect theory: an analysis of decision under risk. Econometrica 47 (2), 263–292. https://doi.org/10.2307/1914185
Khanna, K., Sood, G., 2018. Motivated responding in studies of factual learning. Polit. Behav. 40 (1), 79–101. https://doi.org/10.1007/s11109-017-9395-7
Kou, S.G., Sobel, M.E., 2004. Forecasting the vote: a theoretical comparison of election markets and public opinion polls. Polit. Anal. 12 (3), 277–295. https://doi.org/10.1093/pan/mph019
Krizan, Z., Sweeny, K., 2013. Causes and consequences of expectation trajectories: "High" on optimism in a public ballot initiative. Psychol. Sci. 24 (5), 706–714. https://doi.org/10.1177/0956797612460690
Krosnick, J.A., Narayan, S., Smith, W.R., 1996. Satisficing in surveys: initial evidence. New Dir. Eval. 1996 (70), 29–44. https://doi.org/10.1002/ev.1033
Kunda, Z., 1990. The case for motivated reasoning. Psychol. Bull. 108 (3), 480–498. https://doi.org/10.1037/0033-2909.108.3.480
Lehrer, R., Juhl, S., Gschwend, T., 2019. The wisdom of crowds design for sensitive survey questions. Elect. Stud. 57, 99–109. https://doi.org/10.1016/j.electstud.2018.09.012
Leininger, A., Murr, A.E., Stötzer, L., Kayser, M.A., 2026. Citizen forecasting in a mixed electoral system: the 2021 German federal election as a test case. Int. J. Forecast. 42 (1), 203–215. https://doi.org/10.1016/j.ijforecast.2025.03.007
Leiter, D., Reilly, J.L., Stegmaier, M., 2020. Echoing certainty in uncertain times: network partisan agreement and the quality of citizen forecasts in the 2015 Canadian election. Elect. Stud. 63, 102115. https://doi.org/10.1016/j.electstud.2019.102115
Levine, R., 2007. Sources of bias in voter expectations under proportional representation. J. Elect. Public Opin. Parties 17 (3), 215–234. https://doi.org/10.1080/17457280701617060
Lewis-Beck, M.S., Skalaban, A., 1989. Citizen forecasting: can voters see into the future? Br. J. Polit. Sci. 19 (1), 146–153. https://doi.org/10.1017/S000712340000538X
Lewis-Beck, M.S., Stegmaier, M., 2011. Citizen forecasting: can UK voters see the future? Elect. Stud. 30 (2), 264–268. https://doi.org/10.1016/j.electstud.2010.09.012
Lewis-Beck, M.S., Tien, C., 1999. Voters as forecasters: a micromodel of election prediction. Int. J. Forecast. 15 (2), 175–184. https://doi.org/10.1016/S0169-2070(98)00063-6
Mayerl, J., 2013. Response latency measurement in surveys. Detecting strong attitudes and response effects. Surv. Methods: Insights Field. https://doi.org/10.13094/SMIF-2013-00005
Mayerl, J., Urban, D., 2008. Antwortreaktionszeiten in Survey-Analysen. VS Verlag für Sozialwissenschaften, Wiesbaden. https://doi.org/10.1007/978-3-531-91147-2
Meffert, M.F., Gschwend, T., 2010. Strategic coalition voting: evidence from Austria. Elect. Stud. 29 (3), 339–349. https://doi.org/10.1016/j.electstud.2010.03.005
Meffert, M.F., Gschwend, T., 2011. Polls, coalition signals and strategic voting: an experimental investigation of perceptions and effects. Eur. J. Polit. Res. 50 (5), 636–667. https://doi.org/10.1111/j.1475-6765.2010.01986.x
Meffert, M.F., Huber, S., Gschwend, T., Pappi, F.U., 2011. More than wishful thinking: causes and consequences of voters' electoral expectations about parties and coalitions. Elect. Stud. 30 (4), 804–815. https://doi.org/10.1016/j.electstud.2011.08.001
Miller, M.K., Wang, G., Kulkarni, S.R., Poor, H.V., Osherson, D.N., 2012. Citizen forecasts of the 2008 U.S. presidential election. Polit. Policy 40 (6), 1019–1052. https://doi.org/10.1111/j.1747-1346.2012.00394.x
Mongrain, P., 2021a. Did you see it coming? Explaining the accuracy of voter expectations for district and (sub)national election outcomes in multi-party systems. Elect. Stud. 71, 102317. https://doi.org/10.1016/j.electstud.2021.102317
Mongrain, P., 2021b. A technocratic view of election forecasting: weighting citizens' forecasts according to competence. Int. J. Public Opin. Res. 33 (3), 713–723. https://doi.org/10.1093/ijpor/edab010
Mongrain, P., 2023a. Suspicious minds: unexpected election outcomes, perceived electoral integrity and satisfaction with democracy in American presidential elections. Polit. Res. Q. 76 (4), 1589–1603. https://doi.org/10.1177/10659129231166679
Mongrain, P., 2023b. With a little help from my friends? The impact of social networks on citizens' forecasting ability. Eur. J. Polit. Res. 62 (4), 1320–1346. https://doi.org/10.1111/1475-6765.12576
Morisi, D., Leeper, T., 2024. What influences citizen forecasts? The effects of information, elite cues, and social cues. Polit. Behav. 46 (1), 21–41. https://doi.org/10.1007/s11109-022-09811-4
Murr, A., 2011. "Wisdom of crowds"? A decentralised election forecasting model that uses citizens' local expectations. Elect. Stud. 30 (4), 771–783. https://doi.org/10.1016/j.electstud.2011.07.005
Murr, A., 2015. The wisdom of crowds: applying Condorcet's jury theorem to forecasting US presidential elections. Int. J. Forecast. 31 (3), 916–929. https://doi.org/10.1016/j.ijforecast.2014.12.002
Murr, A., 2016. The wisdom of crowds: what do citizens forecast for the 2015 British general election? Elect. Stud. 41, 283–288. https://doi.org/10.1016/j.electstud.2015.11.018
Murr, A., 2025. Predicciones ciudadanas de las elecciones presidenciales mexicanas, 2000–2024. Polít. Gob. 32 (1). http://www.politicaygobierno.cide.edu/index.php/pyg/article/view/1751.
Murr, A., Lewis-Beck, M.S., 2021. Citizen forecasting 2020: a state-by-state experiment. PS: Polit. Sci. & Polit. 54 (1), 91–95. https://doi.org/10.1017/S1049096520001456
Murr, A., Lewis-Beck, M.S., 2022. Citizen forecasts of the 2021 German election. PS: Polit. Sci. & Polit. 55 (1), 97–101. https://doi.org/10.1017/S1049096521000925
Neundorf, A., Öztürk, A., 9 Dec 2021. Recruiting Research Participants Through Facebook Advertisements: A Handbook. 87rg3v1, https://doi.org/10.31219/osf.io/87rg3
Neundorf, A., Öztürk, A., 2023. How to improve representativeness and cost-effectiveness in samples recruited through Meta: a comparison of advertisement tools. PLoS One 18 (2), e0281243. https://doi.org/10.1371/journal.pone.0281243
Prior, M., Lupia, A., 2008. Money, time, and political knowledge: distinguishing quick recall and political learning skills. Am. J. Polit. Sci. 52 (1), 169–183. https://doi.org/10.1111/j.1540-5907.2007.00306.x
Prior, M., Sood, G., Khanna, K., 2015. You cannot be serious: the impact of accuracy incentives on partisan bias in reports of economic perceptions. Q. J. Polit. Sci. 10 (4), 489–518. https://doi.org/10.1561/100.00014127
Rathje, S., Roozenbeek, J., Van Bavel, J.J., van der Linden, S., 2023. Accuracy and social motivations shape judgements of (mis)information. Nat. Hum. Behav. 7 (6), 892–903. https://doi.org/10.1038/s41562-023-01540-w
Reif, K., Schmitt, H., 1980. Nine second-order national elections – a conceptual framework for the analysis of European election results. Eur. J. Polit. Res. 8 (1), 3–44. https://doi.org/10.1111/j.1475-6765.1980.tb00737.x
Rittmann, O., Neunhoeffer, M., Gschwend, T., 2025. How to improve the substantive interpretation of regression results when the dependent variable is logged. Polit. Sci. Res. Methods 13 (1), 203–211. https://doi.org/10.1017/psrm.2023.29
Searles, K., Ginn, M.H., Nickens, J., 2016. For whom the poll airs: comparing poll results to television poll coverage. Public Opin. Q. 80 (4), 943–963. https://doi.org/10.1093/poq/nfw031
Searles, K., Smith, G., Sui, M., 2018. Partisan media, electoral predictions, and wishful thinking. Public Opin. Q. 82 (S1), 888–910. https://doi.org/10.1093/poq/nfy006
Sjöberg, L., 2009. Are all crowds equally wise? A comparison of political election forecasts by experts and the public. J. Forecast. 28 (1), 1–18. https://doi.org/10.1002/for.1083
Statistisches Bundesamt (Destatis), 2026a. Code: 12211-0100 - Tabelle 'Bevölkerung ab 15 Jahren in Hauptwohnsitzhaushalten: Deutschland, Jahre, Geschlecht, Altersgruppen, Allgemeine Schulausbildung', from https://genesis.destatis.de/datenbank/online/statistic/12211/table/12211-0100 (Retrieved 19 June 2026).
Statistisches Bundesamt (Destatis), 2026b. Code: 12211-1100 - Tabelle 'Bevölkerung ab 15 Jahren in Hauptwohnsitzhaushalten: Bundesländer, Jahre, Geschlecht, Altersgruppen,

Allgemeine Schulausbildung', from https://genesis.destatis.de/datenbank/online/statistic/12211/table/12211-1100 (Retrieved 28 June 2026).
Statistisches Landesamt des Freistaates Sachsen, 2024a. Repräsentative Wahlstatistik, from http://www.wahlen.sachsen.de/landtagswahlen-2024-repraesentative-wahlstatistik.html (Retrieved 28 June 2026).
Statistisches Landesamt des Freistaates Sachsen, 2024b. Wahlergebnisse, from http://www.wahlen.sachsen.de/landtagswahl-2024-wahlergebnisse.php (Retrieved 28 June 2026).
Stiers, D., Dassonneville, R., 2018. Affect versus cognition: wishful thinking on election day. Int. J. Forecast. 34 (2), 199–215. https://doi.org/10.1016/j.ijforecast.2017.12.001
Stoetzer, L.F., Kayser, M.A., Leininger, A., Murr, A.E., 2024. Voters' expectations in constituency elections without local polls. Public Opin. Q. 88 (2), 408–418. https://doi.org/10.1093/poq/nfae015
The Federal Returning Officer, 2025a. Bundestag election 2025, from https://www.bundeswahlleiterin.de/en/bundestagswahlen/2025/ergebnisse/bund-99.html#zweitstimmen-prozente11 (Retrieved 29 September 2026).
The Federal Returning Officer, 2025b. Bundestag election 2025 - results of the representative electoral statistics - Wahlberechtigte, Wählende und Wahlbeteiligung nach Geschlecht und Geburtsjahresgruppen, from https://www.bundeswahlleiterin.de/en/bundestagswahlen/2025/ergebnisse/repraesentative-wahlstatistik.html (Retrieved 19 June 2026).
Thompson-Collart, B., Dufresne, Y., Brie, E., 2024. Can Latin American voters see the future? Citizen forecasting in Argentina. Rev. Latinoam. Opinión Pública 13, e31348. https://doi.org/10.14201/rlop.31348
Thompson-Collart, B., Cadieux, H., Ouellet, C., Dufresne, Y., 2025. Lessons learned: citizen forecasting, candidate resignations, and the 2024 US presidential election. PS: Polit. Sci. & Polit. 58 (2), 312–317. https://doi.org/10.1017/S1049096524000969
Tikochinski, R., Babad, E., 2023. Voters' wishful thinking in an unprecedented event of three national elections repeated within one year: fast thinking, bias, high emotions and potential rationality. Think. Reason. 29 (2), 250–275. https://doi.org/10.1080/13546783.2022.2071990
Tourangeau, R., 1984. Cognitive Sciences and Survey Methods. National Academy Press, Washington, DC.
Zicht, W., Cantow, M., 2025. Die letzten Umfragen zur Bundestagswahl 2025 (Sonntagsfrage). Wahlrecht.de, from https://www.wahlrecht.de/umfragen/archiv/2025.htm (Retrieved 1 July 2026).
Zicht, W., Cantow, M., 2026. Sonntagsfrage zur Landtagswahl in Sachsen. Wahlrecht.de, from https://www.wahlrecht.de/umfragen/landtage/sachsen.htm (Retrieved 30 June 2026).